LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES

A Capstone Project
presented to the faculty of
College of Computer Studies
Camarines Sur Polytechnic Colleges

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Technology

By

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December, 2025

DEDICATION

This capstone project is dedicated first and foremost to God, whose guidance, wisdom, and sustaining grace strengthened us throughout this journey. His presence gave us perseverance, clarity, and purpose from the beginning until the completion of this work.

We also dedicate this study to our families, whose unwavering love, support, and understanding have been our greatest foundation. Their patience, encouragement, and unshakeable belief in our abilities motivated us to keep striving, even in the most challenging moments.

This dedication is likewise extended to our advisers, mentors, and instructors, whose knowledge, expertise, and consistent guidance shaped the direction and quality of this project. Their commitment to our learning has played a significant role in our academic growth.

Lastly, we offer this work to our peers and to all individuals who supported us in any way whether through cooperation, kind words, or simple acts of encouragement. Your presence and support inspired us to complete this study with dedication, integrity, and determination.

ACKNOWLEDGMENTS

The researchers would like to express their heartfelt gratitude to everyone who contributed in any way to the completion of this project.

To **Rosel O. Onesa, MIT**, OIC Dean of the College of Computer Studies, for his unwavering support and guidance, which played a significant role in making this study possible.

To **Shiela Dona S. Manlapaz, MIT**, their project adviser, whose constant guidance, insightful suggestions, and encouragement supported the researchers throughout the entire development of the Intelligent Library Management System.

To **Ms. Marivic L. Ramizares**, a dedicated consultant whose expertise and constructive recommendations significantly strengthened the system and the research work.

To **Ms. Ellamae S. Allamo**, our capstone project grammarian, for the patience, time, and commitment in reviewing and refining the written components of this study. Her careful editing and guidance contributed greatly to the clarity and coherence of the final manuscript.

To the project panelists **Maricris L. Ramizares, MIT, Jeremy Jireh S. Neo MIS, Josefina H. Llagas** thank you for sharing your knowledge, providing meaningful comments, and guiding the researchers toward improving both the system and the research. Your feedback was truly valuable.

To our classmates and friends, the researchers extend their deepest thanks for the support, understanding, and encouragement offered throughout the journey. Your motivation helped us remain resilient from beginning to end.

To our families whose unconditional love, sacrifices, and support have always been the researchers’ source of strength. Your constant guidance and encouragement made this achievement possible.

To **Mr. Jerome L Mamansag**, Administrative Assistant, thank you for your understanding, assistance, and continuous support throughout the process.

To **Yves Aristeo A. Febres**, **RL, MLIS** thank you for your willingness to answer our queries, even beyond your responsibilities. Your help greatly aided the completion of this project.

Finally, the researchers express their deepest gratitude to **God**, for the strength, wisdom, and perseverance granted to them throughout the entire journey.

This accomplishment also belongs to everyone who guided, supported, and believed in the researchers from start to finish.

ABSTRACT

Title:	Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges
Authors:	Ma. Angelica T. Nazaria John Patrick A. Acbang Dince A. Deliva
Number of Pages:	178
School:	Camarines Sur Polytechnic Colleges
Degree Conferred:	Bachelor of Science in Information Technology
Keywords:	Intelligent Library Management System, Library Automation, Data Analytics, Collection Development

This study developed the Intelligent Library Management System to improve library operations at the Camarines Sur Polytechnic Colleges Library by integrating book circulation, reporting, overdue risk assessment, SMS notifications, and analytics into a single digital platform. The system supports data-driven decision-making by providing insights into book usage and resource utilization. The system was evaluated through unit, integration, usability, and performance testing. Unit testing achieved an overall 100% pass rate, with module execution results of 18% for Prescriptive Analytics, 13% for Overdue Risk, and 72% for Book Reports. Integration testing had a 100% success rate as well, and the major modules got 1.00 each. The usability testing showed to be very positive user acceptance with an overall mean of 3.67 (Agree). Performance testing was done through 100 simultaneous requests which had 0 failed requests, average throughput of 9.34 requests per second, and 90% of the responses completed 1067 ms confirming reliability and performance within acceptable limits. Overall, the system is functional, reliable and capable of supporting informed library management decisions.

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CHAPTER 1

INTRODUCTION

This chapter discussed the project context, purpose and description of the project, objectives of the project, significance of the project, scope and limitations of the study and project dictionary of the capstone project, significance of the project, scope and limitations of the study and project dictionary of the capstone project “Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges”

1.1 Project Context

This chapter discussed the project context, purpose and description of the project, objectives of the project, significance of the project, scope and limitations of the study and project dictionary of the capstone project, significance of the project, scope and limitations of the study and project dictionary of the capstone project “Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges”

Education is a cornerstone of societal advancement, fostering personal growth and national development by imparting the knowledge and skills necessary for individuals to navigate complex problems and achieve their goals. It transforms an individual’s way of life, enabling informed decision-making and problem-solving. As a tangible source of knowledge and information, education plays vital role in shaping informed and capable citizens. Furthermore, education provides a platform for individuals to explore their interests and passions, whether through academics, extracurricular activities, or vocational training [1].

In the context of education, knowledge management practices are essential for promoting the analysis and verification of the quality of an institutional environment, as well as the coordination and implementation of activities in an organization. The rapid expansion of knowledge underscores the importance of leveraging information and communication environments to activate the learning process and accommodate diverse student needs [2].

Libraries were essential source for knowledge, learning and community engagement. They provide access to a wide range of sources, support education and research, foster a culture of reading and lifelong learning. They support students in their academic pursuits by providing access to research resources and offering assistance with information literacy skills. Libraries are dynamic institutions that adapt to the evolving educational needs of their communities, serving as vital centers for knowledge, culture, social interaction, and play a critical role in improving the quality of education [3].

The culture of reading is an inherently linked to lifelong learning, fostering intellectual stimulation and mental sharpness as individuals age. Lifelong learning emphasizes continuous personal growth and self-improvement, and reading habits cultivated early in life contribute significantly to this ongoing development [4].\

At Camarines Sur Polytechnic Colleges (CSPC), the library utilizes the KOHA Integrated Library System, an open-source platform widely adopted by academic institutions for cataloging, circulation, and inventory management. This is evidenced by the library's online catalog, which operates on the system. While KOHA provides foundational automation and digital accessibility, much of the data handling especially related to circulation trends, overdue monitoring, and collection development remains underutilized. Monthly statistical reports are still compiled manually, and there is limited capacity for generating data-driven insights that could inform acquisition and resource optimization strategies [5].

Traditional library management methods, still involve repetitive tasks and lack real-time analytics for decision-making. Data regarding the books present in the library including their genres, publication dates, and frequency of use is essential for making informed decisions about acquisitions, collection development, and resource allocation.[6].

1.2 Purpose and Description of the Project

The project aimed to enhance the efficiency and effectiveness of library management at

Camarines Sur Polytechnic Colleges by adopting a data-driven approach to book tracking, circulation monitoring, and resource allocation. Libraries play a critical role in supporting education, research, and lifelong learning; however, traditional library management practices often rely on manual record-keeping, which is time-consuming, error-prone, and insufficient for generating comprehensive data needed for informed decision-making.

To address these limitations, the project proposes the development of a system that applies data analytics to monitor borrowing behavior, track book usage patterns, and support evidence-based collection management. The system includes automated overdue notifications to promote timely returns and reduce circulation delays, as well as real-time monitoring and report generation features that provide meaningful insights into library operations.

1.3 Objectives of the Project

This study aimed to use real-time data on borrowing and returning books in the library, integrating the tracking of used books within the library premises through real-time API integration to generate comprehensive outputs for book collection and data analytics. The objective is to ensure immediate reflection of book-related activities as they occur, enabling dynamic updates and accurate reporting across the system to support efficient library operations and timely decision-making.

Specifically, this study aimed the following:

1. Conduct an investigation into the current practices and issues that pertain to the use, borrowing, and return of books at the CSPC library.
2. To develop and implement a real-time tracking system for the usage of books in the library that has the following modules:

   2.1 Book Report Generation Module

   2.2 Overdue Notification Module

   2.3 Prescriptive Analytics Module

3. Perform black box testing by executing test cases based on system requirements using the following criteria:

   3.1 Integration Testing
   3.2 Unit Testing
   3.3 Usability Testing
   3.4 Performance Testing

4. Deploy the developed system in the CSPC Library environment to evaluate its real-world functionality, stability, and effectiveness in supporting library operations.

1.4 Significance of the Study

The researchers believe that this study benefited the following:

Traditional library management methods, still involve repetitive tasks and lack real-time analytics for decision-making. Data regarding the books present in the library including their genres, publication dates, and frequency of use is essential for making informed decisions about acquisitions, collection development, and resource allocation.[6].

CSPC Library Administration. For administrators, the study offers valuable insights that can guide institutional planning and policy development. Data analytics provides a clear overview of library performance, resource utilization, and user demand, helping administrators allocate budgets more effectively. The intelligent system serves as an evidence-based tool for assessing service quality, supporting decisions that align with the institution’s goals of improving academic support systems.

Students. Students benefited from a more responsive and user-centered library experience. By analyzing borrowing behaviors and resource usage, the system helps ensure that frequently used materials are readily available, reducing waiting times and improving access to essential academic resources. The enhanced library system supports students’ research and learning needs by offering timely and relevant services tailored to their actual usage patterns.

Researchers. The study directly benefits the researchers by enhancing their knowledge and skills in system development, data analytics, and evidence-based decision-making within an academic library context. Through the design, implementation, and evaluation of the intelligent library management system, the researchers gain practical experience in applying theoretical concepts to real-world problems. This contributes to their professional growth, technical competence, and research capability, particularly in the fields of information systems, library science, and data-driven technologies.

Future Researchers. This study contributes to the body of knowledge on intelligent library systems and the application of data analytics in academic institutions. Future researchers can use the findings, methodology, and system model as a foundation for exploring more advanced analytics techniques, integrating predictive models, or expanding the system to include other academic support services. The study serves as a reference point for further innovation in data-driven educational technologies.

1.5 Scope and Limitations of the Project

This project focuses on the design, development, implementation, and evaluation of a real-time, data-driven library management system for the Camarines Sur Polytechnic Colleges (CSPC) Library. The system integrates real-time API-based tracking to ensure immediate reflection of book borrowing and returning activities, enabling dynamic updates, accurate reporting, and timely decision-making. The scope includes investigating existing library practices, developing key functional modules namely the Book Report Generation, Overdue Notification, and Prescriptive Analytics modules and conducting comprehensive black box testing, including functional and non-collection management.

Despite its comprehensive scope, the study is limited to quantitative data retrievable within the Camarines Sur Polytechnic Colleges Library during a defined timeframe, such as borrowing frequency and circulation records. To comply with data privacy policies, the system does not store or display personally identifiable borrower information, nor does it capture qualitative feedback or in-library browsing behavior. The accuracy of analytics outputs depends on the completeness and correctness of recorded data and may be affected by technical issues or incomplete records. Additionally, the system is tailored to the Camarines Sur Polytechnic Colleges Library's operational context and may require modification for use in other institutions. Its successful implementation also relies on the cooperation of library staff and users, as well as the stability of the tools used for real-time data collection and analysis.

1.6 Project Dictionary

This study, titled "Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges" utilizes significant concepts and terminologies that are critical to comprehend the scope, methodology, and goals of the project. Below are a list of terms and their definitions as they pertain to this study:

Book Collection. A book collection is a group of books assembled and organized according to specific themes, subjects, authors, or historical significance, typically maintained by individuals, libraries, or institutions [6]. In this research, book collection refers to the organized repository of printed materials within the library system, which analyzed and managed to ensure accessibility, proper categorization, and support for resource tracking and user needs.

Black Box Testing. Black box testing, a form of testing that is performed with no knowledge of a system's internals, can be carried out to evaluate the functionality, security, performance, and other aspects of an application [8]. In this research, black box testing was employed to assess the library management system's performance by simulating user interactions and verifying that the system responds correctly to various inputs, ensuring reliability and usability without delving delving into the internal codebase.

Data Analytics. The systematic computer analysis of data or statistics in order to draw conclusions, trends, and patterns [9]. It is the process of analyzing circulation data gathered through barcode scanning and existing data on borrowing, in order to make decisions regarding the collection development of the library.

Evidence-Based Collection Development. A data-based process for library collection management and migration, relying on measurement, quality control, planning and regular checkpoints to ensure orderly and efficient implementation [10]. It is particularly concerned with the application of data harvested through barcode technology and analytics in order to maximize selection and book maintenance.

Integration Testing. Checks how individual software modules work together when combined. It identifies issues in how components interact and exchange data [11]. In this study, integration testing ensure that all modules of the library management system such as reporting, analytics, and overdue notifications properly when connected.

Performance Testing. Evaluates how a system behaves under expected workloads, measuring speed, responsiveness, and stability [12]. In this project, performance testing ensures that the library system remains functional and efficient under conditions of high usage or data volume.

Real-Time API. An application program interface through which systems are able to exchange data in real time, allowing real-time information [13].In this case, it retrieves and refreshes book collection information, enabling real-time analytics and outputs for library management.

Unit Testing. Involves checking individual components or functions of the software to ensure they work as intended in isolation. This helps catch errors early in development [14]. In this study, unit testing is employed to verify the accuracy of core modules, including analytics and reporting prior to full integration.

Usability Testing. Evaluates how user-friendly and intuitive a system is by observing how real users interact with it while performing specific tasks [15].]. In this project, usability testing ensures that library staff can navigate and use the system efficiently.

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CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter is a comprehensive collection of local and foreign literature and research aligned with the project objectives and scope, "Leveraging Data Analytics for Intelligent Library Management for Camarines Sur Polytechnic Colleges.". It reviews important concepts, technological innovations, best practices, and existing research concerning library management systems, integration of barcode technology, real-time APIs, and data-driven. By establishing a clear contextual background, this assessment highlights the importance, need, and expected influence of the proposed system, reiterating its value to contemporary library administration and scholarly resource availability.

2.1 Related Literature and System

The review of the current related literature and systems, both foreign and local, gave the researchers a basis for conceptualizing and formulating Leveraging Data Analytics for Intelligent Library Management for Camarines Sur Polytechnic Colleges Integrated Library Management System.

2.1.1 Existing Library Practices and the Need for Automation

Several studies highlight persistent inefficiencies in libraries that rely on manual or semi-automated processes. Padilla [2022], in a study conducted at Cagayan State University–Piat, identified severe operational issues in manual library environments, particularly in book searching, borrowing, and returning. Librarians also reported difficulties in cataloging, inventory control, record retention, and report generation. These challenges indicate structural weaknesses in data accuracy, accessibility, and process reliability critical elements for effective library management [1].

Supporting these findings, Arroyo et al. [2023] developed an automated library management system for Llano High School and demonstrated that automation significantly reduced delays in book retrieval, improved tracking of borrowed materials, and simplified inventory procedures. Features such as real-time transaction updates and automated reporting minimized human error and enhanced operational efficiency. These studies justify the need to investigate existing practices at the CSPC Library as a foundation for system development [3].

Khan and Ayesha [2022] further emphasized that effective Information Management Systems (IMS) must be evaluated based on reliability, data security, usability, compliance with cataloging standards, and system support. While their work focuses on system selection, it reinforces the importance of structured, reliable systems to address the deficiencies observed in manual and outdated library operations [15].

2.1.2 Leveraging Data Analytics to Enhance Library Services

The integration of data analytics has become central to modern library management, enabling evidence-based decision-making and user-centered services. Muazu et al. [2024] highlighted that big data analytics allows libraries to analyze circulation trends, forecast user demand, and optimize collection development. Their findings show that analytics-driven libraries deliver more responsive services and improve resource allocation efficiency[6].

Zong [2023] demonstrated that data mining techniques applied to borrowing patterns and usage logs can support predictive analytics and personalized services in university libraries. These insights enable proactive collection management and budget optimization, aligning closely with the goals of the proposed prescriptive analytics and report generation modules [5].

Earlier studies emphasize the importance of establishing robust analytics infrastructure and continuous evaluation in library environments to support evidence-based and responsive services. Mojjada and Aakundi [2025] demonstrated that analytics tools such as dashboards, predictive models, and metrics analysis enhance service efficiency and strategic decision-making, while related research highlights the role of real-time data processing and visualization in improving the accessibility and usability of library services an approach adopted in this study’s system design

[22].

2.1.3 Evaluation and Uses of Artificial Intelligence in Libraries

Artificial Intelligence (AI) has become a strategic component of modern library systems, supporting both operational automation and evaluative analytics. Asim et al. [2023] found that AI tools such as machine learning–based search, metadata extraction, and intelligent resource allocation improve access, efficiency, and administrative decision-making in university libraries [14].

Shahzad, Khan, and Iqbal [2024] systematically reviewed AI applications in academic libraries and identified predictive analytics, performance monitoring, and digital resource management as key use cases. These applications enable libraries to evaluate circulation trends, forecast demand, and assess service effectiveness functions aligned with the prescriptive analytics module of the proposed system[15].

Mutia et al. [2024] emphasized the importance of user perception in evaluating AI-enabled systems, reporting generally positive attitudes among librarians when AI tools are user-centered and transparent. This supports the inclusion of usability and system system testing in evaluating AI-assisted library applications [16].

2.1.4 Reliability and Quality of Hugging Face Pre-Trained Models

Recent empirical studies have begun to systematically examine the reuse, reliability, and documentation quality of pre-trained models (PTMs) hosted on Hugging Face highlighting both strengths and important caveats. One foundational work is An Empirical Study of Pre-Trained Model Reuse in the Hugging Face Deep Learning Model Model Registry Jiang et al., [2023], published at International Conference on Software Engineering (ICSE 2023). This study interviewed practitioners using Hugging Face and analyzed over 63,000 PTM packages. The authors identified key attributes necessary for safe reuse such as provenance, reproducibility, and portability and documented significant challenges: missing metadata, discrepancies between claimed and actual performance, and potential model risks. Their measurement-based findings show that despite the large number of available models, not all are equally reliable or trustworthy, and reuse requires careful evaluation rather than blind trust [20].

Complementing that, the recent paper Model Hubs and Beyond: Analyzing Model Popularity, Performance, and Documentation Kadasi et al., [2025] offers a broad evaluation of 500 sentiment-analysis models on Hugging Face. The study used nearly 80,000 human annotations and extensive model evaluation to compare "popularity" downloads, likes, stars with actual model performance and documentation quality. Their findings cast doubt on using popularity as a proxy for quality: popularity did not reliably correlate with performance. Moreover, approximately 80% of the evaluated models lacked detailed documentation about training or evaluation processes, and about 88% of model cards overstated model performance. The authors argue that model card documentation (metadata, evaluation details, limitations) is often incomplete or misleading, which undermines trustworthiness and can lead to problematic reuse [21].

These studies demonstrate that while Hugging Face enables broad reuse of pre-trained models, which reduces effort and cost compared to training from scratch it also poses nontrivial risks related to model accuracy, transparency, and reproducibility. They suggest that researchers and developers must treat models from the hub critically: verifying metadata, re-evaluating performance on one's own benchmarks or data, and applying caution especially when models are used in sensitive or high-stakes contexts. In addition, proper documentation and version control are essential to track model updates and prevent unintended changes in behavior over time. Establishing standardized evaluation and reporting practices can further improve trust and comparability across reused models. Ultimately, responsible adoption of pre-trained models requires balancing efficiency gains with rigorous validation and ethical accountability. Moreover, fostering transparency through clear disclosure of training data sources, limitations, and intended use can help mitigate misuse and support more informed decision-making by model adopters.

2.1.5 Black Box Testing

These studies demonstrate that while Hugging Face enables broad reuse of pre-trained models, which reduces effort and cost compared to training from scratch it also poses nontrivial risks related to model accuracy, transparency, and reproducibility. They suggest that researchers and developers must treat models from the hub critically: verifying metadata, re-evaluating performance on one's own benchmarks or data, and applying caution especially when models are used in sensitive or high-stakes contexts.

Black box testing is widely used to validate system behavior without examining internal code structures. Fernandes and Lina [2021] emphasized its importance in verifying the functional correctness of automated library applications, particularly through techniques such as Boundary Value Analysis. Their work highlights the role of user feedback in identifying functional gaps and improving system reliability [10].

Recent studies extend black box testing beyond functional validation to include performance evaluation using benchmark tools. Apache Bench (ab) is an established command-line benchmarking utility that measures HTTP request handling, concurrency, and response time by generating controlled loads against a system. Official documentation describes how ab sends repeated requests and reports performance metrics such as requests per second and latency, making it suitable for performance testing in a black box context. Performance testing surveys also list Apache Bench as a practical tool for quick load and latency, making it suitable for performance testing in a black box context. Performance testing surveys also list Apache Bench as a practical tool for quick load and throughput evaluation, supporting its use for performance measurement in system evaluation 24]. Ahn and Lee [2021] and Jiang et al. [2021] further expanded black box testing to include predictive load modeling and robustness testing, emphasizing stability and scalability under stress condition[11].

These studies collectively justify the use of black box testing covering unit, integration, system, usability, and performance testing as the primary evaluation approach for the proposed Intelligent Library System

2.2 Synthesis of the State-of-the-Art

Recent studies consistently focus on the use of automation, data analytics, artificial intelligence, and systematic testing to improve library management systems. Similarities across the literature show a shared recognition that traditional, manual, or semi-automated library operations suffer from inefficiencies such as delayed circulation, weak inventory control, and unreliable reporting, as identified by Padilla [1] and Arroyo et al. [3], while Shahzad, Khan, and Iqba[15] similarly emphasize the need for reliability, usability, and standards compliance.

In terms of analytics, Muazu et al. [6], Zong [5], and Mojjada and Aakundi [22] agree that circulation data and usage patterns are valuable for forecasting demand and supporting evidence-based collection development; however, these studies often assume the availability of structured and timely data.

AI-focused research by Asim et al. [14], and Mutia et al. [16] similarly highlights the potential of AI for predictive analytics and decision support but differs in its emphasis on challenges such as infrastructure readiness, staff training, and user acceptance. Evaluations of Hugging Face pre-trained models by Jiang et al. [20] and Kadasi et al. [21] align in cautioning against blind model reuse due to issues in documentation quality and performance transparency, yet differ in scope, with one focusing on reuse risks and the other on popularity performance mismatches. Across these works, black box testing and usability evaluation are commonly recommended as quality assurance mechanisms [10, 11, 17]. Unlike previous studies that examine automation, analytics, AI, or testing independently, the present study differs by integrating real-time data collection, prescriptive analytics, AI-assisted insights, and comprehensive testing within a single system tailored to the operational context of the CSPC Library.

2.3 Gap Bridged by the Study

Although prior studies have demonstrated the benefits of library automation and analytics-based management, the reviewed literature reveals a notable gap in the integration of real-time analytics within academic libraries operating under limited digital maturity. Studies by Padilla [1] and Arroyo et al. [3] confirm that many academic institutions continue to rely on manual or partially automated systems, leading to inefficiencies in circulation, cataloging, and report generation. While these studies advocate for automation, they primarily focus on basic transactional improvements and do not address the need for systems capable of capturing and processing book-related activities in real time to support continuous, data-driven decision-making.

Further gaps are evident in the application of data analytics and intelligent decision support in library environments. Zong [5] and Muazu et al. [6] demonstrated how circulation data and usage patterns can inform collection development and resource allocation; however, their approaches generally rely on well-structured and timely datasets. This assumption overlooks the operational realities of institutions like the Camarines Sur Polytechnic Colleges (CSPC) Library, where data collection is often delayed, fragmented, or manually recorded. As a result, the full potential of analytics particularly prescriptive analytics that generate actionable recommendations remains underutilized in similar institutional contexts.

Additionally, existing literature on system evaluation and intelligent technologies highlights limitations in how quality assurance and AI are implemented in practice. Iqbal et al. [7] and Lampaza et al. [9] emphasized the importance of usability, reliability, and performance testing but largely treated evaluation as a post-development activity rather than an integral component of system design and operation. Meanwhile, studies by Asim et al. [14], Shahzad et al. [15], and Mutia et al. [16] recognized the transformative potential of Artificial Intelligence in library services, while also identifying barriers related to infrastructure, training, and organizational readiness.

This study bridges these gaps by proposing and implementing a real-time, analytics-driven system tailored to the operational context of the CSPC Library. By integrating real-time data collection through API-based mechanisms, automated reporting, prescriptive analytics, and built-in evaluation through black box, usability, and performance testing, the system operationalizes concepts presented in the literature within a localized and practical setting. In doing so, the study advances existing research by demonstrating how intelligent, data-informed library services can be effectively deployed in resource-constrained academic institutions, narrowing the gap between theoretical capability and real-world implementation.

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CHAPTER 3
TECHNICAL BACKGROUND

The proposed system utilized a web-based platform enhanced with integrated data analytics to optimize collection management and support data-driven decision-making. By incorporating dynamic web interfaces and advanced visualization tools, the system facilitates comprehensive tracking, reporting, and analysis of usage patterns. These capabilities strengthen library operations and inform strategic collection development aligned with user needs.

3.1 Overview of Current Technologies Used in the System

Modern libraries increasingly leverage digital technologies to improve operational efficiency, accuracy, and service delivery. This project adopts real-time API integration, data analytics, and artificial intelligence to automate key library processes and support evidence-based decision-making. Through live data streams and intelligent usage analysis, the system enables precise, efficient, and responsive management of library resources.

Web development frameworks provide structured tools and standardized architectures that facilitate the development and maintenance of web-based systems. These frameworks enhance development efficiency by offering reusable components and built-in functionalities that support scalability and long-term system maintainability.

Flask was utilized as the backend web framework responsible for handling application logic and API-based communication. As a lightweight and modular micro-framework, Flask facilitates the development of real-time API endpoints that manage book circulation data, analytics requests, and system interactions. Its flexibility allows seamless integration with data analytics and AI components, enabling efficient request handling, dynamic data updates, and scalable system expansion within the operational context of the CSPC Library.

Data analytics plays a central role in transforming raw circulation and transaction data into meaningful insights. Python is used as the primary analytics programming language, enabling systematic data processing, statistical analysis, and automation of analytical workflows. Through Python-based data analytics libraries and scripts, borrowing frequency, return behavior, and usage trends are analyzed to support report generation, pattern identification, and evidence-based decision-making for collection development. This approach provides administrators and librarians with accurate, timely, and actionable information that supports resource optimization and responsive library services.

To enhance analytical capabilities, the system integrates artificial intelligence through the Hugging Face API, specifically utilizing the DialoGPT pre-trained model. DialoGPT is employed to support analytical interpretation and recommendation generation based on historical book usage patterns and circulation data. Through controlled API-based interaction, the model assists in producing prescriptive insights related to collection management, such as identifying frequently borrowed materials and suggesting potential acquisition or retention actions. By leveraging a cloud-hosted, pre-trained language model, the system avoids the need for extensive local model training and high computational resources, making the solution practical and efficient for institutional environments with limited infrastructure while still enabling intelligent, data-informed library management.

3.2 Resources

This section outlines the essential resources required for the development and implementation of the system proposed in the study, Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges. The successful deployment of this system depends on a range of hardware and technological components, including computing devices, networking infrastructure, and data processing tools. These resources form the foundation for data acquisition, statistical analysis, and the generation of actionable insights for collection management. Selecting appropriate and reliable technologies is critical to ensuring the system's capacity to support data-driven decision-making and enhance the efficiency of library operations.

3.2.1 Hardware Specifications

This section details the hardware resources essential for the development phase of the proposed system. Emphasis is placed on computing devices, networking infrastructure, and peripheral equipment required to support activities such as coding, debugging, and system testing. The selected hardware specifications prioritize performance, reliability, and compatibility to ensure seamless execution of development tasks and alignment with the system's technical requirements. These resources form the foundation for building a robust and scalable solution capable of supporting the intended functionalities throughout the development lifecycle.

Table 1: Hardware Specification Table

HARDWARE TOOL	SPECIFICATION
Processor	1.6 GHz or Faster processor/Intel Core i3 or Higher
RAM	8 GB RAM minimum
Storage	256GB SSD or Higher
Input Device	Standard keyboard and mouse
Output Device	Monitor
Network Device (Router)	100 Mbps LAN ports, supports IEEE 802.3

The hardware specifications in Table 1 meet the minimum requirements for standard development tools such as Visual Studio Code (Microsoft, 2024) and XAMPP (Apache Friends, 2024). A configuration with an Intel Core i3 processor, 8 GB of RAM, and a 256 GB SSD is sufficient for front-end and back-end development using HTML, CSS, JavaScript (Mozilla, 2024), PHP, and Python (Python Software Foundation, 2024). Additionally, the selected router specifications follow the IEEE 802.3 Ethernet standard (IEEE, 2022). This configuration enables stable connectivity for API testing using tools such as Postman, seamless access to cloud-based version control systems (e.g., Git repositories), and consistent performance when hosting local servers during the development phase.

3.2.2 Software Specifications

This section focuses on identifying the key software elements required to connect data-capturing mechanisms with analytics tools, enabling libraries to transform raw collection data into actionable intelligence for evidence-based decision-making. The table below provides a detailed overview of these software requirements, highlighting their contributions to effective data processing, analytics, and collection management workflows.

Table 2: Software Specification Table

Software Tool	Specification
Operating System	Windows 10 and above
Database Server	MySQL via XAMPP (version 3.3.0+)
Web Server	Apache via XAMPP (for Windows)
Front End Language	HTML5 (For structuring web pages) CSS (for styling) JavaScript (ES6+, for interactivity)
Front End Framework	Bootstrap (v5.3.0, for responsive and user-friendly design)
Visualization Library	Chart.js (v4.4.0, for rendering charts and graphs for usage statistics)
Development Tool	Visual Studio Code (for coding and debugging front-end files)
Communication Tool	AJAX (via JavaScript Fetch API, for asynchronous backend communication)
API Framework	Flask (v2.3.2 for real-time API endpoints)
Transform Model	DialoGPT

The software requirements outlined in Table 2 are specifically designed to support the "Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges" system to enable effective data management, user interface, and analytics for library operations. The Operating System (Windows 10 and above) provides a stable platform for deployment. MySQL stores book and API data. Database Server, responsible for book collection data, whereas Apache via XAMPP is the Web Server, which runs the web-based interface for remote access. Frontend structure, presentation, and user interaction of web interfaces is performed by HTML5, CSS3, and JavaScript (ES6+), with Bootstrap (v5.3.0) providing a responsive, user-focused look. Chart.js (v4.4.0) provides visualization of usage data, Visual Studio Code provides coding and debugging, and AJAX (through Fetch API) provides real-time, seamless interaction with the backend. Together, all these tools provide a powerful, scalable infrastructure for barcode-based library management and evidence-based collection development.

3.2.3 Program Specifications

This section presents the essential software components required during the development phase of the proposed system. These tools are integral to enabling data analytics capabilities and converting raw library data into actionable intelligence that supports evidence-based decision-making. The selected software solutions facilitate efficient data processing, analytical modeling, and the development of interactive user interfaces. The accompanying table provides a detailed overview of each software component, highlighting its specific function within the system architecture and its contribution to analytics workflows and interface design.

Functional Requirements

Functional requirements serve as the fundamental basis for evaluating the system’s overall effectiveness in fulfilling the specific needs of users and the organization. These requirements clearly define the expected behaviors, operations, and outputs for each module, providing a detailed benchmark against which system performance can be measured. Through functional testing, the system is carefully examined to ensure that it performs as intended, processes data accurately, responds appropriately to user inputs, and executes tasks reliably across all modules. Successfully meeting these functional requirements demonstrates that the Intelligent Library Management System operates efficiently, supports seamless user interaction, and delivers all intended library management services effectively.

Table 3: Functional Requirements

Functional Requirements	Description
Book Report Generation Module	Generates detailed reports on library operations, including book circulation, borrowing patterns, and collection usage. This module transforms raw transaction data into actionable insights, supporting evidence-based decision-making and resource allocation. It fulfills the objective of developing a real-time tracking system for book usage.
Overdue Notification Module	Automatically identifies overdue books and sends timely notifications to borrowers. This module ensures efficient monitoring of borrowed materials and promotes timely returns, reducing delays and supporting smooth library operations. It addresses the objective of improving borrowing efficiency and circulation management.
Prescriptive Analytics Module	Analyzes circulation and usage data to provide actionable recommendations for collection management. By leveraging real-time analytics and AI-driven insights, this module enables strategic decision-making, optimizes resource allocation, and aligns library operations with user needs. It fulfills the objective of integrating data-driven intelligence into library management.

This table presents the core functional modules of the Intelligent Library Management System and describes how each module supports the objectives of the study. The listed modules such as the Prescriptive Analytics Module, Book Report Generation Module, and Overdue Notification Module define the system's primary operational capabilities in managing library data, automating circulation processes, and supporting data-driven decision-making. Each module was implemented and evaluated to ensure that it performs its intended function, processes real-time data accurately, and contributes to the overall efficiency, reliability, and effectiveness of library operations at Camarines Sur Polytechnic Colleges .Collectively, these modules demonstrate how the system integrates intelligent automation and analytical tools to enhance library service delivery, improve resource utilization, and support informed administrative planning.

Non-Functional Requirements

Non-functional requirements define the quality attributes of the system that determine its usability, performance efficiency, and reliability. These requirements evaluate how well the Intelligent Library System operates under expected conditions and support the assessment of system stability and user experience.

Table 4: Non-Functional Requirements

Non-Functional Requirements	Description
Integration	Ensures seamless interaction with external services and existing institutional systems to enable reliable data exchange and interoperability.
Usability	Focuses on providing an intuitive and user-friendly interface that supports efficient task completion with minimal learning effort.
Performance	Ensures timely system response, efficient processing of transactions, and stable operation under normal and peak usage conditions.

This table summarizes the non-functional requirements evaluated through integration, usability and performance testing. The results indicate that the system provides seamless interaction with external and institutional services, supports an intuitive and user-friendly interface, and maintains timely and stable performance under simulated workloads using Apache Bench. Additionally, all defined features operate correctly and consistently, confirming that the system meets the established quality, reliability, and performance standards. Overall, the evaluation demonstrated that the system is robust, scalable, and suitable for deployment in a real-world operational environment.

3.2.4 System Logical Diagram

This section presents the System's Logical Diagram, which illustrates the conceptual structure of the system and the logical relationships among its major components. The diagram provides a high-level view of how data flows between users, core modules, and supporting services, emphasizing functional interactions rather than physical implementation. It serves as a reference for understanding system behavior, process coordination, and the overall logical design that supports the system's objectives.

System Architecture Diagram

System Architecture refers to the overall structure of a software system how its different parts are organized, how they interact, and how data flows between them. It acts like a blueprint that shows the components (such as user interfaces, APIs, databases, and external services) and how they work together to deliver the system's functionality.

In our study, the System Architecture of the Leveraging Data Analytics Intelligent Library Management System outlines how students, librarians, and thesis admins interact with the system through a React-based interface. Their requests are handled by various internal components and services, including a keyword generator module and a MySQL database for storing and managing data. The system also integrates external tools like the Hugging Face API to provide intelligent book recommendations.

This architectural design served as a roadmap throughout development, providing a clear understanding of module interactions within the system. By visualizing relationships and data flow, it helped identify dependencies, optimize processes, and ensure consistent performance. The emphasis on scalability, security, and efficiency enabled the system to handle increasing user demands while protecting sensitive data. Overall, the design reduced development risks and contributed to a more organized, maintainable, and future-ready system that supports long-term institutional needs.

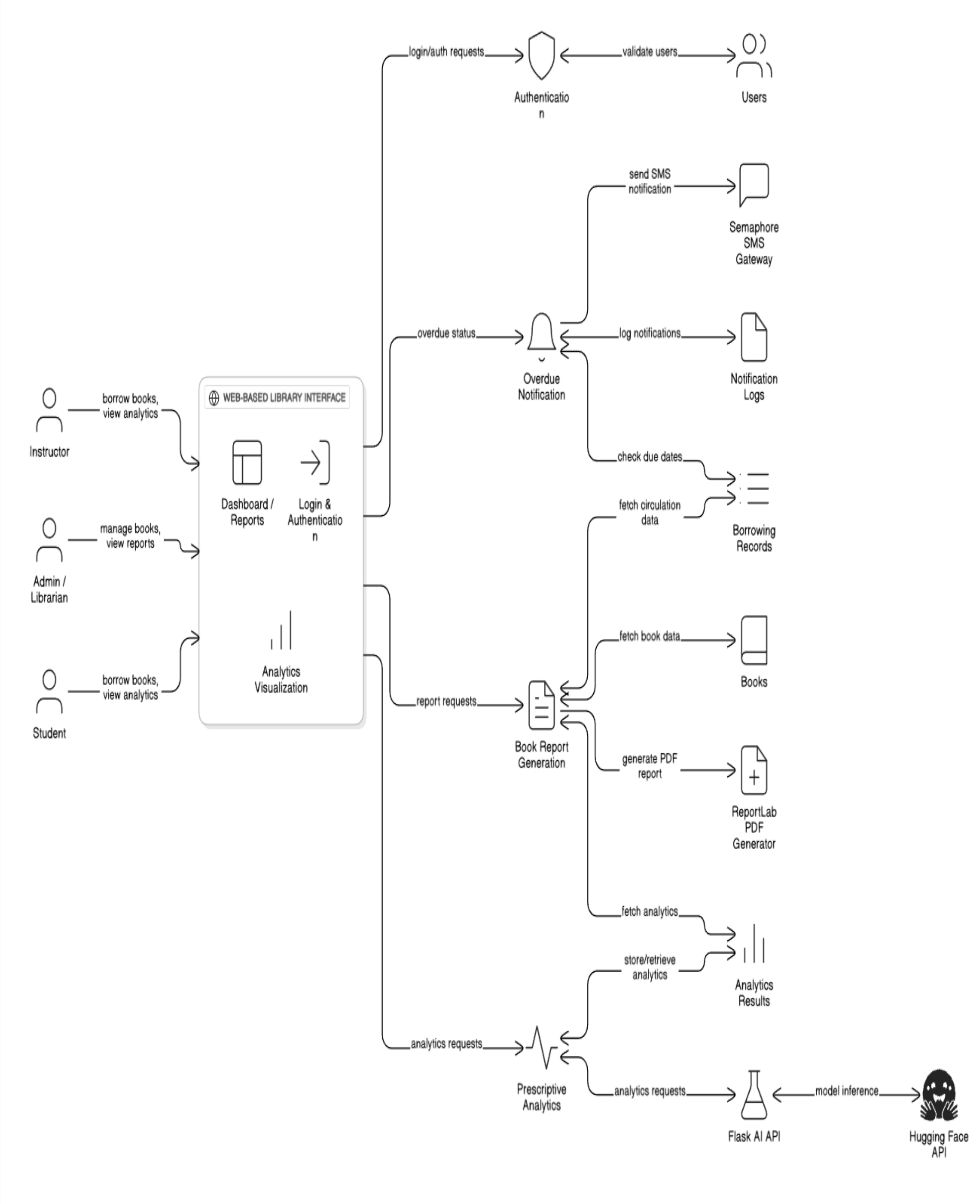

Figure 1: <u>System Architecture</u>

The system architecture for the Intelligent Library Management System is centered on a web-based library interface that serves instructors, librarians, and students for login, borrowing, reporting, and analytics visualization. Through this interface, users submit login/authentication requests, borrowing and return actions, report requests, and analytics queries, which are routed to the appropriate backend services for processing.

Authentication requests are handled by an authentication component that validates user credentials against the Users data store, ensuring that only authorized users can access system features. Overdue status checks trigger the Overdue Notification module, which inspects borrowing records for due dates, sends SMS alerts via the Semaphore SMS Gateway, and records each message in the Notification Logs. Report requests from the dashboard are processed by the Book Report Generation module, which fetches circulation and book data, then produces formatted PDF reports using the ReportLab PDF generator.

Analytics requests from the web interface are sent to the Prescriptive Analytics module, which reads and writes Analytics Results and forwards advanced inference requests to a Flask-based AI API connected to the Hugging Face AI service. This integration supports model-driven recommendations and risk analyses that are returned to the web interface as visual dashboards. Overall, the architecture illustrates how user interactions flow through authentication, notifications, reporting, and AI-powered analytics components, all orchestrated around a central web-based interface for CSPC library operations.

Flowchart

The flowchart provides a detailed and structured representation of the borrowing and returning process used by the Camarines Sur Polytechnic Colleges Library. It begins with user authentication and progresses through a series of logical steps that guide the user through accessing the dashboard, viewing reports, and interacting with the system based on their access privileges. If a user lacks data access, the system prompts a request for access, which then triggers a notification to the administrator. The admin reviews the request and decides whether to grant or deny access. If approved, the user proceeds to view reports; if denied, the process ends or loops back for further action. Once access is granted, users can select the type of report they wish to generate. The system then processes the request and offers an option to export the report. Depending on the user's choice, the report is either exported or the session concludes.

This workflow is visually organized using standard flowchart symbols diamonds for decision points and rectangles for process steps making the transitions between stages clear and intuitive. The diagram emphasizes how the system handles both successful and unsuccessful transactions, ensuring that each action is validated and recorded appropriately. It also highlights the importance of administrative oversight in managing user permissions and maintaining data integrity. By presenting the procedure in a linear yet flexible format, the flowchart supports a comprehensive understanding of the library's operational logic, illustrating how user interactions are managed, how errors are addressed, and how accurate book availability is maintained throughout the borrowing and returning cycle.

Moreover, the flowchart serves as a critical tool for system analysis and development by mapping the sequence of operations and decision points within the library management process. By visually illustrating interactions between users and administrators, it helps stakeholders identify bottlenecks, inefficiencies, and possible error points. The inclusion of clear decision nodes and structured process steps ensures scenarios such as denied access or report generation errors are handled properly, enhancing reliability and accountability. Additionally, the flowchart supports training and onboarding through a step-by-step representation of procedures. It also aids system maintenance and future enhancements by showing module relationships and areas suitable for automation or optimization.

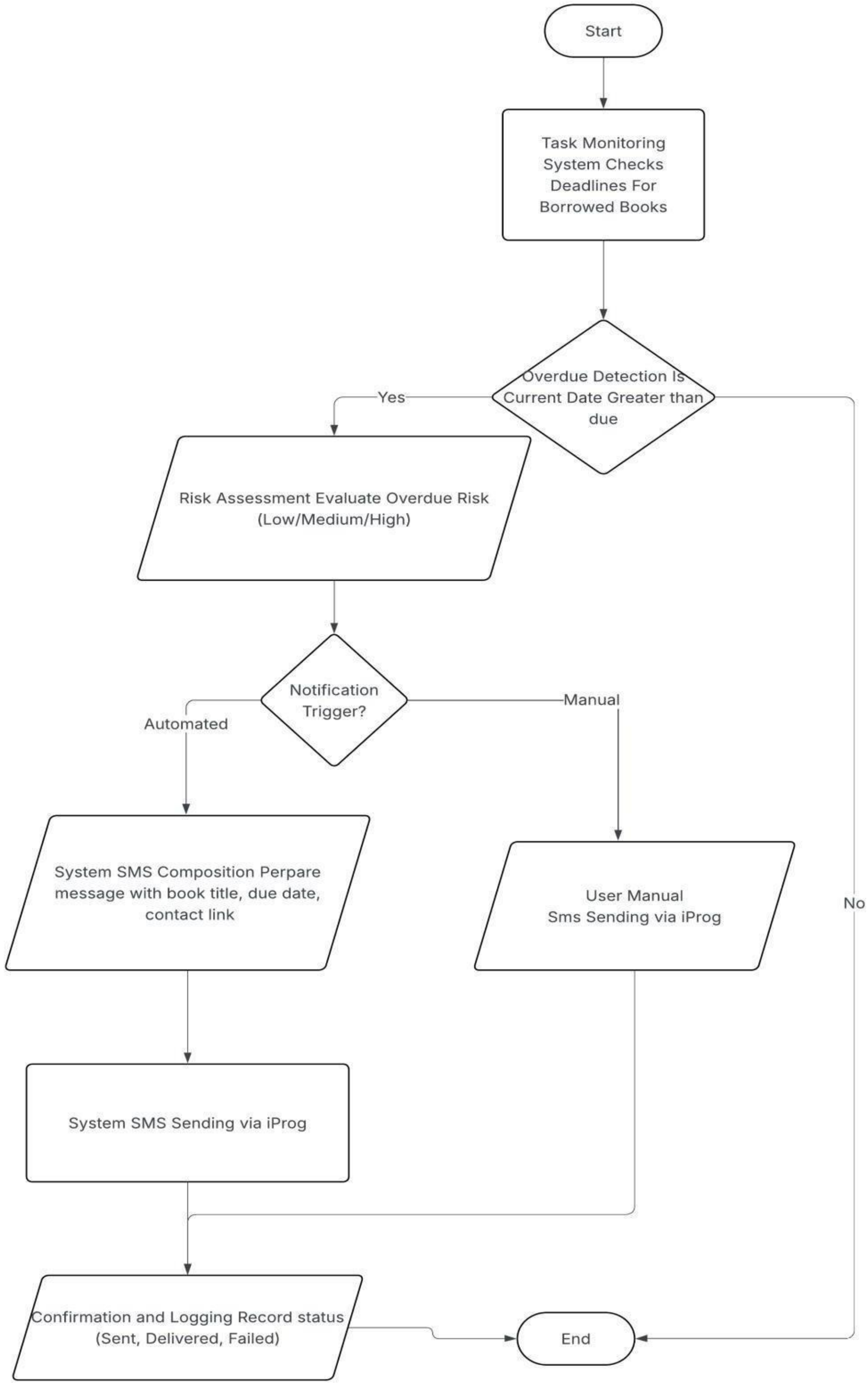

FIGURE 2: <u>Overdue Notification Module</u>

Figure 2 Overdue Notification Module shows the automated workflow of the Intelligent Library Management System of Camarines Sur Polytechnic Colleges in managing overdue library materials. The system retrieves borrowing records, evaluates due dates to identify overdue items, and generates reminder notifications that are sent to borrowers primarily through SMS alerts. All notification activities are recorded in the database for monitoring and follow-up, helping reduce manual work, improve borrower compliance, and enhance overall circulation efficiency in the CSPC Library.

The use of SMS alerts as the primary notification channel allows the system to reach borrowers quickly and effectively, increasing the likelihood of prompt returns. Since mobile phones are widely accessible, SMS notifications provide a practical and reliable communication method for reminding both students and employees of their overdue obligations. This approach supports improved borrower compliance and helps reduce the number of long-overdue materials.

Additionally, the logging of all notification activities in the database enables continuous monitoring and follow-up by library administrators. These records can be used for reporting, trend analysis, and policy evaluation, such as identifying frequent offenders or determining peak periods of overdue incidents. Overall, the Overdue Notification Module contributes to enhanced circulation efficiency, reduced workload for library personnel, and improved management of library resources within Camarines Sur Polytechnic Colleges.

The Overdue Notification Module automates the identification of overdue items and sends SMS alerts, reducing human error and ensuring timely borrower communication. Notification logs provide analytics for monitoring overdue patterns and improving circulation policies. By minimizing manual tasks, the system allows staff to focus on management activities, enhancing operational efficiency, borrower compliance, and overall library resource management at CSPC. This automation ultimately supports a more proactive and data-driven approach to managing overdue materials and maintaining an organized library system.

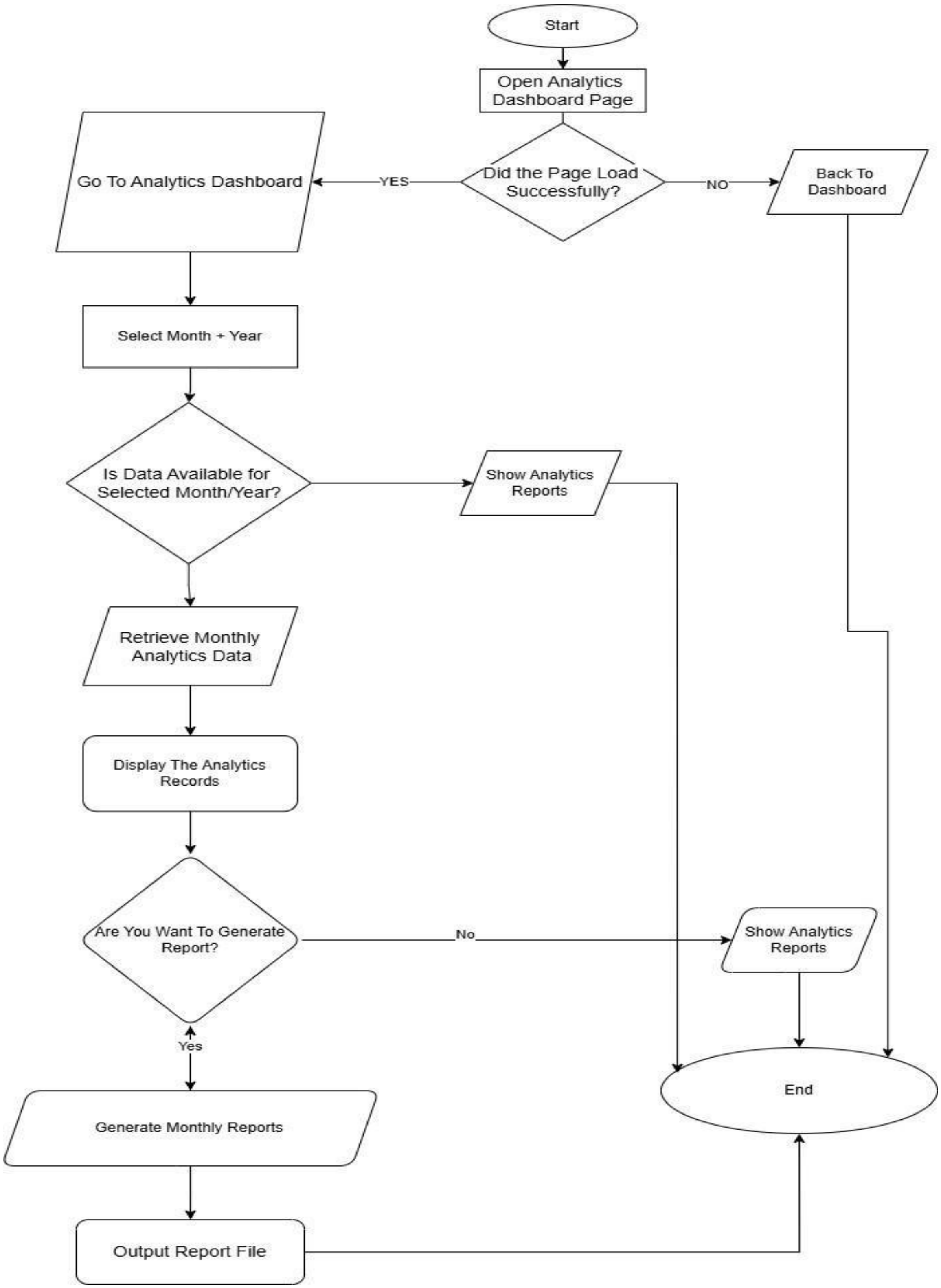

Figure 3: <u>Prescriptive Analytics</u>

Figure 3 Prescriptive Analytics illustrates the workflow of the analytics component of the Intelligent Library Management System of Camarines Sur Polytechnic Colleges, which supports data-driven decision-making through monthly analysis and report generation. The process begins when the user accesses the analytics dashboard, where the system verifies successful page loading before allowing interaction. The user selects a specific month and year, and the system checks the availability of analytics data for the selected period. Once available, the system retrieves and displays the analytics records, allowing the user to review circulation trends and performance indicators. If the user opts to generate a report, the system processes the data and outputs a monthly analytics report; otherwise, the process concludes after displaying the analytics results. This module enables librarians and administrators to efficiently analyze library data and generate actionable reports to support planning and collection management decisions.

3.3 Testing Plan

Black box testing is a software testing approach that focuses on validating system functionality without examining the internal structure, source code, or implementation details of the application. In this method, the system is tested based on predefined requirements, input conditions, and expected outputs, ensuring that each feature performs as intended from the user's perspective. This approach is particularly suitable for evaluating complex systems where the primary concern is whether the system meets functional and non-functional specifications rather than how the processes are internally executed.

Another important aspect of black box testing is its ability to uncover discrepancies between user expectations and actual system behavior. Because test cases are designed from requirements and real-world usage scenarios, this approach helps identify missing functionalities, incorrect outputs, interface issues, and performance problems that might not be apparent through code-based testing alone. Additionally, black box testing can be conducted by testers who are not involved in development, which promotes objectivity and reduces bias, ultimately contributing to higher software quality and improved user satisfaction.

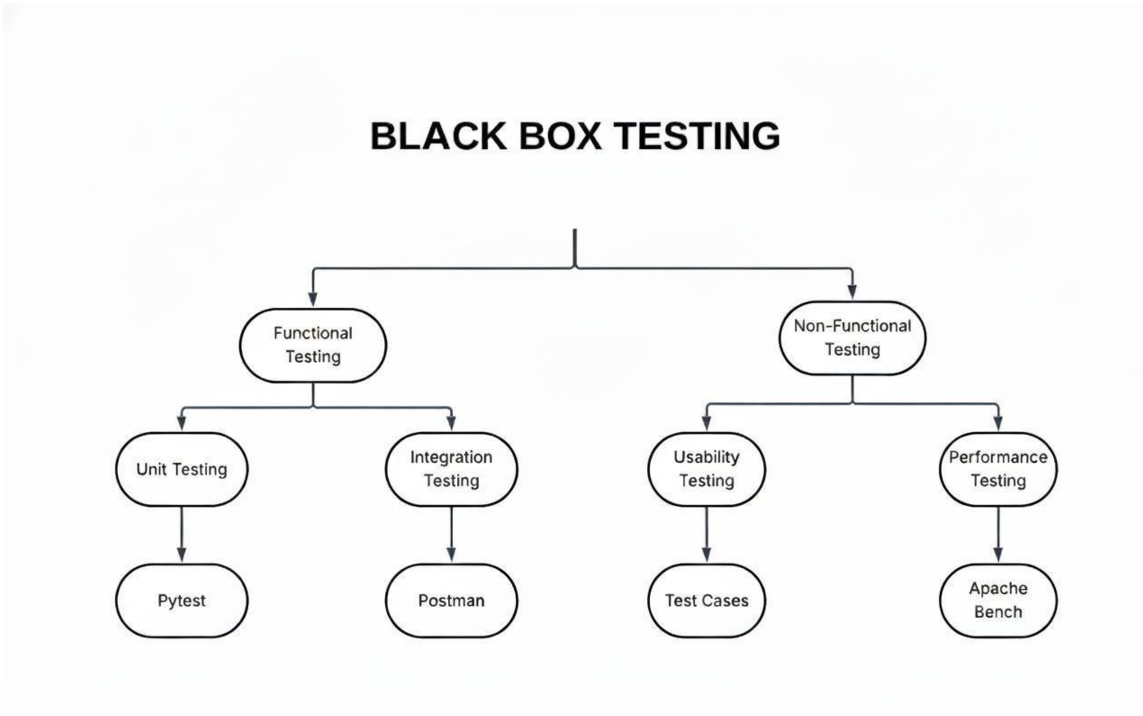

Figure 4. <u>Testing Plan</u>

3.3.1 Types of Testing

To ensure system reliability, functionality, and alignment with project objectives, the development phase incorporates a comprehensive testing strategy.

This includes:

Integration Testing. Integration testing verifies the seamless interaction between system modules, including front-end interfaces, backend services, and API endpoints. It ensures that data flows correctly across components and that interdependencies function as expected.

Unit Testing. Unit testing focuses on validating individual components or functions in isolation to ensure each one performs its intended operation accurately. It checks the correctness of small, discrete units of code such as methods, scripts, or modules before they are integrated with other system parts, helping to detect errors early and improve overall system reliability.

Usability Testing. This phase assesses the intuitiveness, accessibility, and overall user experience of the interface. It ensures that the system is easy to navigate and effectively supports the workflows of key users, particularly library staff and administrators.

Performance Testing. Performance testing measures the system's scalability and responsiveness under simulated workloads. It evaluates the system's ability to handle concurrent operations and real-time analytics, ensuring consistent performance under stress.

3.3.2 Testing Tools and Framework

To ensure the robustness, reliability, and usability of the proposed library management system, a range of specialized testing tools and frameworks is employed throughout the development process. Each tool supports a specific testing activity unit, integration, system, usability, and performance aligning with the project's goal of delivering an efficient, data-driven, and user-centered solution.

Pytest is used to automate unit testing for the Python-based analytics components of the system. It validates the accuracy, correctness, and stability of individual functions and modules in isolation, ensuring that each unit performs its intended operation before integration. Pytest's fixtures, assertions, and detailed reporting enhance early error detection and help maintain reliability during ongoing development.

Postman is utilized to simulate HTTP requests and evaluate interactions between the web interface, PHP backend, and Flask-based analytics endpoints. It verifies the seamless flow of data across components, ensuring proper request–response behavior, API functionality, authentication processes, and data consistency during integration testing.

Usability test cases are employed to evaluate the user experience of the system, particularly for library staff, administrators, and analytics users. These test cases assess interface clarity, ease of navigation, accessibility, and alignment with user expectations. Feedback gathered from testers helps refine the design and improve operational efficiency.

Apache Bench is used to perform performance and load testing on the system's API endpoints and server components. It measures response times, throughput, and server behavior under varying levels of traffic, ensuring that the system can handle expected workloads while maintaining responsiveness, stability, and reliability.

Notes

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CHAPTER 4

METHODOLOGY, RESULTS AND DISCUSSION

This chapter discusses the system's SCRUM approach, which involves the planning, design, development, sprint review, and sprint retrospective phases. It presents the information gathered, organized, analyzed, and evaluated throughout the development process. These activities were essential in establishing a structured flow of information for the creation of the Leveraging Data Analytics for Intelligent Library Management System. The chapter also includes the database model design and system interface, which served as essential guides in ensuring a systematic and efficient development process for the proposed system.

4.1 Planning

The planning phase marks the first stage in the SCRUM software development process, where project objectives, requirements, and development strategies are defined. During this phase, a Sprint Planning meeting is conducted to establish project goals, assign roles for the Scrum Master and stakeholders, and organize the software development team. System requirements are gathered through discussions between the development team and the client to ensure that both functional and non-functional specifications are addressed. This includes identifying the necessary hardware and software resources, as well as selecting appropriate development approaches to efficiently build the system.

4.1.1 Requirement Gathering and Stakeholder Consultation

During the planning phase, the development team collaborated closely with librarians, library staff, administrators, and students to understand operational challenges and user needs. Through questionnaires and interviews, the team identified current issues in library operations, including inefficiencies in cataloging, book tracking, and resource usage monitoring. This iterative consultation helped the team define system requirements for the Leveraging Data Analytics for

Intelligent Library System, ensuring that the design is aligned with real-world library conditions and user expectations.

4.1.2 Validation and Testing Plan

To ensure that the system meets its intended functionality, a Black Box Testing strategy was planned and integrated into each sprint. This testing approach validated system performance from the user perspective without examining internal code, focusing on input-output behavior and adherence to system requirements. Coupled with ongoing stakeholder feedback, this iterative testing process ensured that the system not only addresses user needs but also maintains reliability, usability, and data accuracy throughout development.

4.2 Research Design

The researchers used a quantitative research design to evaluate the performance and effectiveness of the proposed Intelligent Library System for Camarines Sur Polytechnic Colleges. A survey was conducted using a structured questionnaire that was answered by both students and library staff. The purpose of the questionnaire was to measure user satisfaction, ease of use, and the system's impact on improving access to library resources. Through this, the researchers were able to gather statistical data on how the system addressed the limitations of the manual library management process, particularly in areas such as borrowing efficiency, overdue notifications, and collection development.

In addition to the survey, the researchers employed Black Box Testing to ensure that the system functioned correctly. This type of testing focused only on the input and output without examining the internal code. Several kinds of Black Box Testing were carried out to validate the system's performance. Functional Testing was used to verify that modules such as data visualization, report generation, and overdue notifications worked as intended. System Testing confirmed that all modules operated seamlessly together. Usability Testing determined whether the interface was user-friendly for both staff and students. Performance Testing was conducted to evaluate how well the system performed under heavy usage.

By combining survey results with Black Box Testing, the researchers were able to validate both the usefulness and reliability of the proposed system. The quantitative data gathered from users provided evidence of satisfaction and effectiveness, while the technical testing confirmed that the system met performance and functional standards. This approach demonstrated that the Intelligent Library Management System is not only efficient in theory but also practical and reliable in real-world application.

4.2.1 Research Method

In this section we utilized a quantitative method based on a questionnaire completed by users and librarians. The research instrument utilized, the data collection process, the sampling method, the participants and respondents, and the statistical tests applied to examine the data were also discussed.

Instrument

This part explains how the researchers collected data for the proposed Intelligent Library Management System. The main tool was a questionnaire distributed to students and library staff of Camarines Sur Polytechnic Colleges.

Questionnaire. The questionnaire was developed based on the System Usability Scale (SUS) testing standards to evaluate the functionality, integration, and usability of the Intelligent Library System. It was designed to gather structured feedback from staff, librarians, administrators, employee and students regarding specific test cases and system performance.

Each section contained specific test cases designed to validate different aspects of the system. Integration testing focused on verifying the seamless exchange of data among modules such as Reporting, and Overdue Notifications. System testing evaluated performance, security, accuracy, and backup functions to ensure that the system operated reliably under normal conditions. The usability testing adopted items based on the System Usability Scale (SUS), which utilized a 5-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5). These items measured users' perceptions of ease of use, interface design, accessibility, responsiveness, and overall satisfaction.

Test Cases. A test case is a sequence of actions or conditions developed to verify whether a specific component or function of a software application performs correctly as intended. It defines the procedures to be executed, the input data to be applied, and the expected results to confirm that each module operates according to the system specifications. Test cases are an essential part of the software testing process, serving as the basis for evaluating the system's performance, accuracy, and reliability.

4.2.2 Data Gathering Procedure

This section describes how the researchers gathered relevant data used for the assessment and improvement of the proposed Intelligent Library System . Multiple data collection methods were employed to ensure accurate evaluation of the system's functionality, integration, and usability.

Interview. The researchers conducted an informal interview with the administrator, to obtain insights into their experiences and expectations regarding the current library system. Prior to the interview, a guide was prepared containing questions related to common library concerns such as book borrowing, returning, overdue notifications, and collection management. The information gathered from these interviews helped the researchers understand user needs and preferences, which served as the basis for developing the features and functionalities of the proposed system.

Test Cases. The researchers utilized test cases as a structured approach to evaluate the usability of the system and determine whether its features operated in a manner that was clear, intuitive, and aligned with user expectations. Each test case included a defined objective, a sequence of user actions, input conditions, and the expected system response. These test cases focused on assessing ease of navigation, clarity of interface elements, responsiveness of user interactions, and overall user satisfaction when performing key tasks such as viewing analytics dashboards, generating reports, and receiving overdue notifications. The results of the usability- based test case evaluations provided systematic and measurable feedback, which guided refinements to the interface design and interaction flow prior to system deployment.

Evaluation Forms. The researchers utilized evaluation forms as a structured method for gathering feedback from respondents regarding the functionality, performance, and usability of the Intelligent Library Management System (ILMS). The forms were designed based on the results of the conducted test cases and incorporated items aligned with the System Usability Scale (SUS). Each question assessed key aspects such as system features, interface design, efficiency, and user satisfaction. The evaluation forms served as a systematic tool for identifying areas that required improvement and for validating the overall effectiveness of the developed system.

System Usability Scale (SUS). To further evaluate the user experience, the researchers administered the System Usability Scale (SUS) questionnaire to selected respondents. The SUS, is a standardized instrument consisting of ten statements rated on a five-point Likert scale from Strongly Disagree (1) to Strongly Agree (5). It was used to assess the perceived usability, efficiency, and overall satisfaction with the library system. The responses were converted into a composite score ranging from 0 to 100, which was then interpreted to determine the system's usability level. The SUS results provided valuable feedback that helped the researchers ensure the system's interface and functions were user-friendly and effective.

4.2.3 Sampling Technique

The researchers employed a purposive sampling technique in selecting participants who were directly involved in the operations and use of the library system. This method ensured that only individuals with relevant knowledge and experience such as librarians, staff, and students were included in the evaluation. Purposive sampling was deemed appropriate for this study as it allowed the researchers to gather data from respondents who could provide meaningful insights into the system's functionality, usability, and overall performance.

4.2.4 Respondents

The respondents of this study played a vital role in evaluating the usability and functionality of the system. Their participation provided empirical data and practical feedback that guided the researchers in validating whether the system meets both operational requirements and user expectations within the CSPC Library. The respondents represented key stakeholder groups who directly interact with or benefit from the library system, ensuring a balanced and comprehensive system evaluation.

For this study, respondents were grouped as librarians, staff, students, and employees, each offering distinct perspectives. Librarians evaluated operations like cataloging and circulation, while staff assessed usability and interface accessibility. Students and employees reviewed transactions and notifications. Participation in test cases and questionnaires helped identify strengths, limitations, and areas for improvement, supporting a balanced assessment of overall system performance.

Table 5: Frequency Distribution of the Respondents

Respondents	Population	Sample Size	Frequency	Percentage
Staffs	12	12	12	18.46
Librarian	6	6	6	9.23
Students	14,000	389	35	53.85
Employee	300	190	10	15.38
IT expert	2	2	2	3.08
Total	14,322	599	65	100.00

Table 5 presents the frequency and percentage distribution of the respondents involved in the system evaluation. Out of the 65 total respondents, the majority were students, accounting for 53.85% of the sample, reflecting their role as the primary users of the system. This was followed by staff members at 18.46% and employees at 15.38%, who provided operational and user-level feedback. Librarians comprised 9.23%, contributing professional and managerial perspectives, while IT experts represented 3.08%, offering technical validation of system functionality and reliability. This distribution demonstrates a balanced mix of end users, operational staff, and technical evaluators, strengthening the credibility of the findings and supporting the system's readiness for institutional deployment.

Frequency and Percentage. The researchers analyzed the demographic profile of respondents using Frequency and Percentage to determine the proportional representation of each group in the study. The percentage for each category was computed by dividing its frequency by the total number of respondents and multiplying the result by 100.

Formula:

$$P^{-} = \frac{f}{n} \ x \ 100$$

Where:

P=Percentage

f=Frequency

n = Total number of respondents

4.2.5 Statistical Analysis Tools

To analyze the gathered data, the researchers employed a combination of statistical techniques designed to organize, summarize, and interpret the results effectively. These techniques enabled the researchers to evaluate the system's performance, analyze user responses, and ensure that the findings were reliable and meaningful. The primary tools used in this study include the Arithmetic Mean and Slovin's Formula.

Arithmetic Mean. Commonly referred to as the *average*, is a statistical measure used to determine the central value of a dataset. It is computed by dividing the sum of all observed values by the total number of observations. In this study, the arithmetic mean was utilized to evaluate the responses gathered from system users and testers, particularly in assessing the functionality, usability, and performance of the Intelligent System.

By using the arithmetic mean, the researchers were able to simplify the interpretation of survey ratings and test results, making it easier to identify patterns, user satisfaction levels, and areas where the system may require further refinement.

Formula:

$$\bar{X} = \frac{\sum X_i}{n}$$

Where:

$\bar{X}$=Arithmetic Mean

X_i=Data Set Values

n= Number of Values in the Data Set

Slovin's formula. A statistical tool used to determine the appropriate sample size when the total population is known. It ensures that the selected number of respondents is sufficient to achieve a reliable and accurate representation of the population while maintaining a controlled margin of error. This method is particularly useful in large populations, such as the academic community of Camarines Sur Polytechnic Colleges.

In this study, Slovin's formula was applied to calculate the exact number of respondents needed for the system evaluation. Using this formula ensured that the conclusions drawn from the respondents' feedback were statistically valid and representative of the larger population.

Formula:

$$n = \frac{N}{1 + Ne^2}$$

Where:

n= Sample Size

N= Total Population

e= Margin of Error

Likert Scale. A five-point Likert Scale was used to measure respondents' attitudes and perceptions toward the developed system. This scale allowed respondents to express varying degrees of agreement or disagreement with statements related to system usability, functionality, and efficiency. The computed weighted mean for each indicator served as the basis for interpreting the level of acceptability of the proposed system.

Scale	Range	Verbal Interpretation	Description
5	4.21 – 5.00	Strongly Agree	The respondents strongly agree that the system meets the criteria and performs excellently.
4	3.41 – 4.20	Agree	The respondents agree that the system is acceptable and functions well based on the criteria.
3	2.61-3.40	Neutral	The respondents neither agree nor disagree; the system is moderately acceptable but may need improvements.
2	1.81-2.60	Disagree	The respondents disagree; the system does not adequately meet the criteria and requires enhancements.
1	1.00-1.80	Strongly Disagree	The respondents strongly disagree; the system performs poorly and needs major revisions.

4.2.6 Identified Problems and Proposed Solution

During the assessment and collaboration with the library staff and administrators of Camarines Sur Polytechnic Colleges (CSPC), the researchers identified several key issues that hindered the efficiency and reliability of the existing library management operations. The primary concern was the continued use of manual data management, which resulted in time-consuming processes, data inaccuracies, and difficulties in maintaining organized records. In addition, the underutilization of circulation data limited the library's ability to make informed decisions regarding book acquisitions and collection development. The absence of automated overdue notifications also caused delays in communication with borrowers, leading to inefficient circulation management and reduced resource availability.

To address these problems, the researchers proposed the development of an Intelligent Library System that integrates automation and data analytics. The system introduces automated data processing with real-time API integration to replace manual operations, ensuring accuracy and efficiency in record handling. It also incorporates data visualization and prescriptive analytics modules to transform raw circulation data into actionable insights for evidence-based decision-making. Furthermore, an automated notification module was designed to deliver timely reminders through email or SMS, while real-time monitoring features provide librarians and administrators with live updates on borrowing trends and system activity. Collectively, these solutions aim to enhance operational efficiency, promote data-driven management, and improve overall library services within CSPC.

Table 6 presents the identified problems encountered during the planning and assessment phase of the study, along with their corresponding descriptions and proposed solutions. This table provides a clearer understanding of the existing challenges in the library's operations and how the proposed system aims to address them through automation, data analytics, and real-time monitoring.

Table 6: Identified Problems and Proposed Solutions

Identified Problems	Proposed Solutions
Manual Data Management	Automated Data Processing: Implement a standalone system with real-time API integration to automate data collection and report generation, replacing manual processes. This solution leverages Python (Flask) and MySQL to streamline data handling and reduce errors.
Underutilized Circulation Data	Data Analytics Integration: Develop modules such as the Data Visualization Module (using Chart.js) and Prescriptive Analytics Module (using Statsmodels) to analyze circulation data, enabling informed decisions on acquisitions and archiving based on usage patterns.
Delayed Overdue Notifications	Automated Notification System: Introduce an Overdue Notification Module with email/SMS integration to alert users of overdue items, improving circulation efficiency and resource availability.
Limited Real-Time Insights	Real-Time Monitoring: Incorporate real-time API endpoints to provide live updates on borrowing trends and system usage, supporting dynamic adjustments to library operations.

It also presents the identified problems and corresponding solutions related to the current management and operation of the library's circulation and data analytics processes. Based on the findings, it was evident that the system still relied heavily on manual procedures, which resulted in several operational challenges. These challenges include manual data management that often leads to inconsistencies, the underutilization of circulation data that limits evidence-based decision-making, delayed overdue notifications that hinder efficient circulation flow, and limited access to real-time insights needed for timely interventions. Such limitations contribute to inefficiencies, missed opportunities for data-driven improvements, and reduced effectiveness in supporting library users and administrators.

To address these issues, the proposed enhancements introduce an integrated, automated system designed to modernize and streamline library operations. The solution includes automated data processing through a standalone system with real-time API integration, enabling faster and more accurate data collection and report generation. Additionally, the integration of data analytics supported by modules such as the Data Visualization Module (using Chart.js) and Prescriptive Analytics Module (using Statsmodels) maximizes the use of circulation data for informed decisions on collection development, acquisitions, and archiving. The system also incorporates an automated notification feature that sends timely email or SMS alerts to users with overdue items, improving circulation efficiency. Furthermore, real-time monitoring via API endpoints provided immediate insights into borrowing trends and system performance, allowing library staff to make dynamic adjustments to operations. By shifting from manual to automated and analytics-driven processes, the system is expected to enhance accuracy, responsiveness, and overall effectiveness, ultimately improving the services experienced by both library users and administrators.

4.2.7 Operational Feasibility

Operational feasibility referred to how effective the proposed Intelligent Library Management System was in addressing the current problems of library operations and in taking advantage of new opportunities brought by data analytics. This included both students and library staff, and examined whether they were willing and able to use the system in their daily academic and administrative activities. It also evaluated whether the system met essential library service needs, especially in resolving issues such as difficulty in locating resources, lack of timely overdue reminders, and the absence of real-time usage reports.In this section, the researchers studied how the proposed system responded to these challenges and how effective it was in providing a reliable and data-driven solution. The system streamlined borrowing and returning processes, automated overdue notifications, and generated visual reports that could support evidence-based collection development. To more clearly demonstrate operational feasibility, the researchers also made use of diagrams such as fishbone diagrams and functional decomposition diagrams to illustrate how the system worked, how it was used by both students and staff, and how it simplified the overall library management process.

Fishbone Diagram

A fishbone diagram (also called an Ishikawa or cause-and-effect diagram) is a visual tool that maps and categorizes potential causes of a specific problem to help identify its root causes [2]. In the case of the proposed Intelligent Library Management System, the fishbone diagram was used to identify the common issues experienced by students and library staff in the traditional library process. The figure illustrated the cause-and-effect relationship of problems such as difficulty in locating resources, lack of timely overdue reminders, delays in accessing records, and limited support for data-driven decision-making.

This served as a guide in creating better solutions to improve the system. Through the fishbone analysis, the researchers were able to identify the root causes of inefficiencies and design features that directly addressed them. It also helped the team understand which issues affected users the most. Because of this, the proposed system was developed with solutions that fit the needs of both students and staff, ensuring a more efficient, reliable, and data-driven library management process. Additionally, the analysis provided a structured approach to prioritizing system improvements based on their impact and feasibility. This ensured that development efforts were focused on critical areas that would deliver the greatest benefits to overall library operations.

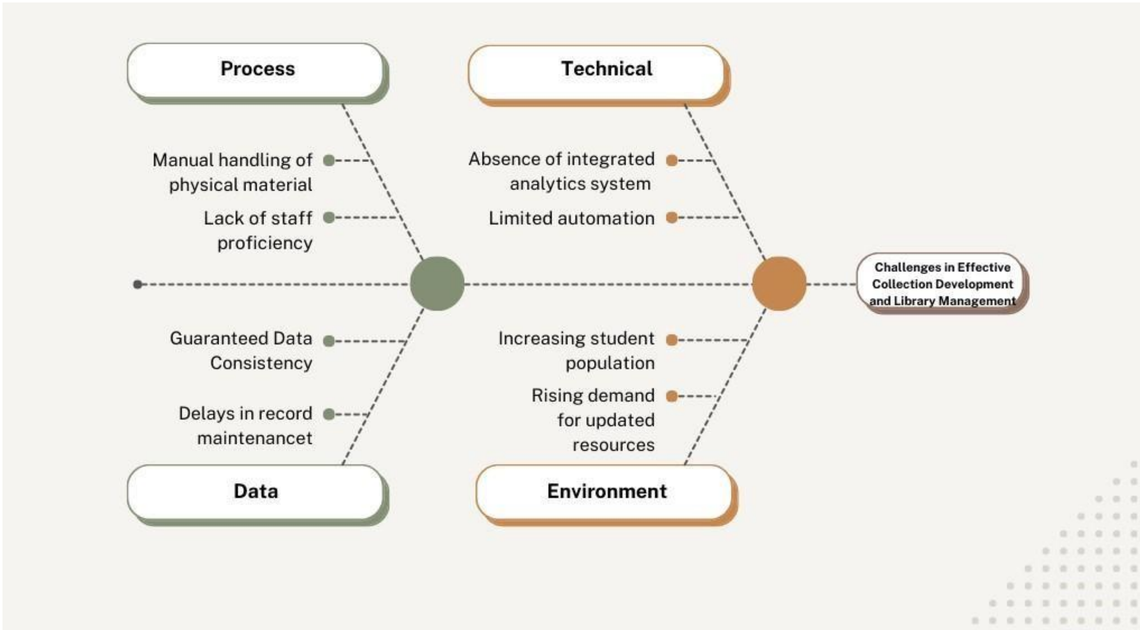

Figure 5: <u>Fishbone Diagram</u>

Figure 4 illustrates the problems and causes related to collection development and library management. The head of the fishbone diagram represents the main problem, while the branches show the reasons why the problem occurs. Each branch refers to different areas that affect the system such as process, technology, data, and environment.

In the process, it was found that issues included manual tracking of borrowed and returned books, delays in updating records, and inconsistent evaluation of book usage. Under technology, the absence of an integrated analytics system, outdated systems, and limited automation hinder efficiency. In terms of data, incomplete borrowing statistics, unreliable usage reports, and difficulty in extracting insights were identified. Lastly, environmental challenges included an increasing student population and a rising demand for updated resources.

This diagram is important because it clearly presents the root causes behind the challenges faced by the library. With the help of the fishbone diagram, the researchers were able to identify problem areas that require attention and highlight the importance of adopting a data-driven approach to improve collection development and library services.

4..2.8 Functional Decomposition Diagram

The Functional Decomposition Diagram (FDD) breaks down the Intelligent Library Management System into three major functional modules: Book Report Generation, Overdue Notification, and Prescriptive Analytics. Starting from the overall system at the top, the diagram branches into these core areas and then into specific tasks such as collecting circulation data and sending SMS notifications.

For instance, the Book Report Generation module focuses on transforming raw circulation data into meaningful reports that support decision-making. The Overdue Notification module automates the process of identifying overdue materials and delivering timely SMS reminders to borrowers, reducing manual workload for library staff. Meanwhile, the Prescriptive Analytics module analyzes historical and current data to generate actionable recommendations, such as improving circulation policies or optimizing resource allocation.

In addition, the FDD helps the development team clearly visualize the relationships and dependencies between modules and their sub-processes. This allows for better planning, efficient task allocation, and streamlined integration of system components, reducing development errors and ensuring that each module operates seamlessly within the overall system architecture.

Furthermore, the FDD serves as a valuable reference during system testing and evaluation. By mapping each function and its corresponding tasks, the diagram guides the creation of test cases, facilitates performance monitoring, and ensures that all system functionalities meet the operational requirements of the CSPC Library. Ultimately, it provides a structured framework that enhances development, implementation, and long-term maintenance of the system. It also helps developers quickly identify functional gaps or redundancies, allowing timely refinements before full deployment. In addition, the FDD supports future system enhancements by providing a clear foundation for modifying or expanding existing functions without compromising system stability.

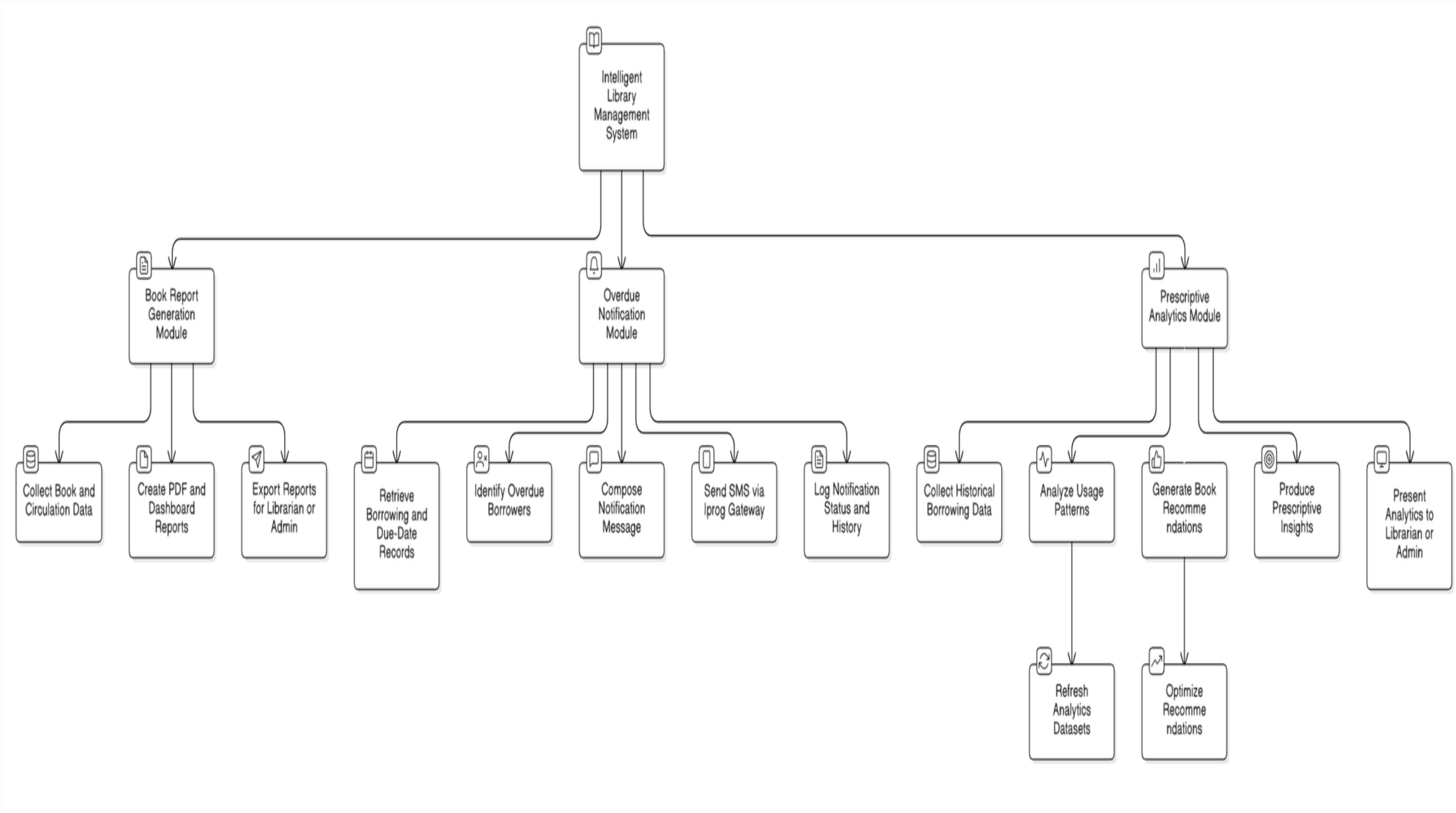

FIGURE 6: <u>Functional Decomposition Diagram</u>

In this figure, the Functional Decomposition Diagram provides a detailed, tree-like view of how the Intelligent Library Management System transforms raw operational data into reports, notifications, and analytics-based decisions for the CSPC Library. At the highest level, the system node represents the complete system, which then splits into three principal modules: the Book Report Generation Module, the Overdue Notification Module, and the Prescriptive Analytics Module. Each of these modules is further decomposed into concrete, executable sub-functions that show step-by-step how the system delivers value to librarians, administrators, and borrowers.

On the reporting side, the Book Report Generation Module begins with collecting book and circulation data, then proceeds to create PDF and dashboard reports tailored to library needs, and finally exports these reports for use by librarians or admins during planning and evaluation. The Overdue Notification Module illustrates the full lifecycle of reminder messages: it retrieves borrowing and due-date records, identifies overdue borrowers, composes appropriate notification messages, sends SMS alerts through the gateway, and logs notification status and history for auditing and follow-up. This decomposition clarifies how the system automates what would otherwise be repetitive manual tracking and communication tasks.

The Prescriptive Analytics Module shows how the system supports data-driven decision-making. It collects historical borrowing data, analyzes usage patterns, and generates book recommendations that can then be refined by refreshing datasets and optimizing recommendation logic. The module also produces prescriptive insights such as which subjects to prioritize for acquisition and presents these analytics clearly to librarians or admins for collection development and policy decisions. By mapping these activities as connected sub-functions, the diagram makes it easier to see how analytics operations are integrated with day-to-day library work.

Overall, the FDD not only enumerates what the system does but also highlights how responsibilities are grouped, how data flows between stages, and where automation reduces human workload. This structured breakdown improves understandability for developers and stakeholders, supports modular implementation and testing, and provides a clear reference when planning enhancements or troubleshooting specific parts of the Intelligent Library Management System.

4.2.9 Schedule Feasibility

The project was found to be schedule feasible as the system was developed and deployed within the approved timeframe using a Scrum-based Agile approach. Development activities were organized into sprint cycles with clear milestones for analytics, reporting, and notification modules, allowing tasks to be completed incrementally and on time. Regular sprint planning, reviews, and testing ensured early issue identification and timely adjustments, preventing delays during deployment. Overall, the structured development process and continuous coordination with library stakeholders enabled the successful completion and deployment of the system within the project schedule.

Gantt Chart

These diagrams serve as tools for organizing a project by outlining the sequence in which tasks must be completed. The Gantt chart provides a visual representation of the project's progress, showing the timeline of activities and the duration of each task. It clearly illustrates which tasks can be performed simultaneously and which ones depend on the completion of previous tasks before they can begin. Additionally, the Gantt chart helps identify potential bottlenecks, overlaps, and tasks that may have been overlooked, ensuring a more accurate and efficient project schedule.

Figure 6 serves as a visual roadmap that guides the researchers through the system development process, illustrating the sequential steps, key decision points, and interactions among different development phases. By presenting the workflow in a structured and organized manner, it helps the team understand the overall progression from planning and design to implementation and testing. This roadmap ensures that development activities are executed systematically, facilitates proper resource allocation, minimizes potential errors, and provides a clear reference for monitoring progress, ultimately supporting the successful and efficient completion of the system.

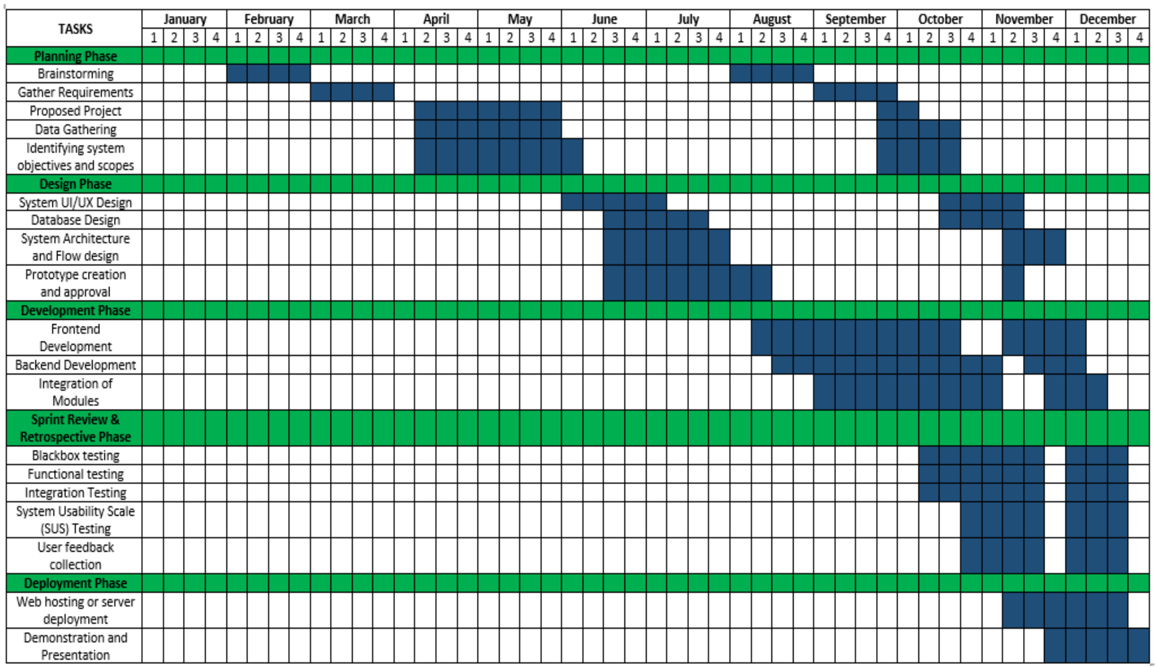

FIGURE 7: Gantt Chart Diagram

The Gantt chart presents the project timeline from January to December, showing the structured and sequential development of the system. The project begins with the Planning Phase, covering brainstorming, requirements gathering, proposal preparation, data gathering, and scope definition. This is followed by the Design Phase, which includes UI/UX design, database design, system architecture, and prototype approval. Core system development is carried out during the Development Phase through frontend, backend, and integration activities. The Sprint Review and Retrospective Phase focus on black box testing, functional and integration testing, System Usability Scale (SUS) evaluation, and user feedback to ensure system quality. Finally, the Deployment Phase covers server hosting, system demonstration, and presentation, confirming that all activities were completed in an organized manner within the approved schedule.

4.2.10 Requirement Modelling

Requirements modeling is the process of identifying, documenting, and organizing the essential inputs, processes, and outputs of the system to ensure that it meets the needs of all stakeholders. This technique is important in system development because it provides a structured way of analyzing how information enters the system, how it is processed, and what results are delivered to users. For the Intelligent Library System of Camarines Sur Polytechnic Colleges, requirements modeling was used to align the system design with both functional and non-functional requirements. It served as the basis for defining user roles such as students, librarians, and administrators, and for clarifying how key modules such as book cataloging, borrowing and returning, inventory tracking, user authentication, and data analytics interact to support the efficient operation and management of the library.

Input

Users must log in using their assigned library system accounts to ensure secure, role-based access. Students can search for books, check availability, request borrowing, renew loans, and view their transaction history. Librarians can input new book records, update catalog information, manage borrowing and returning transactions, and record inventory movements. Administrators can encode system configurations, manage user accounts, view analytics dashboards, and oversee overall library operations. The system also accepts inputs such as barcode scans, book details, user data, feedback, and circulation logs, ensuring that all relevant information is accurately captured for processing.

Process

All inputs are processed and stored in a centralized database. When a student borrows a book, the system validates the request, updates inventory status, records loan details, and logs the transaction. Returned books are scanned, processed, and automatically updated in the system to reflect availability. Catalog updates are processed in real time to ensure accurate metadata and indexing. Data analytics modules process circulation records, usage patterns, overdue statistics, and inventory trends to generate insights for evidence-based collection development. These automated processes minimize manual workload, reduce errors, and enhance efficiency in managing library operations.

Output

After successfully logging into the system, users are directed to dashboards tailored to their assigned roles. Administrators can access analytics dashboards, monitor library operations, manage user accounts, generate circulation and inventory reports, and review overall collection performance. Authorized staff members can view book availability, manage catalog records, process borrowing and returning transactions, monitor inventory status, and generate summaries such as overdue lists and circulation logs. Their access is limited only to the specific modules and features that the administrator has assigned to them, ensuring proper role-based control within the system. The system produces real-time outputs such as alerts, book status updates, transaction receipts, inventory summaries, and data analytics visualizations, ensuring that authorized users receive accurate, timely, and role-specific information needed for efficient and intelligent library management

Performance

The system's performance is evaluated by its ability to handle multiple concurrent users, maintain real-time responsiveness, and process library transactions and analytics efficiently. Functional and non-functional testing ensures the stability, speed, and accuracy of system operations. The Intelligent Library System is designed to remain reliable during peak usage periods such as enrollment season or exam weeks allowing continuous access to catalog data, borrowing and returning services, and analytics modules without delays or system interruptions. The system ensures quick data retrieval, efficient processing of transactions, and consistent performance across authorized devices used by administrators and designated library staff.

Control

Control and security protocols are strictly implemented to protect sensitive library data.

Administrators hold full control over system configurations, user management, analytics access, and catalog administration. Librarians and Staff are granted access based on the permissions explicitly granted to them by the Administrator. Students do not have direct system access; their interaction is limited to receiving automated SMS notifications for transaction updates, such as overdue book alerts. The system incorporates Role-Based Access Control (RBAC), authentication, audit logs, and data validation to ensure information integrity and prevent unauthorized modifications. These control mechanisms maintain accuracy, confidentiality, and security within the Intelligent Library Management System.

4.2.11 Data and Process Modeling

The researchers employed data and process modeling to establish a clear structure and workflow for the Intelligent Library Management System. This systematic approach commenced with identifying core data elements such as user information, book records, borrowing history, and notification logs which provided the foundation for effective database organization and defining system inputs and outputs. For process clarity, flowcharts were used to visualize operational logic, mapping out key actions like login procedures, overdue checks, and report generation. This focused modeling ensured the resulting system was developed with a logical structure, maintained functional integrity, and effectively addressed real-world library management requirements.

Context Diagram

The Context Diagram illustrates the overall interaction between the Leveraging Data Analytics for Intelligent Library Management System in the CSPC and its external entities. The system is positioned at the center, representing the core platform that processes library transactions, analytics requests, reporting functions, and notification services, while defining the system's boundaries and its relationship with users and supporting modules.

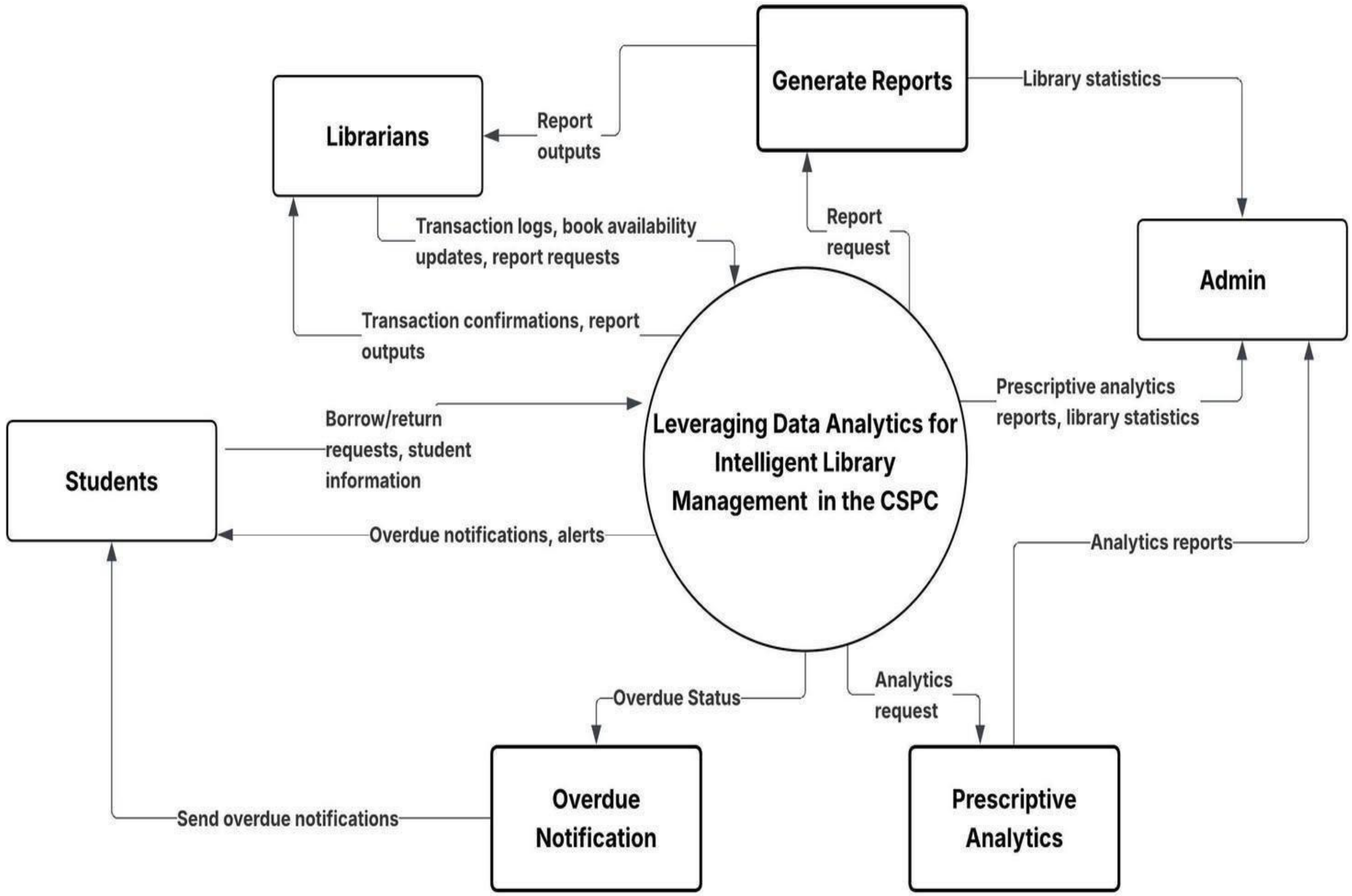

FIGURE 8: <u>Context Diagram</u>

The Context Diagram for the capstone project titled “Leveraging Data Analytics for Intelligent Library Management in the CSPC” presents a clear and simplified view of how the proposed system interacts with its primary users and supporting functional modules. As student developers, this diagram was created to visualize the overall structure of the system, identify key external entities, and understand how information flows between the system and its users within the library environment.

As illustrated in the diagram, the Intelligent Library Sytem serves as the central platform that interacts directly with Students, Librarians, and Administrators. Students communicate with the system by submitting borrow and return requests and receiving overdue notifications and alerts. Librarians interact with the system by managing transaction logs, updating book availability, and requesting reports, while receiving transaction confirmations and report outputs in return. Administrators interact with the system by requesting analytics and receiving library statistics, analytics reports, and prescriptive insights that support monitoring and decision-making.

The diagram also highlights key supporting system components, including Generate Reports, Prescriptive Analytics, and Overdue Notification modules. These components work in coordination with the central system to process analytics requests, produce library statistics and reports, generate prescriptive analytics outputs, and deliver automated overdue notifications to students. Each module contributes to the system’s ability to provide data-driven insights and improve operational efficiency.

This visual representation helped clearly define the system’s boundaries by distinguishing between user interactions and internal system processes. It also clarified the types of data exchanged within the system, such as borrowing and return requests, transaction confirmations, report outputs, analytics requests, library statistics, and overdue status notifications.

Overall, the Context Diagram served as an essential planning and communication tool during system development. It supported system organization, guided architectural decisions, and enabled clear explanation of the project to instructors, classmates, and stakeholders. By presenting the system at a high level, the diagram ensured that the developed solution is cohesive, well-structured, and aligned with the operational needs of the CSPC Library.

Data Flow Diagram

The Data Flow Diagram (DFD) in our study helps us visualize how information moves through Intelligent Library System. It shows the flow of data between users like librarians and students and system processes such as book recommendations, resource management, and report generation. During development, the DFD served as a guide to understand how inputs are received, how data is processed, and where outputs are delivered.

By mapping out these flows, we were able to design a system that handles tasks efficiently and securely. For example, when a student requests a book recommendation, the DFD shows how that request is processed through the system and how the result is returned. This helped us identify key components, avoid data bottlenecks, and ensure smooth communication between modules. Overall, the DFD was a valuable tool in building a well-structured and reliable system.

The DFD also facilitated better collaboration among the development team by providing a common visual reference for understanding system processes. It allowed team members to clearly see how each module interacts with others, reducing misunderstandings and improving coordination during development. By having a visual representation of data flows, the team was able to prioritize critical processes, allocate tasks effectively, and ensure that the system architecture supported all user requirements.

The DFD guided system evaluation by identifying key points for validation, error handling, and security, ensuring data integrity and efficient, secure, and user-friendly operation of the Intelligent Library Management System. This structured analysis also helped evaluators verify that data flows correctly between processes, minimizes redundancies, and aligns system functions with real-world library operations and user requirements.

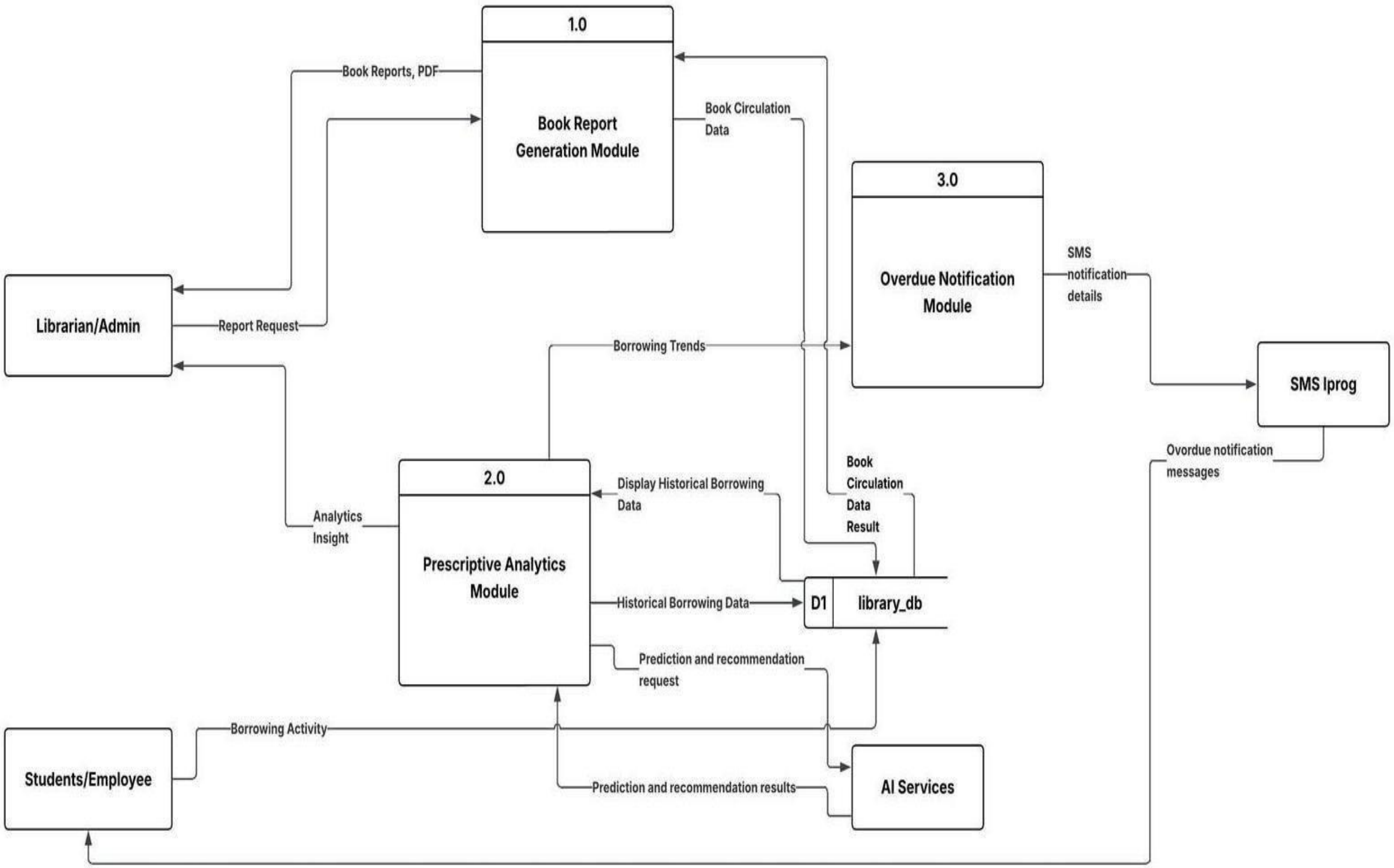

FIGURE 9: : <u>Data Flow Diagram Leve 0</u>

Figure 8 presents the Level 0 Data Flow Diagram (DFD) of the Intelligent Library System , illustrating the primary processes of the system and how data flows between external entities, internal modules, and supporting services. This diagram provides a high-level view of the system's major functional components and defines how information is processed within the system.

As shown in the diagram, the external entities interacting with the system include the Librarian/Admin, Students/Employees, SMS Ipro, and AI Services. The Librarian/Admin initiates report requests and receives book reports in PDF format and analytics insights generated by the system. Students and Employees interact with the system through borrowing activities, which serve as a primary input for analytics and notification processes.

The system is composed of three main processes at Level 0. The Book Report Generation Module (Process 1.0) receives book circulation data from the library database and processes report requests from the Librarian/Admin to generate book reports. The Prescriptive Analytics Module (Process 2.0) processes historical borrowing data retrieved from the library database and borrowing activity data from users. It generates analytics insights, borrowing trends, and prediction and recommendation results, which are then relayed to the Librarian/Admin and stored in the database.

The Overdue Notification Module (Process 3.0) uses book circulation data results from the library database to determine overdue status. It generates SMS notification details, which are sent to the SMS Iprog service for delivery of overdue notification messages to Students and Employees.

A centralized data store, labeled library_db (D1), serves as the system's main repository for book circulation data and historical borrowing records. This data store supports all three major processes by providing the necessary information for report generation, analytics processing, and overdue notification.

Overall, the Level 0 DFD illustrates how the system integrates reporting, analytics, and notification functionalities into a cohesive system. It highlights the flow of data among users, system processes, external services, and the database, demonstrating how the system supports intelligent, data-driven library management and automated user communication.

4.2.12 Object Modeling

An object model provides a crucial graphical representation of the objects, actions, and attributes within a system. This modeling technique is integral to Object-Oriented Analysis and Design (OOAD), serving as a blueprint that depicts the system's core components and their complex relationships. The visual clarity of the model is essential for understanding the system's internal processes and how various functionalities interact, effectively guiding the entire development process.

Use Case Diagram

The Use Case Diagram for the Leveraging Data Analytics Intelligent Library Management System provides a focused, high-level overview of system functionality and user interaction. It clearly delineates the roles and responsibilities of the two primary human actors: the Admin/Librarian and the Students/Employees, while highlighting integration points with key external components like AI Services and the SMS Gateway. This helps us understand the system’s functionality from the user's perspective and ensures that all key interactions are clearly defined.

Furthermore, the Use Case Diagram assists the development team in identifying and prioritizing critical system functionalities, ensuring that all required features, such as report generation, book recommendations, and overdue notifications, are properly linked to user actions. By presenting the system from the user’s perspective, it allows the researchers to anticipate potential workflow issues, reduce ambiguities during development, and improve system design. Overall, the diagram ensures that the Intelligent Library Management System operates efficiently, aligns with the operational needs of librarians and students, and delivers a reliable, user-friendly experience.

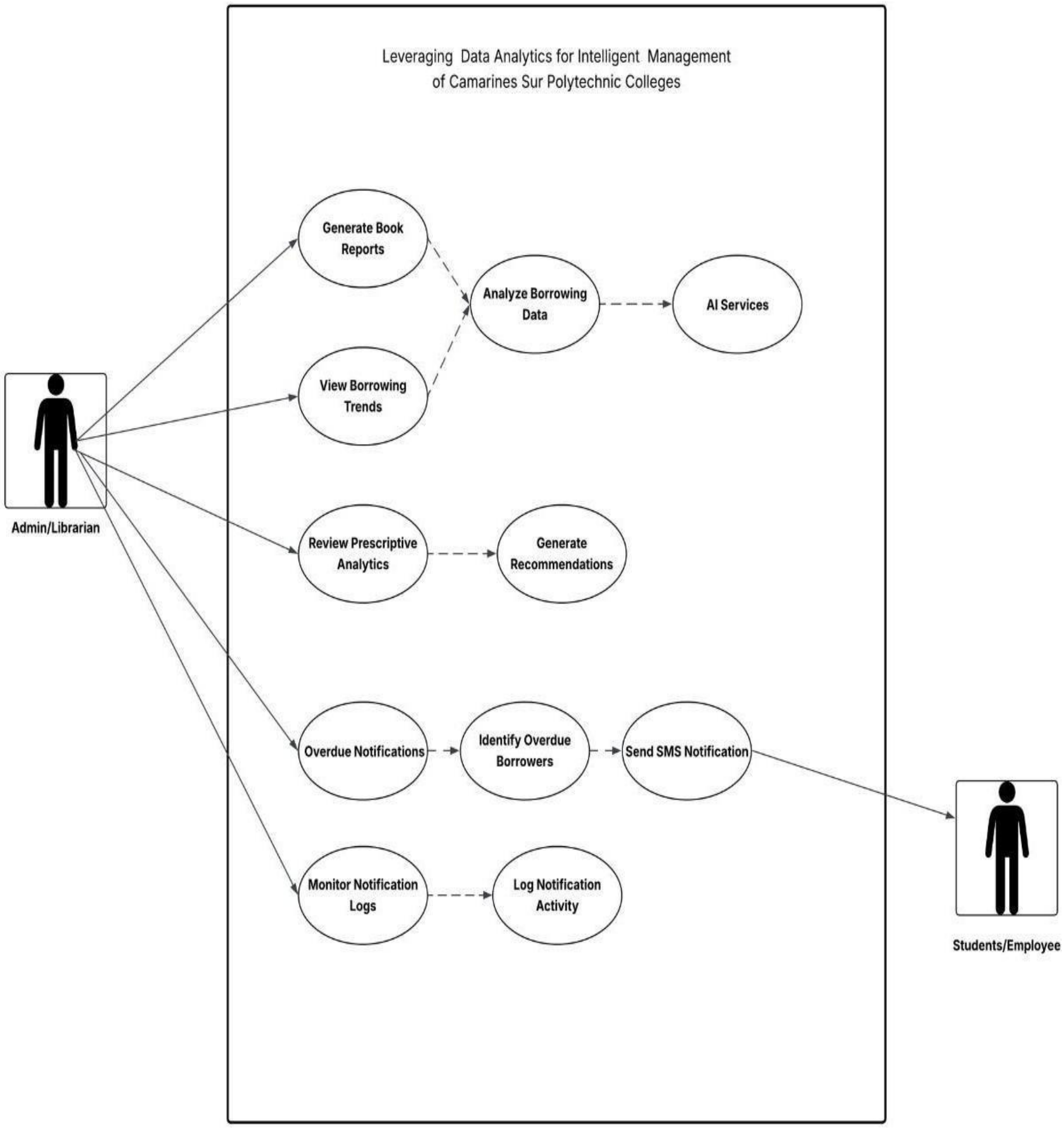

FIGURE 10: <u>Use Case Diagram</u>

This Use Case Diagram for the "Leveraging Data Analytics for Intelligent Library Management" system illustrates how the CSPC Library utilizes analytics, reporting, and notification functions to support data-driven operations. The system involves two main actors: Admin/Librarian and Students/Employees. The Admin/Librarian accesses the system to generate book reports, view borrowing trends, and analyze borrowing data, which form the foundation of the system's analytical capabilities. These processes enable the review of prescriptive analytics and the generation of recommendations, supported through integration with AI services to assist in evidence-based collection development and decision-making.

The Notification Module manages overdue monitoring and communication. The Admin/Librarian initiates overdue notifications, allowing the system to identify overdue borrowers and automatically send SMS notifications to Students and Employees. All notification activities are recorded through notification logs for monitoring and accountability. Overall, the diagram demonstrates how reporting, analytics, and automated notifications are systematically integrated to enhance operational efficiency, improve decision-making, and support intelligent library management at CSPC.

Sequence Diagram

The Sequence Diagram in our study plays a crucial role in illustrating how the Intelligent Library Management System handles specific tasks in a step-by-step, logical manner. It clearly depicts the flow of interactions between users, such as librarians and students, and system components, including external services like the Hugging Face API and the SMS Gateway. During development, this diagram helped the researchers understand the exact order of processes, such as when a student requests a book recommendation, when a librarian updates records, or when overdue notifications are sent. By mapping out these sequences, the team was able to design a system that responds accurately and efficiently to user actions. This ensures smooth, reliable performance, minimizes errors, facilitates troubleshooting, and enhances the overall usability and effectiveness of the library management system. Moreover, the sequence diagram supports better coordination among developers by providing a shared understanding of system behavior during runtime. It also serves as a valuable reference during system testing and maintenance by helping identify process bottlenecks and ensuring consistent interaction flows as the system evolves.

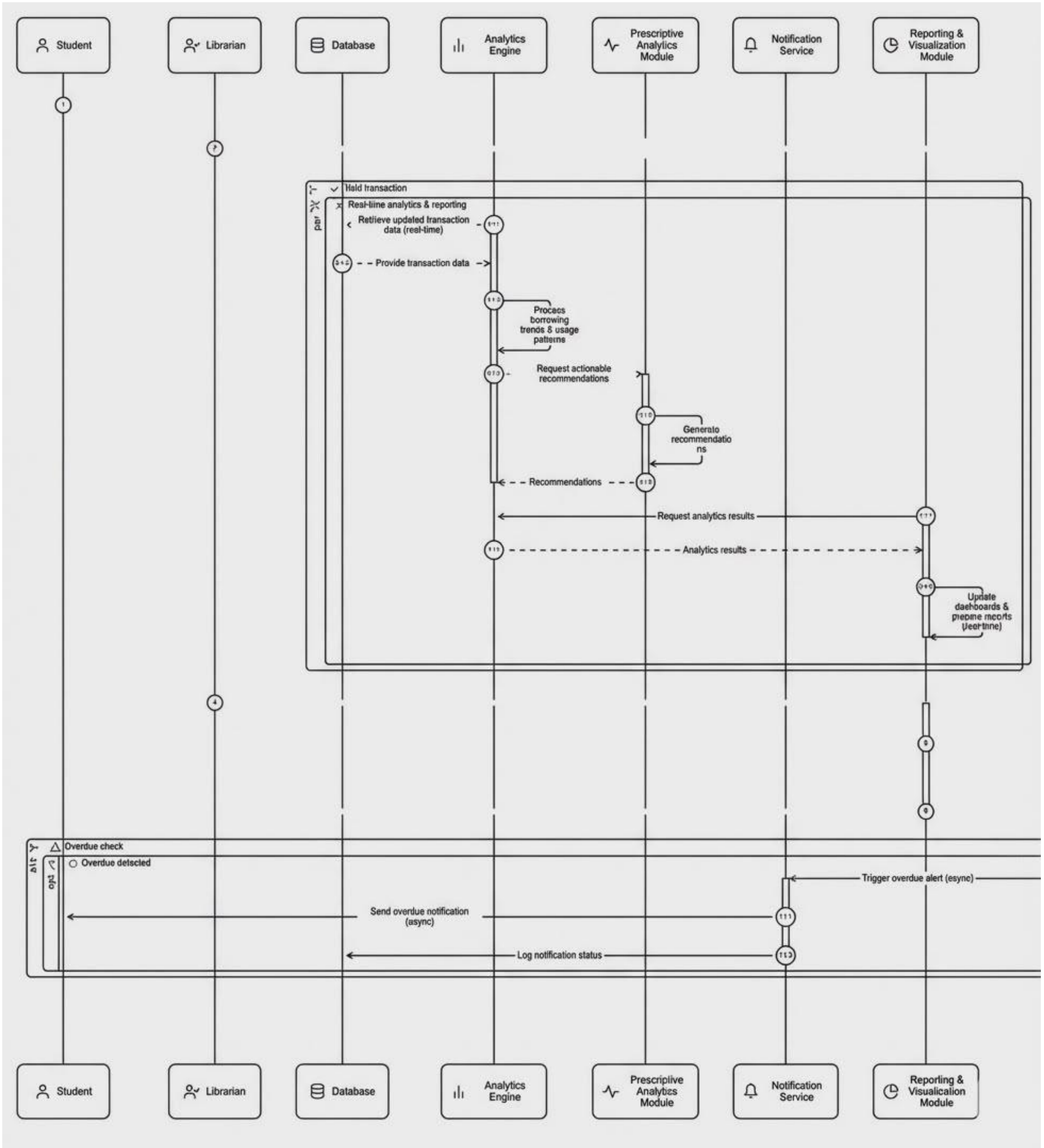

FIGURE 11: Sequence Diagram

Figure 11 presents the Sequence Diagram for Intelligent Library System, illustrating how system components interact to process analytics, generate recommendations, and deliver notifications. The sequence begins when the Librarian initiates system access for analytics and reporting purposes. The system first validates transactions and retrieves real-time borrowing data from the Database, ensuring that all records used for analysis are up to date and accurate.

The retrieved data is forwarded to the Analytics Engine, where borrowing trends and usage patterns are processed. Based on these analytics results, the Prescriptive Analytics Module is triggered to generate actionable recommendations that support data-driven library decision-making. The analytics results and recommendations are then sent to the Reporting and Visualization Module, which updates dashboards and prepares analytical reports for librarian review. In parallel, the system performs an overdue check, and when overdue transactions are detected, the Notification Service asynchronously sends SMS notifications to Students, while logging notification status for monitoring and accountability.

Overall, the sequence diagram demonstrates the coordinated flow of data among analytics, reporting, and notification services. It highlights the system's ability to support real-time analytics, prescriptive recommendations, and automated notifications, ensuring efficient library operations, timely communication with borrowers, and informed decision-making for collection management

4.3 Designing

The design phase of the Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges focused on establishing a structured framework that ensures the system meets all functional and non-functional requirements identified during the analysis stage. This phase involved the creation of the system architecture, database schema, data flow diagrams, and user interface mockups that illustrate how information is processed, stored, and visualized within the system.

The system design centers on three primary user groups librarians, faculty, and administrators to ensure an interface and feature set appropriate for each role. For librarians, the design highlights modules for book cataloging, collection monitoring, and updating inventory records. Faculty users are provided access to view available academic resources and observe usage trends relevant to their teaching and research needs. Administrators oversee the overall operations of the system, including user management, analytics dashboards, and generating comprehensive reports for evidence-based collection development.

Core design principles such as usability, scalability, security, and data accuracy were integrated throughout the design process. These principles ensure that the system is easy to navigate, capable of handling increasing library data over time, secure in managing bibliographic and user-related records, and reliable in producing meaningful analytics insights. Wireframes and layout prototypes were developed to visualize the system's workflow and ensure that navigation is clear, efficient, and aligned with the operational needs of CSPC's library.

Through this systematic and user-centered design approach, the ILMS is ensured to be functional, secure, and adaptable supporting the institution's shift toward data-driven decision-making in library management and collection development.

4.3.1 Data Design

This phase involves developing the structured data models that organize and define how information is stored and processed within the Intelligent Library Management System for Camarines Sur Polytechnic Colleges. It follows an iterative approach in which various data design options such as entity structures, relationships, and data flow arrangements are created, reviewed, and refined to ensure they accurately support the system's analytical and library management functions. The primary objective of this phase is to finalize the database organization and data specifications, ensuring that all informational requirements are fully addressed before proceeding to the system's full-scale development.

4.3.2 Output and User-Interface Design

The output and user-interface design of the Intelligent Library Management System were developed to provide accurate, timely, and accessible information for all types of users within Camarines Sur Polytechnic Colleges. Examples of system outputs include book inventory records, usage and circulation summaries, collection analytics, user activity logs, and data-driven reports that support evidence-based collection development. To access these outputs, the system features a clean and well organized dashboard where modules are logically arranged, navigation is intuitive, and essential functions can be easily reached. The design incorporates consistent color schemes, readable typography, and responsive layouts to ensure smooth accessibility across desktops, laptops, and other devices used by library staff and administrators. By combining a user-friendly interface with reliable analytical outputs, the system enhances operational efficiency, supports data-informed decision-making, and promotes improved management of library resources within CSPC.

4.3.3 Forms

An effective form serves as a crucial communication point between users and the system, allowing them to provide information accurately and access features with ease. In the Intelligent Library System , forms are designed to be clear, intuitive, and user-friendly, guiding users smoothly through each required step whether they are typing in their credentials, selecting options, or interacting with system controls. A well-structured form minimizes user confusion, reduces errors, and ensures that data collected is accurate and reliable. By presenting information in an organized and accessible format, the form becomes a bridge between the user and the system, turning what could be a complex process into a simple and seamless interaction.

Figure 12 illustrates the Login Page of the system, which functions as a secure gateway for authorized users such as staff, librarians, and administrators. This page is designed to ensure that only authenticated users can access the system by requiring valid login credentials. The inclusion of clearly labeled username and password fields promotes ease of use while maintaining security standards. Additionally, the login mechanism helps protect sensitive data and system functionalities by preventing unauthorized access, thereby supporting effective user management and overall system integrity.

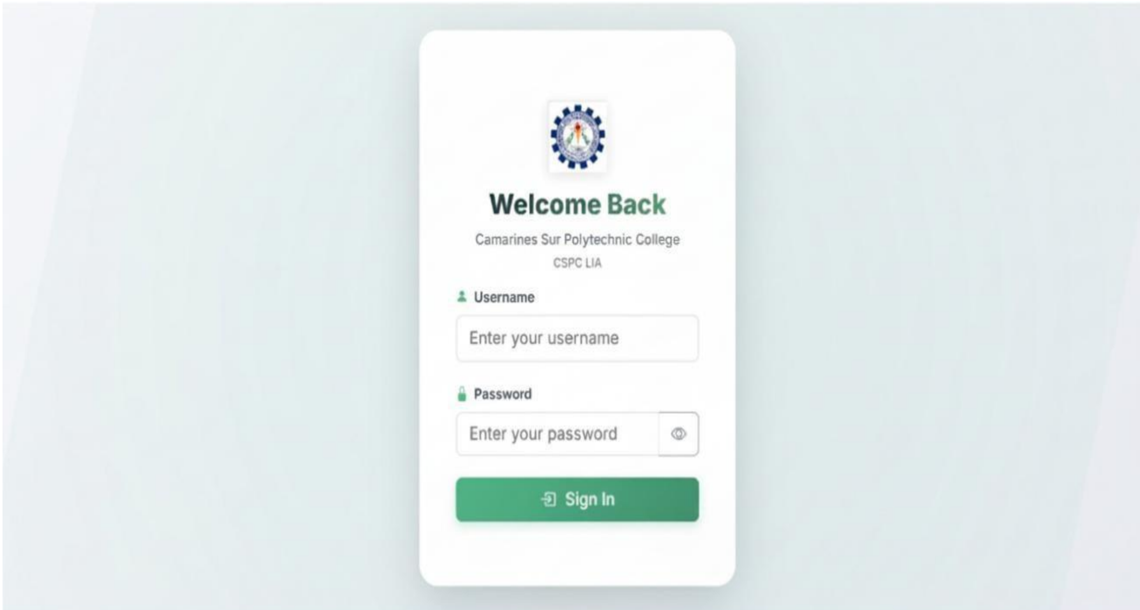

Figure 12: Login Page

The login interface is designed to support easy navigation and role-based access, ensuring that each user type, Librarian, or Administrator is properly authenticated before entering the system. With its modern design and intuitive flow, the login form enhances user experience and ensures a secure and efficient entry into the system platform.

Figure 13 presents the Library Analytics Dashboard of the system, highlighting monthly statistics related to returned books. The dashboard displays key performance indicators, including Total Returned Books, Average Processing Time, Active Categories, and an Efficiency Score, providing administrators with a clear and concise overview of library return operations. It also features configurable controls for selecting month and year, generating reports, and setting optional signatories. This functionality allows for flexible analysis, thorough documentation, and effective monitoring of library activities, supporting evidence-based decision-making and improving overall operational efficiency and management within the library. It also helps administrators track trends and make quick, informed decisions.

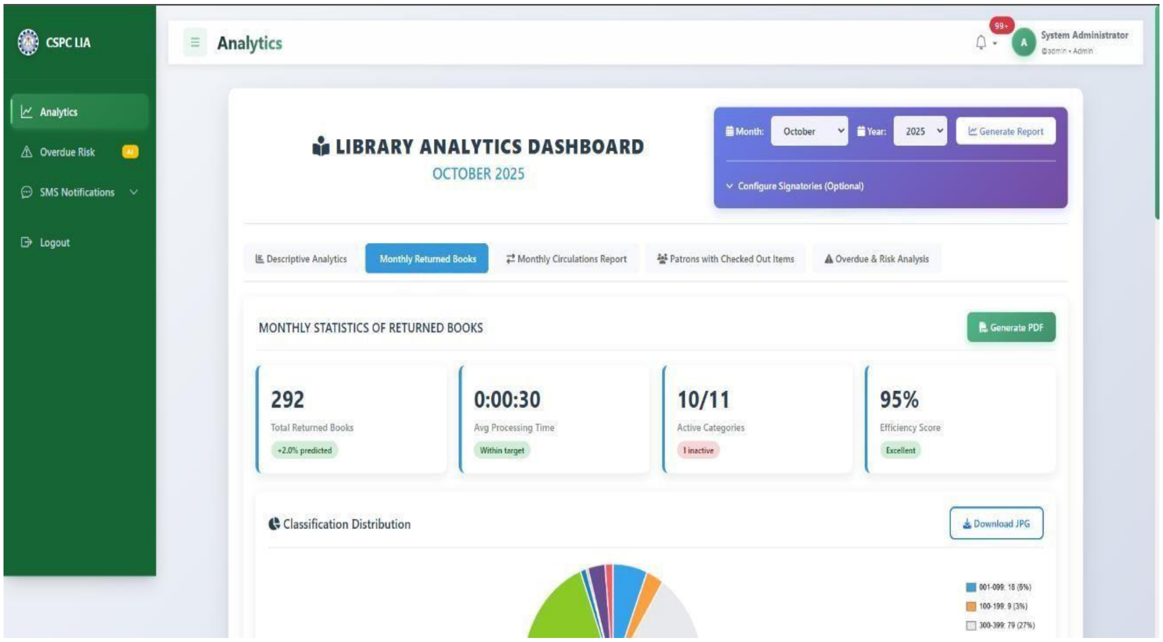

Figure 13: <u>Library Analytics Dashboard</u>

The Library Analytics Dashboard is designed to support descriptive analytics by visually summarizing monthly returned book transactions and operational efficiency within the library system. It presents essential indicators such as the total number of returned books, average processing time per transaction, category activity status, and an overall efficiency rating. The dashboard also includes a Classification Distribution chart that illustrates the proportion of returned books across classification ranges, enhancing insight into collection usage patterns. Built-in features for PDF and image report generation facilitate formal reporting and record-keeping. With its structured layout, clear visual hierarchy, and intuitive controls, the dashboard enables administrators to efficiently monitor library return activities and supports evidence-based decision-making within the system platform.

Figure 14 presents the Overdue Risk Analysis page of the Intelligent Library System, which uses AI-powered prediction to classify borrowed items into High, Medium, Low, and Safe risk levels. The dashboard summarizes circulation risk through item counts and percentages, while detailed listings of high- and medium-risk items provide essential borrower and due-date information. Built-in SMS notification features enable administrators to proactively intervene and reduce overdue occurrences.

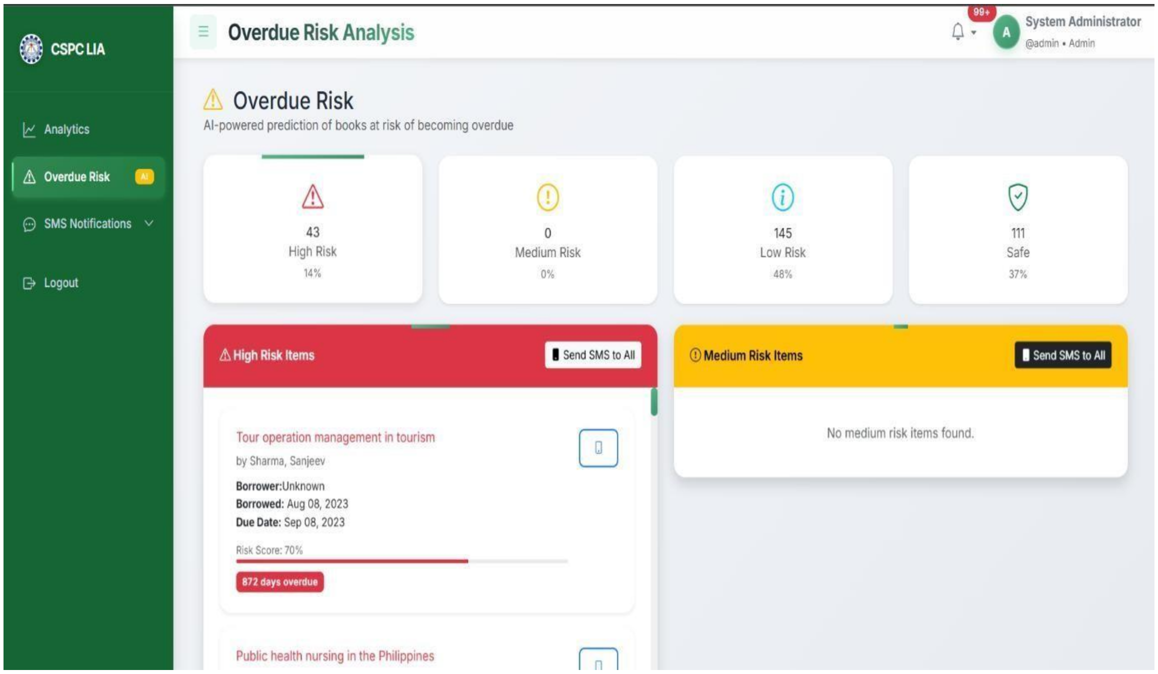

Figure 14: <u>Overdue Risk Analysis</u>

The Overdue Risk Analysis interface is designed to support predictive analytics by visually identifying and categorizing borrowed items based on their likelihood of becoming overdue. It employs clear numerical indicators and percentage-based summaries to communicate risk distribution across High Risk, Medium Risk, Low Risk, and Safe categories. Color-coded alerts and segmented panels emphasize items requiring urgent attention, while detailed item cards provide actionable insights for follow-up. The inclusion of SMS notification functionality allows administrators to directly communicate with borrowers associated with high-risk items, enabling timely intervention. Through its structured layout, real-time risk assessment, and integrated communication tools, the interface supports evidence-based circulation management and reduces potential overdue incidents within the system platform.

Figure 15 shows the Send Custom SMS page of the Intelligent Library System, which allows system administrators to manually compose and send SMS notifications to students or employees. The interface provides structured input fields for recipient type, recipient selection, phone number, and message content, ensuring accurate and controlled message delivery. Built-in character limits and validation guidance support effective and compliant SMS communication.

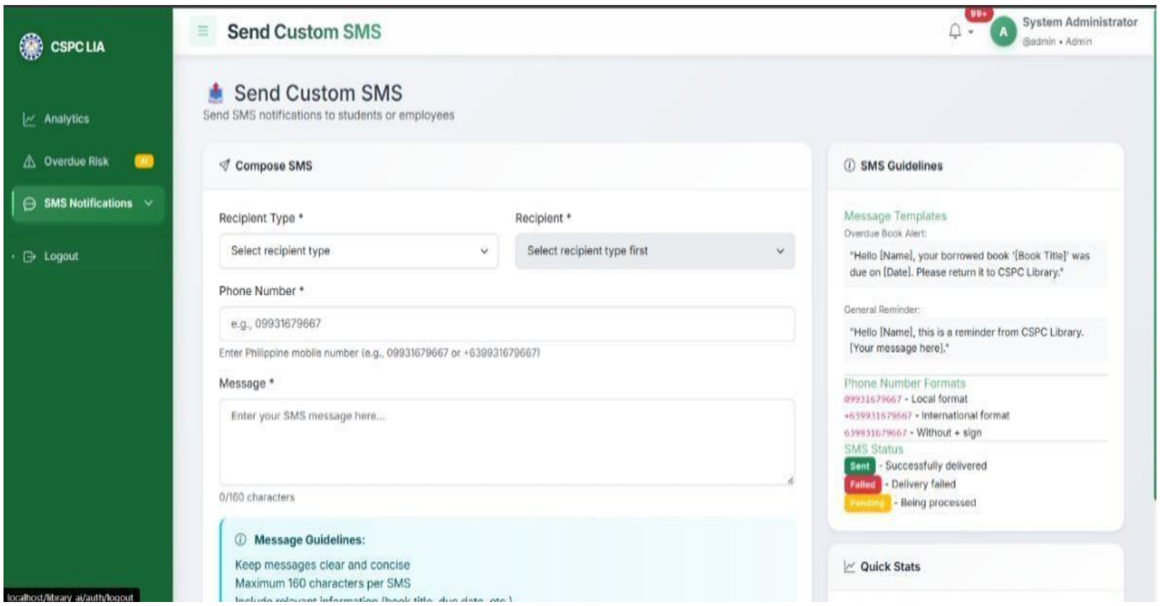

Figure 15: <u>Custom SMS</u>

The Send Custom SMS interface is designed to facilitate direct and customizable communication between the library administration and its users. It supports targeted messaging through selectable recipient types and predefined message templates, while built-in guidelines ensure proper formatting and adherence to SMS length constraints. Supplementary panels display message templates, phone number format standards, and SMS delivery status indicators, enhancing usability and reducing delivery errors. Through its intuitive layout and integrated validation, the interface supports timely notifications and strengthens communication-driven library management within the system platform. This interface allows library staff to efficiently reach specific users with relevant information, improving engagement and compliance. It also minimizes errors and delays in communication, ensuring that important updates are delivered promptly and reliably.

4.3.4 System Architecture

The System Architecture serves as the blueprint detailing the basic structure and core component interactions of the system. This design procedure translates user requirements into a technical system, encompassing both the fixed structure and the dynamic behavior of the software. It clearly describes the key software modules and their interrelationships, ensuring they collectively provide the desired functionality within a web-based environment.

Security

This system addresses its security objective by ensuring controlled access, data protection, and reliable operation across the Notification, Analytics, and Reports modules. Secure user authentication is implemented to allow only authorized personnel to access system functionalities relevant to their responsibilities.

The Notification Module restricts alert creation and delivery to authorized users to prevent unauthorized or misleading messages. The Analytics Module secures access to circulation data and analytical outputs to ensure that insights used for decision-making remain accurate and protected. The Reports Module limits report generation and export functions to authorized users, safeguarding analytical summaries and administrative outputs.

The system adheres to the principles of confidentiality, integrity, and availability by enforcing access restrictions, validating data inputs and outputs, and ensuring stable system access during regular library operations. Secure data handling mechanisms protect information exchanged between modules, supporting reliable analytics, accurate reporting, and trustworthy notifications. These measures collectively ensure that the system fulfills its security objective while supporting data-driven library management at CSPC. Additionally, the implementation of these security measures enhances user confidence in the system, ensuring that librarians and administrators can depend on its functionality for informed decision-making and efficient daily operations.

4.4 Development

The Intelligent Library System for Camarines Sur Polytechnic Colleges is a web-based platform designed to modernize and enhance library operations through the integration of data analytics. The system automates essential processes such as book cataloging, collection monitoring, usage tracking, and analytics generation to support evidence-based collection development. By centralizing library data into a unified platform, the system ensures accuracy, efficiency, and accessibility of information for librarians, faculty, and administrators. It minimizes manual record-keeping, reduces errors, and provides real-time analytical insights that help the institution make informed decisions regarding resource acquisition, allocation, and improvement of library services.

The back-end of the system serves as the core engine responsible for processing requests, managing data, and ensuring secure communication between the user interface and the system database. The system was developed using PHP (CodeIgniter 4) due to its robust MVC architecture, scalability, and ability to support modular, well-structured coding practices. This framework enables efficient routing, request handling, and server-side processing while maintaining clean and maintainable code.

For data storage and management, MySQL was utilized as the primary relational database. The database stores structured information such as book records, user accounts, circulation logs, usage analytics, and system activity data. MySQL enables efficient querying and indexing, ensuring that large volumes of library data can be retrieved, analyzed, and visualized quickly and securely.

To facilitate smooth communication between system components, RESTful APIs were implemented, enabling real-time updates in modules such as collection analytics, usage tracking, and report generation. Security features including authentication, authorization, and role-based access control (RBAC) were incorporated to protect sensitive library records and restrict system functionalities based on user roles. Input validation, encrypted data transactions, and secure session handling further reinforce system security.

Asynchronous communication using AJAX was also integrated to enhance responsiveness, allowing dynamic updates such as loading analytics dashboards or retrieving filtered book records without requiring full-page reloads. The system's architecture was designed with scalability in mind, enabling future integration of advanced analytics, machine learning recommendations, and extended library services.

4.4.1 Sprint Reviewing and Retrospective

The development of the system followed the Agile methodology, which emphasizes iterative and incremental progress through sprint cycles. At the end of each sprint, Sprint Review and Sprint Retrospective activities were conducted to evaluate completed features, gather feedback from stakeholders, and identify areas for improvement. These practices supported continuous system enhancement and ensured alignment with user and institutional requirements.

4.4.2 Sprint

A sprint is a time-boxed development period focused on delivering specific system functionalities. For the system, each sprint addressed modules such as Book Cataloging, Collection Monitoring, Usage Tracking, Analytics Dashboard, and Report Generation. During each sprint, the development team defined tasks, prioritized user stories, and set achievable goals. This incremental approach enabled early issue detection, progressive refinement, and the delivery of functional system components at the end of each iteration.

4.4.3 Daily Scrum

Daily Scrum meetings were conducted to synchronize development activities and monitor progress throughout each sprint. During these short meetings, team members reported on completed tasks, upcoming activities, and any obstacles encountered. This practice facilitated coordination among backend, frontend, and analytics developers, ensuring smooth module integration and adherence to sprint timelines.

4.4.4 Sprint Review

At the conclusion of each sprint, a Sprint Review was conducted to present newly developed system features to stakeholders, including librarians, faculty representatives, and administrators. Demonstrations included functionalities such as real-time analytics, book record updates, and usage trend visualizations. Stakeholder feedback gathered during these sessions was documented and used to guide feature enhancements and prioritization in subsequent sprints.

4.4.5 Sprint Retrospective

Following the Sprint Review, the development team conducted a Sprint Retrospective to evaluate the sprint process internally. Discussions focused on identifying successful practices, challenges encountered, and areas requiring improvement, such as testing workflows, code optimization, or team coordination. The insights gained from the retrospective helped improve development efficiency, collaboration, and output quality in succeeding iterations.

4.4.6 Test Plan

This section outlines the objectives, scope, and strategy of the software testing process for the Intelligent Library System of Camarines Sur Polytechnic Colleges. The main goal of testing was to validate that the system satisfies all functional and non-functional requirements, performs reliably under expected workloads, and effectively supports the institution’s library operations. The testing methodology implemented was black-box testing, which focused on evaluating the system’s external behavior and user interface without examining the internal code structure.

The testing covered several key areas including unit testing, integration testing, system testing, usability testing and performance testing. Test cases were formulated based on the system’s primary processes such as book cataloging, user management, collection usage tracking, analytics generation, and report creation to ensure that each module performed according to its intended behavior.

Table 7. UNIT TESTING RESULT

Integrated Modules	Result	Interpretation
Prescriptive Analytics	18%	**Passed**
Overdue Risk	13%	**Passed**
Book Reports	72%	**Passed**
Overall	100%	**Passed**

The unit testing results presented in Table 7 indicate that all integrated analytical modules of the system successfully passed their respective test cases. The Prescriptive Analytics module achieved an 18% coverage, the Overdue Risk module recorded 13%, and the Book Reports module attained the highest coverage at 72%. Despite the variation in test coverage percentages, all modules met their expected outcomes and obtained a “Passed” result, demonstrating correct execution of their defined unit-level functions.

The passing results confirm that each module functioned correctly when tested independently. The Overdue Risk module accurately processed borrowing rules to identify potential overdue items, while the Prescriptive Analytics module generated appropriate system-driven recommendations based on circulation data. The Book Reports module showed extensive test coverage, reflecting its broader functional scope in data aggregation and report generation. Overall, the results indicate that the system conforms to its functional requirements at the unit testing level.

These findings are consistent with software testing principles emphasizing unit testing as a critical step in validating individual components before system integration. Atlassian highlights that successful unit testing improves software reliability by detecting logical errors early in development. Similarly, Fernandes and Lina [2021] noted that black box unit testing enhances confidence in analytics-driven library systems. The correct performance of the Overdue Risk and

Prescriptive Analytics modules also aligns with studies by Zong [2023] and Muazu et al. [2024], which stress that reliable predictive and prescriptive outputs depend on well-tested computational units.

The successful unit testing suggests that the system is built on a stable technical foundation. Reliable analytical modules reduce the likelihood of errors during integration and deployment, supporting accurate overdue prediction, reporting, and decision support for library administrators at CSPC.

Unit testing was conducted on individual modules in isolation and did not evaluate inter-module interactions or real-time data dependencies. Therefore, these results alone cannot guarantee full system reliability without integration, system, and performance testing.

In summary, the unit testing results demonstrate that the system meets its functional requirements at the unit level. The successful validation of Prescriptive Analytics, Overdue Risk, and Book Reports modules provides a strong basis for subsequent testing phases and supports the system's readiness for higher-level evaluation.

Table 8: INTEGRATION TESTING RESULT

Integrated Modules	Result	Status
Authentication Module	1.00	Passed
Analytics Module	1.00	Passed
Overdue Risk	1.00	Passed
SMS Module	1.00	Passed
Overall	1.00	Passed

The integration testing results presented in Table 8 show that all integrated modules of the Intelligent Library System successfully passed the defined integration test cases. The Authentication Module, Analytics Module, Overdue Risk Module, and SMS Module each obtained a result rating of 1.00, indicating a Passed status for all modules. These results confirm that the system components functioned correctly and consistently when integrated, demonstrating seamless interaction and reliable data flow within the overall system environment.

The results confirm that data flow and communication between modules were successfully implemented. The Authentication Module reliably handled user access and credential validation, enabling secure interaction with system services. The Analytics and Overdue Risk modules accurately processed and exchanged circulation data without data loss or processing errors. Additionally, the SMS Module correctly received system triggers and delivered notifications as expected. These outcomes indicate that the integrated modules operated cohesively and adhered to their defined interaction requirements.

The results are consistent with established software testing principles, which emphasize integration testing as a critical phase for detecting interface defects and data flow issues that may not be identified during unit testing. According to Atlassian [2024]**,** integration testing validates interactions between system modules and ensures that combined components operate correctly within an integrated environment. Similarly, Myers et al. [2011] emphasized that black box–based integration testing is essential for verifying externally observable system behavior, particularly in modular software systems where correctness depends on accurate interaction among independently developed components.

The successful integration testing indicates that the system is capable of supporting end-to-end workflows, including secure access, analytics processing, overdue risk evaluation, and automated SMS notifications. This enhances system reliability and readiness for deployment in the CSPC Library environment.

Integration testing was conducted under controlled test conditions and may not fully capture performance issues arising from high user load or real-time operational constraints. Further system and performance testing are necessary to assess scalability and robustness

Overall, the integration testing results confirm that the system modules interact correctly and reliably. The 100% pass rate across all integrated modules demonstrates system stability and provides a strong foundation for subsequent system-level and performance evaluations.

Table 9: USABILITY TESTING RESULTS

Metric	Result	Interpretation
User Satisfaction	4.0	Agree
User Interface Design	3.0	Neutral
Accessibility	3.76	Agree
System Responsiveness	4.1	Agree
Error and Feedback Display	3.48	Neutral
Overall	3.67	Agree

The results of the usability evaluation demonstrate that the developed system effectively supports the operational needs of the library, particularly in terms of performance, accessibility, and overall user satisfaction. The high mean score for System Responsiveness (4.1) indicates that the system can efficiently handle routine library transactions, minimizing delays during peak hours and supporting reliable service delivery in real-world library environments.

The favorable ratings for User Satisfaction (4.0) and Accessibility (3.76) further suggest that the system is easy to learn and practical for daily use. These results indicate that users, including those with limited technical background, can navigate the system with minimal assistance, contributing to reduced reliance on technical staff and improved operational independence among library personnel.

Despite these strengths, the evaluation identifies areas that require improvement. The User Interface Design received a neutral rating (3.0), implying that the visual layout and consistency may not be fully intuitive, particularly for first-time users. Similarly, the neutral rating for Error and Feedback Display (3.48) suggests that system messages during errors or incorrect inputs may lack sufficient clarity, which could affect user confidence and interaction efficiency.

Overall, the findings confirm that the system is functionally sound and ready for deployment, as it meets its core usability objectives. However, further refinement of the user interface and feedback mechanisms such as clearer error messages, improved visual consistency, and guided prompts is recommended to enhance long-term usability, user acceptance, and effectiveness in actual library operations.

```
Server Software:        Apache/2.4.58
Server Hostname:        localhost
Server Port:            80

Document Path:          /library_ai/
Document Length:        24767 bytes

Concurrency Level:      10
Time taken for tests:   10.707 seconds
Complete requests:      100
Failed requests:        0
Total transferred:      2521600 bytes
HTML transferred:       2476700 bytes
Requests per second:    9.34 [#/sec] (mean)
Time per request:       1070.672 [ms] (mean)
Time per request:       107.067 [ms] (mean, across all concurrent requests)
Transfer rate:          230.00 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    1   0.9      0       6
Processing:   462  654 355.1    561    3655
Waiting:      460  652 355.2    559    3655
Total:        462  655 355.1    562    3655
WARNING: The median and mean for the initial connection time are not within a normal deviation
        These results are probably not that reliable.

Percentage of the requests served within a certain time (ms)
  50%    562
  66%    580
  75%    605
  80%    652
  90%   1067
  95%   1089
  98%   1459
  99%   3655
 100%   3655 (longest request)
PS C:\xampp\htdocs\library_ai>
```

Figure 16: Performance Testing Result

The results show the system performs well under typical conditions, with no sign of performance degradation or system failures. The stable performance of these modules suggests that the system is designed to meet operational demands without significant delays or issues. The warning about connection time variability indicates that while the system performs well under expected loads, some aspects of connection handling (especially in high variability scenarios) might require attention.

The findings align with best practices in performance testing for large-scale information systems, where APIs, data retrieval, and notification services are key to ensuring operational continuity. Previous studies emphasize the importance of response time and throughput in evaluating system reliability, particularly for library systems that manage large datasets and real-time interactions. The system demonstrates compliance with these guidelines by achieving consistent performance metrics across its key services.

For the CSPC Library, the results suggest that the system enhanced operational efficiency. Faster transaction processing, real-time analytics, and timely notifications support better decision-making and improve the user experience for both library staff and students. This consistent performance also implies a reduced likelihood of service interruptions, enabling the library to depend on the system for daily operations without significant bottlenecks.

The testing primarily focused on typical operational conditions, and did not evaluate the system under extreme traffic or prolonged stress scenarios. Furthermore, there was no in-depth scalability testing to assess how the system might perform under heavy loads or variable network environments. The absence of these tests means that the system's behavior under peak or atypical conditions is not yet fully understood.

In summary, the performance testing confirms that the system provides stable and reliable service under typical load conditions. The system demonstrates acceptable response times, stable throughput, and high request completion rates across its core modules. While these results establish a solid foundation for deployment, further stress testing and scalability analysis are recommended to ensure the system's robustness under peak loads or more complex usage scenarios.

4.4.7 Deployment

Within the SDLC framework and under the Scrum methodology, the deployment phase represents the final stage where the Intelligent Library System is released for operational use within Camarines Sur Polytechnic Colleges. The process involved preparing the production environment, configuring the server and database, and migrating all validated development data into the live system. After deployment, the system was thoroughly tested to ensure proper functionality, secure role-based access, and complete data integrity. Users including librarians, faculty members, and administrators—were guided on how to navigate the platform and utilize its features for cataloging, monitoring, and analytics.

Post-deployment monitoring and continuous feedback collection were conducted to detect potential issues and ensure system optimization. This phase provided CSPC with a centralized, efficient, and data-driven platform that enhances library operations and supports evidence-based collection development. The deployment marks the transition from a development setting to a live, production ready environment. Although it signifies the end of the primary development cycle, the system remains open for incremental updates and improvements depending on user feedback and evolving library needs.

Deployment plays a crucial role in ensuring that all developed functionalities become accessible to the intended users and that the system delivers a seamless experience. A feedback mechanism was established to gather insights during real-world use, helping guide future refinements. This practice reflects the iterative and adaptive principles of the Scrum methodology, ensuring that the system continues to evolve and provide reliable, meaningful insights for CSPC's library operations.

During the sprint review, the development team demonstrated completed features and modules to stakeholders, including librarians, administrators, and faculty representatives. The review focused on showcasing functionalities such as book cataloging, collection monitoring, usage tracking, and analytics dashboards. Feedback gathered during this session enabled the identification of areas for enhancement to ensure that the system continues to meet institutional requirements and user expectations.

Deployment Diagram

A deployment diagram visually represents the physical structure of the Intelligent Library Management System by illustrating how its software components are distributed across hardware resources and how these components communicate with one another. It provides a clear overview of the system's runtime environment, allowing stakeholders to understand where and how the system operates within the library's infrastructure. This diagram serves as a bridge between system design and actual implementation, ensuring that technical requirements align with operational needs.

The deployment diagram for the Intelligent Library Management System was developed after the system successfully passed all testing phases, indicating that it is ready for actual use by librarians, faculty members, and administrators. Creating the diagram at this stage confirms that the system's components have been validated and can function reliably within the intended environment. Its inclusion demonstrates that the system is prepared for deployment and capable of supporting daily library activities without compromising performance or security.

Deployment diagrams play a critical role in system modeling by explaining the relationship between software and the hardware that supports it. According to Unified Modeling Language (UML) specifications, deployment diagrams illustrate how run-time components are configured and identify the physical nodes where software processes and services are executed. These nodes may include servers, client workstations, databases, and network devices, while communication links represent the data flow and protocols used for system interaction.

The deployment diagram maps software components to hardware nodes, showing how the system functions in a real operational setting. It supports analysis, maintenance, and scalability, ensuring secure access, efficient data processing, and effective library management at CSPC.

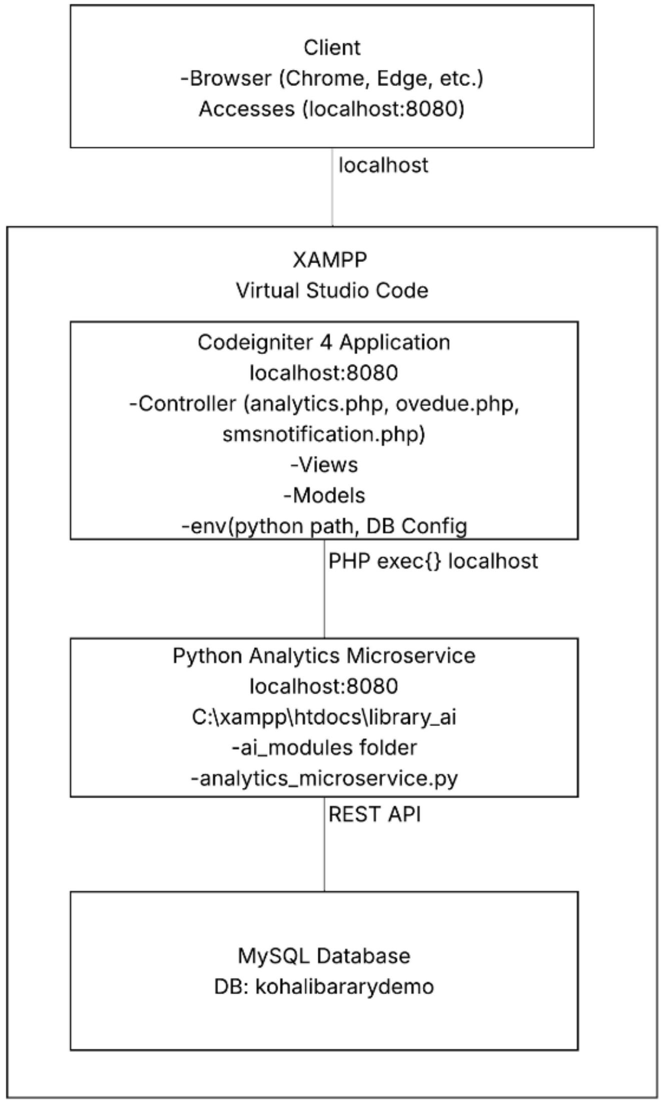

Figure 17: <u>Deployment Diagram</u>

Figure 16 illustrates the Deployment Diagram of the Intelligent Library System developed for Camarines Sur Polytechnic Colleges. The diagram presents the core components of the system and shows how the client interface, web application, analytics microservice, and database interact within a local development environment.

Users, such as administrators and authorized library staff, access the system through standard web browsers (e.g., Google Chrome or Microsoft Edge). The client sends requests to the application via localhost (port 8080**)**, allowing direct interaction with the system hosted on the local machine.

The web server environment is implemented using XAMPP, which provides the Apache server and PHP runtime. Within this environment, the CodeIgniter 4 application serves as the main system backbone. It consists of controllers (such as analytics.php, overdue.php, and smsnotification.php), views for user interface rendering, models for database interaction, and environment configuration files that define database credentials and the Python execution path. The application handles core library operations, request processing, and presentation logic.

For advanced analytics and data processing, the CodeIgniter application communicates with a Python Analytics Microservice hosted locally at localhost:8080. This microservice resides in the directory C:\xampp\htdocs\library_ai and contains the analytics_microservice.py script along with its supporting AI modules. Communication between the PHP application and the Python service is achieved through PHP exec() calls and REST API requests, enabling the system to perform analytics-related tasks efficiently.

The MySQL database, named kohalibrarydemo, functions as the system's primary data repository. It stores essential records such as book information, user accounts, circulation transactions, and analytical data. Both the CodeIgniter application and the Python microservice interact with the database to ensure consistent data retrieval and processing.

Overall, the deployment structure demonstrates a modular and well-integrated system architecture. By combining a PHP-based web application, a Python analytics microservice, and a relational database within a local server environment, the system ensures efficient data processing, reliable system performance, and a scalable foundation for future enhancements.

Notes

[1] Alex Kumah and Evans Forkuo-Minka. 2024. *Cause-and-Effect (Fishbone) Diagram: A Tool for Analyzing Root Causes.* National Library of Medicine, PMC. Retrieved from https://pmc.ncbi.nlm.nih.gov/articles/PMC11077513/

[2] Shcherbyna, A. 2023. *Breaking It Down: Decomposition Techniques for Better Software Development*. Medium, Mar. 4. Retrieved from https://medium.com/@khdevnet/breaking-it-down-decomposition-techniques-for-better-software-development-43d8d1048793

CHAPTER 5

SUMMARY, FINDINGS, CONCLUSION AND RECOMMENDATIONS

This chapter presents the summary of the study, the major findings based on the objectives, the conclusions drawn from the results, and the recommendations for further enhancement and institutional adoption of the Intelligent Library System with Data Analytics for Camarines Sur Polytechnic Colleges (CSPC).

5.1 Summary

The study entitled "Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges" was conducted to modernize and improve the library's operational processes by integrating data analytics, automated cataloging, and intelligent usage monitoring. The system was designed to address challenges in manual book recording, limited access to usage insights, inconsistent collection evaluation, and time-consuming report generation. Through the system, essential library services including cataloging, user management, collection tracking, and analytics reporting were centralized into a single digital platform to support evidence-based collection development.

The researchers adopted the Agile Scrum methodology, allowing iterative development, continuous testing, and regular stakeholder feedback. The system was developed using CodeIgniter 4 (PHP) and MySQL to ensure a secure, scalable, and efficient backend infrastructure. The system underwent black-box testing, including integration testing, system testing, usability testing, and user acceptance evaluation. Respondents included librarians, faculty members, and administrators who used the system to assess functionality, usability, and reliability. Data were analyzed using descriptive statistics such as frequency, percentage, and weighted mean.

Overall results showed that the system successfully automated core library operations, improved the accuracy of book data and usage tracking, and strengthened decision-making through detailed analytics. The findings also aligned with related studies, such as those by Ahmed [2021] and Cualbar [2025], which reported that integrating automation and analytics improves operational efficiency, transparency, and organizational decision-making.

5.2 Findings

Based on the objectives presented in Chapter 1, the following findings were determined:

1. The investigation of existing library practices at the CSPC Library revealed that the continued use of manual and semi-manual processes in cataloging, borrowing, returning, and monitoring book collections resulted in operational inefficiencies, fragmented data, limited tracking of book usage trends, and delays in generating timely reports for collection development and administrative decision-making.

2. The study successfully developed and implemented a real-time, data-driven Intelligent Library Management System composed of the Book Report Generation Module, Overdue Notification Module, and Prescriptive Analytics Module. The system enabled accurate real-time tracking of book usage, improved data accuracy, automated key library operations, and supported evidence-based collection management. In addition, the integration of DialoGPT enhanced user interaction by providing AI-assisted, conversational responses for library-related inquiries and system guidance, thereby improving accessibility, responsiveness, and overall user experience within the library system.

3. The study successfully developed and implemented a real-time, data-driven Intelligent Library Management System comprising the Book Report Generation, Overdue Notification, and Prescriptive Analytics modules. System testing confirms that it meets the defined functional and non-functional objectives.

3.1 **Integration Testing:** Achieved a perfect score of 1.00 across all integrated modules, confirming seamless interaction and real-time data flow between reporting, notification, and analytics components.

3.2 **Unit Testing:** Recorded an overall 100% pass rate, with 72% for Book Reports, 18% for Prescriptive Analytics, and 13% for Overdue Risk, verifying correct functionality of individual modules.

3.3 **Usability Testing:** Obtained an overall mean score of 3.67 (Agree); System Responsiveness (4.10) and User Satisfaction (4.00) were high, while User Interface Design (3.00) and Error Feedback (3.48) indicate minor areas for improvement. This confirms the system is accessible and user-friendly.

3.4 **Performance Testing**: Processed 100 requests with zero failures, achieving 9.34 requests/sec and a mean response time of 1070.67 ms, confirming reliable and efficient system performance.

Overall, these results demonstrate that the system provides accurate real-time tracking, automated library operations, AI-assisted user support, and reliable performance, fully supporting the study’s specific objectives.4. Deployment of the system in the CSPC Library environment demonstrated stable and effective real-world performance. Findings indicated high levels of user satisfaction in terms of usability, efficiency, and system interaction, confirming the system’s effectiveness in supporting daily library operations.

5.3 Conclusion

Based on the findings of the study, the researchers conclude the following:

1. Manual library operations significantly hinder efficient cataloging, monitoring, and reporting processes. Addressing these limitations requires the adoption of an automated and data-driven system capable of producing accurate, timely, and actionable information.

2. The developed Intelligent Library Management System effectively automated core library processes and provided real-time insights through its integrated modules. The system demonstrated its capability to support intelligent decision-making for collection development and resource management. The integration of DialoGPT further enhanced the system by enabling intelligent, conversational interaction that assisted users in accessing library information and system features more efficiently, demonstrating that conversational AI can improve usability, responsiveness, and user engagement in an intelligent library environment.
3. Comprehensive black box testing validated that the ILMS met its functional and performance requirements. The system consistently operated according to specifications, demonstrating reliability, usability, and acceptable performance levels across different testing conditions.
4. The successful deployment and positive user evaluation of the system confirm that the system enhances operational efficiency, reduces librarian workload, and supports sustainable library management practices within the CSPC Library.

5.4 Recommendations

Based on the conclusions derived from the study, the following recommendations are made for future enhancement and system expansion:

1. The CSPC Library may conduct periodic assessments of library workflows to continuously identify emerging operational challenges that can be addressed through system enhancements.
2. Additional functional modules, such as predictive demand forecasting and acquisition recommendation systems, may be integrated to further strengthen evidence-based collection development.

3. Regular regression testing and performance monitoring should be conducted to maintain system reliability as usage volume increases.
4. Continuous system maintenance, security updates, and scalability planning are recommended to ensure long-term sustainability and adaptability of the system

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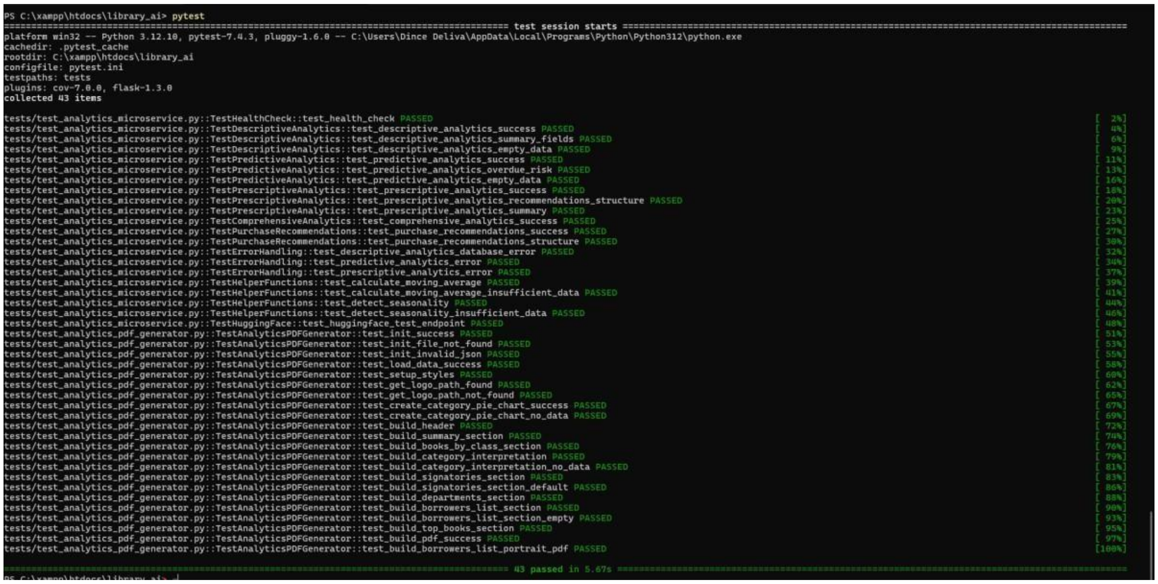
APPENDICES

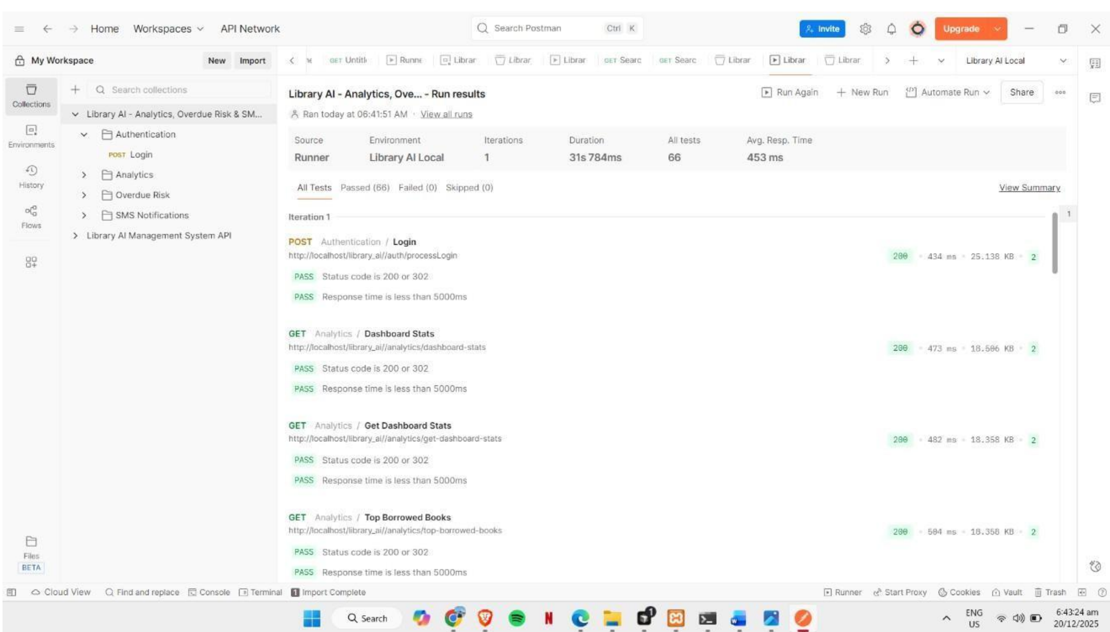
APPENDIX A

EVALUATION TOOL

UNIT TESTING



INTEGRATION TESTING



USABILITY TESTING

LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES

Profile Respondents

Name: (optional)____

Role in Relation to the Library

☐ Administrator

☐ Librarian/ Staff

☐ Student/ Employee

This section aims to evaluate the overall performance, functionality, and reliability of the Intelligent Library Management System based on your experience using its various features and modules. Your responses will help assess how well the system operates, how consistently its components work together, and how effectively it supports library processes such as borrowing, cataloging, and analytics. Please rate each statement according to your experience with the ILMS during testing using the 5-point Likert scale provided.

Scale:

1 – Strongly Disagree

2 – Disagree

3 – Neutral

4 – Agree

5 – Strongly Agree

Usability Testing

Statement	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
I would like to use this system frequently..					
I found the system unnecessarily complex.					

I think that I would need the support of a technical person to be able to use this system.					
The system exhibits noticeable delays or lag during use.					
The integration between modules requires excessive manual adjustment or correction.					

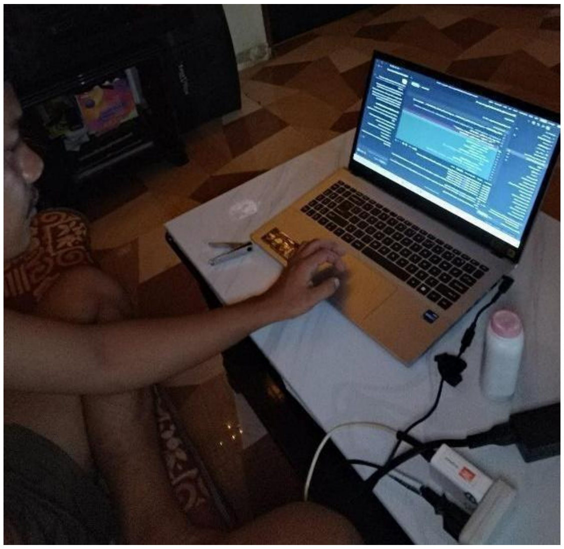
PERFORMANCE TESTING

```
Server Software:        Apache/2.4.58
Server Hostname:        localhost
Server Port:            80

Document Path:          /library_ai/
Document Length:        24767 bytes

Concurrency Level:      10
Time taken for tests:   10.707 seconds
Complete requests:      100
Failed requests:        0
Total transferred:      2521600 bytes
HTML transferred:       2476700 bytes
Requests per second:    9.34 [#/sec] (mean)
Time per request:       1070.672 [ms] (mean)
Time per request:       107.067 [ms] (mean, across all concurrent requests)
Transfer rate:          230.00 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    1   0.9      0       6
Processing:   462  654 355.1    561    3655
Waiting:      460  652 355.2    559    3655
Total:        462  655 355.1    562    3655
WARNING: The median and mean for the initial connection time are not within a normal deviation
        These results are probably not that reliable.

Percentage of the requests served within a certain time (ms)
  50%    562
  66%    580
  75%    605
  80%    652
  90%   1067
  95%   1089
  98%   1459
  99%   3655
 100%   3655 (longest request)
PS C:\xampp\htdocs\library_ai>
```

APPENDIX B

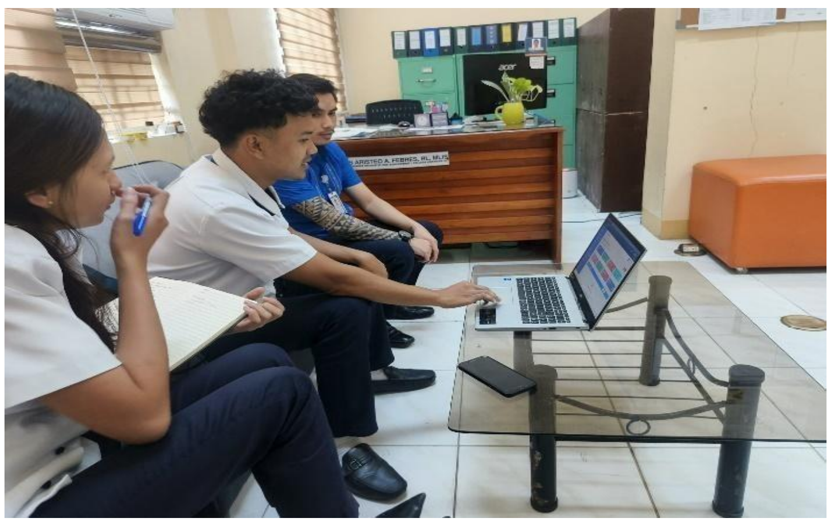
DOCUMENTATION

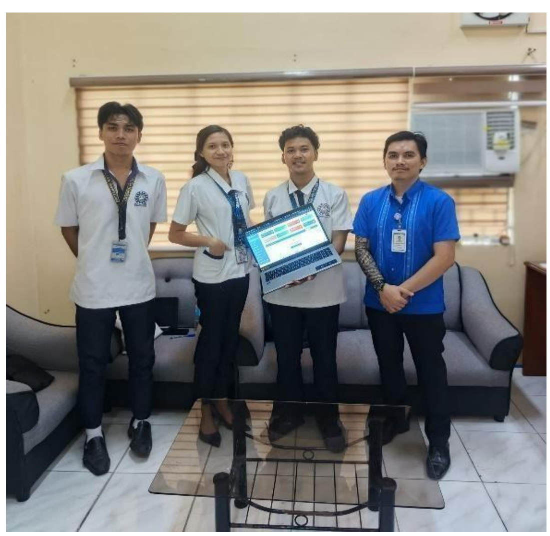
SYSTEM DEVELOPMENT



System Testing with the administrator, librarians and staffs.

System Testing with the client and gathering recommendations for system improvement.



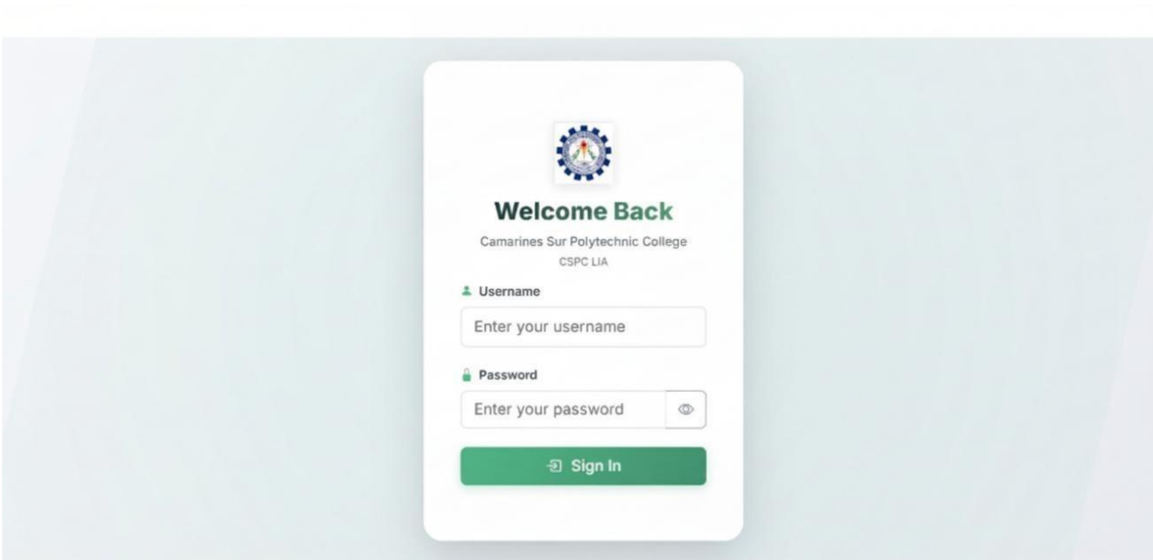


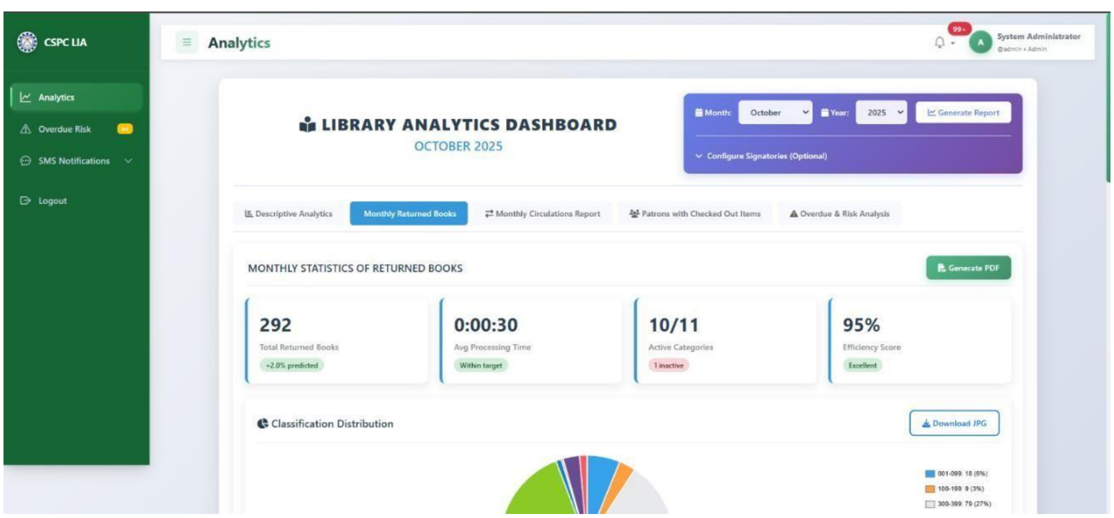
TITLE DEFENSE

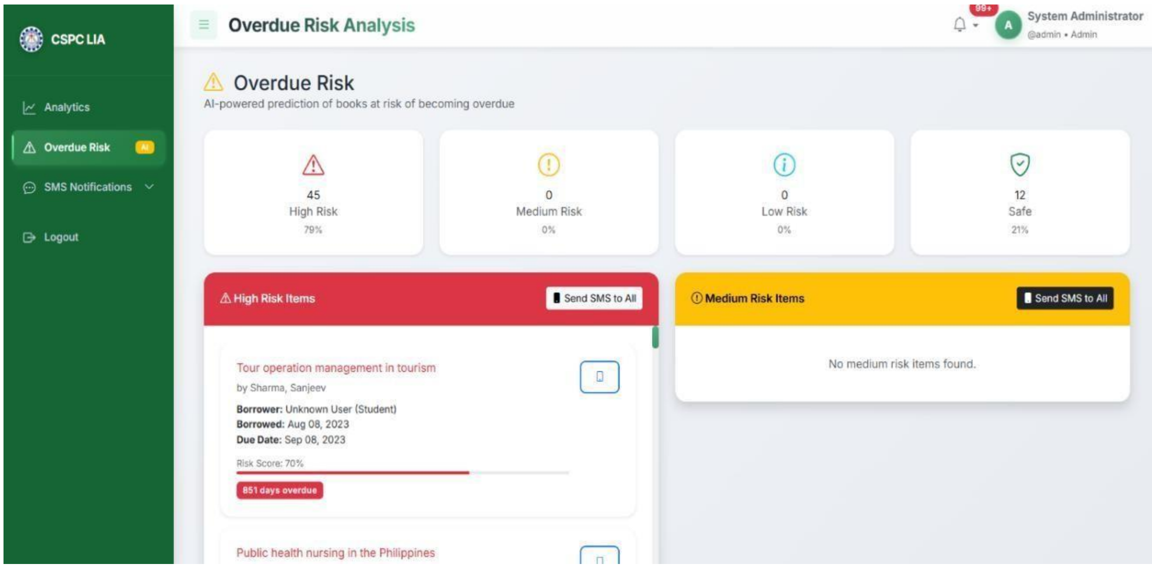
FINAL ORAL DEFENSE

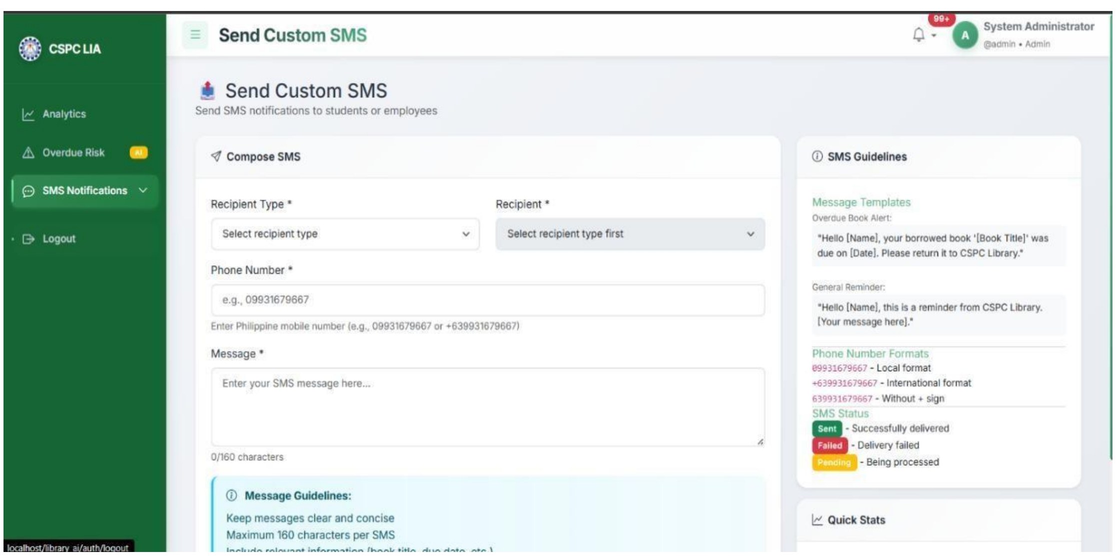
APPENDIX C

SAMPLE INPUT/OUTPUT/REPORTS



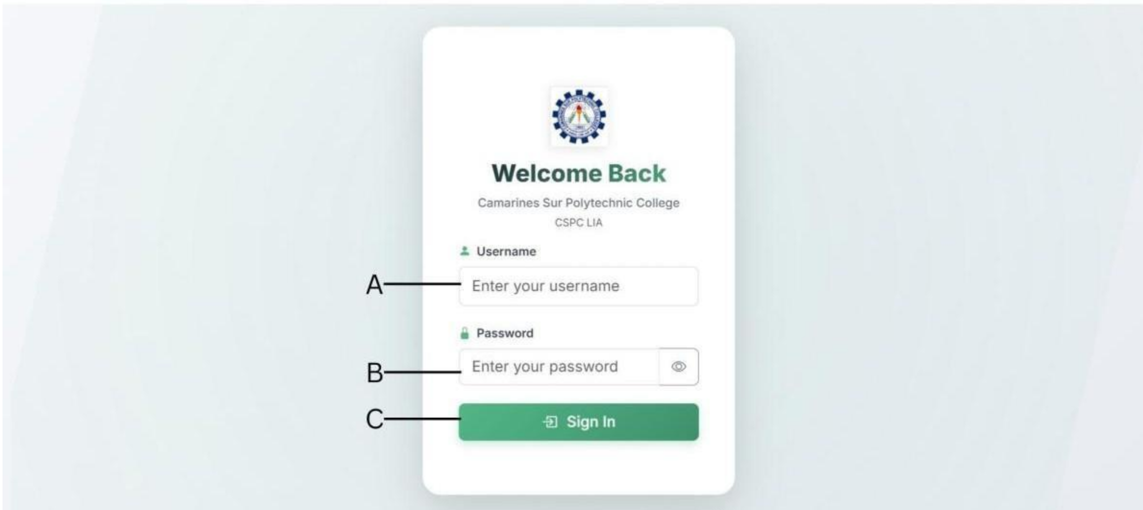



CSPC LIA
Analytics
Overdue Risk
AI
SMS Notifications
Logout
Overdue Risk Analysis
99+
System Administrator
@admin • Admin
Overdue Risk
AI-powered prediction of books at risk of becoming overdue
45
High Risk
79%
0
Medium Risk
0%
0
Low Risk
0%
12
Safe
21%
High Risk Items
Send SMS to All
Medium Risk Items
Send SMS to All
No medium risk items found.
Tour operation management in tourism
by Sharma, Sanjeev
Borrower: Unknown User (Student)
Borrowed: Aug 08, 2023
Due Date: Sep 08, 2023
Risk Score: 70%
851 days overdue
Public health nursing in the Philippines


CSPC LIA
Analytics
Overdue Risk
AI
SMS Notifications
Logout
localhost/library_ai/auth/logout
Send Custom SMS
99+
System Administrator
@admin • Admin
Send Custom SMS
Send SMS notifications to students or employees
Compose SMS
Recipient Type *
Select recipient type
Recipient *
Select recipient type first
Phone Number *
e.g., 09931679667
Enter Philippine mobile number (e.g., 09931679667 or +639931679667)
Message *
Enter your SMS message here...
0/160 characters
Message Guidelines:
Keep messages clear and concise
Maximum 160 characters per SMS
SMS Guidelines
Message Templates
Overdue Book Alert:
"Hello [Name], your borrowed book '[Book Title]' was due on [Date]. Please return it to CSPC Library."
General Reminder:
"Hello [Name], this is a reminder from CSPC Library. [Your message here]."
Phone Number Formats
09931679667 - Local format
+639931679667 - International format
639931679667 - Without + sign
SMS Status
Sent - Successfully delivered
Failed - Delivery failed
Pending - Being processed
Quick Stats

APPENDIX D

USER'S GUIDE



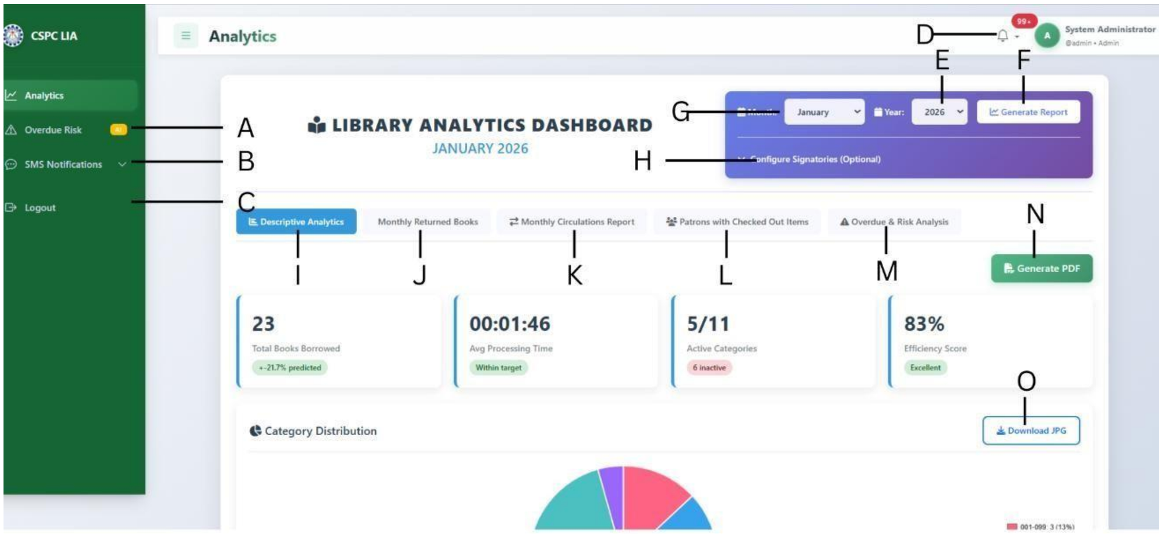
The Login Page allows authorized users to securely access the system using their registered credentials.

A. Username Field – Enter your assigned username provided by the CSPC Library or System Administrator. This identifies the user account attempting to access the system.

B. Password Field – Enter your corresponding password. Click the eye icon to toggle password visibility and verify your input if needed.

C. Sign In Button – Click to authenticate your credentials and log in to the system. If the credentials are valid, the user is redirected to the system dashboard based on their assigned role.

This login process ensures controlled access to the system, protects sensitive library data, and enforces role-based security for administrators, librarians, and authorized personnel.



The Analytics Page provides data-driven insights into library usage, circulation trends, and operational efficiency. It helps administrators and librarians monitor performance, identify risks, and generate reports for decision-making.

A. Analytics Menu - Provides access to the Analytics module from the system's main navigation panel.

B. Overdue Risk Menu - Redirects users to analytics related to overdue materials and risk assessment.

C. SMS Notifications Menu - Allows access to notification analytics and messaging-related features.

D. Menu Toggle Icon - Allows users to expand or collapse the side navigation menu for better workspace visibility.

E. System Notification Icon - Displays system alerts and notifications relevant to the user.

F. User Profile Menu - Shows the logged-in user role (e.g., System Administrator) and provides access to account settings and logout options.

G. Month Selector - Enables users to select the specific month for which analytics data that are displayed.

H. Year Selector - Allows users to choose the year corresponding to the analytics data.

I. Generate Report Button - Generates an analytics report based on the selected month and year.

J. Configure Signatories Option - Allows users to configure or assign report signatories prior to report generation.

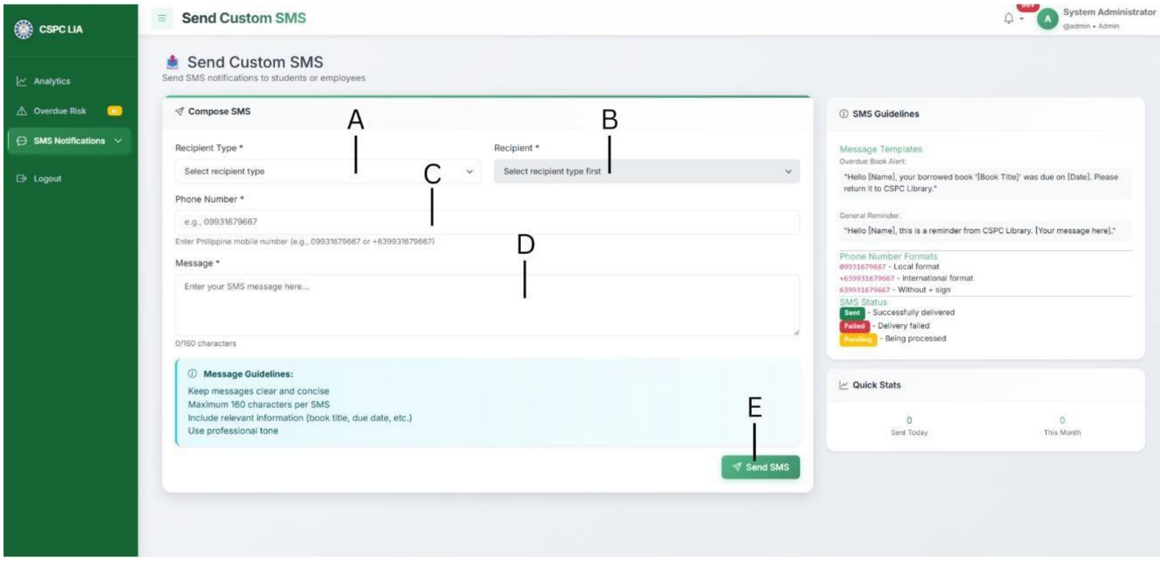
K. Descriptive Analytics Tab - Displays an overall summary of library performance metrics for the selected period.

L. Monthly Returned Books Tab - Shows statistics related to books returned within the selected month.

M. Monthly Circulations Report Tab - Provides data on borrowing and circulation activity.

N. Patrons with Checked-Out Items Tab - Lists analytics related to patrons currently holding borrowed materials.

O. Overdue & Risk Analysis Tab - Presents overdue trends and risk indicators to help identify potential issues.



The Send Custom SMS page allows the System Administrator or Librarian to send customized SMS notifications to students or employees for reminders, announcements, and overdue notices.

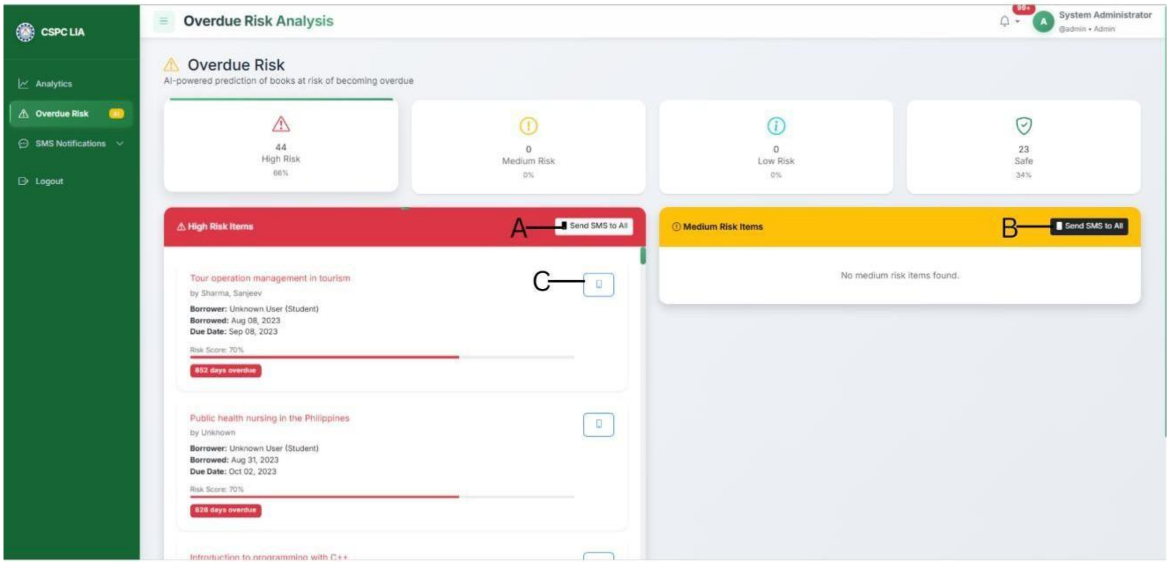
A. Recipient Type Dropdown – Click to choose the type of recipient (e.g., Student or Employee). This enables the system to load the appropriate recipient list.

B. Recipient Field – Click to select the specific recipient after choosing the recipient type. The list is filtered based on the selected recipient category.

C. Phone Number Field – Enter the recipient's Philippine mobile number using the correct format (e.g., 0993XXXXXXX or +63993XXXXXXX).

D. Message Field – Type the custom SMS message content. The character counter indicates the maximum allowed length.

E. Send SMS Button – Click to send the composed message. The system processes the request and updates the SMS delivery status.



The Overdue Risk Analysis page provides AI-powered predictions that identify library materials at risk of becoming overdue. This feature assists librarians and administrators in proactively managing circulation and reducing overdue incidents through early intervention and automated notifications.

A. Send SMS to All (High Risk Items) - Allows administrators to send SMS notifications to all borrowers with high-risk items. This function supports proactive reminders to reduce overdue returns.

B. Send SMS to All (Medium Risk Items) - Enables bulk SMS notifications for borrowers with medium-risk items. This feature is useful for preventive communication before items become overdue.

C. Individual Item Notification Button - Allows the administrator to send an SMS notification for a specific overdue or high-risk item.

APPENDIX E

COMMUNICATION LETTERS

CAPSTONE PROJECT AGREEMENT

This Agreement is made and entered into the May 22, 2025 by and between Team of **Code Crafter** a group og 3rd year Bachelor of Science Information Technology(BSIT) students from **Camarines Sur Polytechnic Colleges,** herein referred to as the "Developer",

-and-

CSPC Library Administration, located at **Nabua, Camarines Sur**, represented by **Yves Aristeo A. Febres**, referred to as the **"Client"**.

1. **Project Description**
   The developers agrees to design, develop and implement a **Barcode-Based Library Collection Analytics System** titled
   "From Shelves to Statistics: Leveraging Barcode Technology for Evidence-Based Collection Development Through Data Analytics of Book Collection."
   The goal is to modernize and optimize the CSPC library's collection management and usage analytics by:

- Generating real-time reports and data visualizations
- Supporting evidence-based collection development decisions

2. **Scope of Work**
   The Developer shall provide the following services:
   - Development of a web-based barcode-integrated book tracking system
   - Real-time API data integration for book borrowing and return monitoring
   - Automated overdue notification/reminder feature
   - Generation of digital reports of book titles per course
   - Visualization dashboards for book usage trends and borrowing frequency
   - Secure database and backup system for records management

3. **Project Duration**
   The implementation of the system starts on __________ and is expected to be completed by __________.

4. **Responsibilities of the Developer**
   - Deliver the system and all outlined modules
   - Conduct system testing, debugging, and optimization
   - Provide technical assistance and support during implementation

5. **Responsibilities of the Client**

- Provide access to existing library records and infrastructure details
- Collaborate in testing and evaluation of system features
- Assign personnel to participate in training and system onboarding

6. **Confidentiality**
Both parties agree to maintain the confidentiality of all sensitive library data (e.g., borrowing history, user records) throughout the development and deployment process.

7. **Termination**
Either party may terminate the agreement with written notice and valid reason, subject to mutual agreement and documented acknowledgment.

8. **Acceptance**
Upon successful delivery, deployment, and validation of the system, the Client agrees to sign an **Acceptance Form** confirming that the project meets the agreed-upon specifications.

IN WITNESS WHEREOF, the parties have executed the Agreement as of the date first written above.

For the Developer:

MA. ANGELICA T. NAZARIA
Project Head

For the Client:

YVES ARISTEO A. FEBRES, RL, MLIS
Unit Head, Learning and Development, CL III

SHIELA DONA S. MANLAPAZ, MIT
Capstone Project Adviser

APPENDIX F

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: MAY 28, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **MARICRIS L. RAMIZARES** hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is: **MARICRIS L. RAMIZARES - Panel Chair**

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges: **MS. ROSEL O. ONESA, MIT - Dean**

2. The confidential information disclosed under this Agreement is described as: **Contents of the TCP by:**

   Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang

   which is entitled:

"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges".

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in

connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
MARICRIS L. RAMIZARES, MIT	**MS. ROSEL O. ONESA, MIT**
Expert	**CSPC Representative**

Non-Disclosure Agreement Form

EFFECTIVE DATE: MAY 28, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **JOSEFINA H. LLAGAS** hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   JOSEFINA H. LLAGAS, Panel

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang

   which is entitled:

"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges".

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in

connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
JOSEFINA H. LLAGAS, MIT	**MS. ROSEL O. ONESA, MIT**
Expert	**CSPC Representative**

Non-Disclosure Agreement Form

EFFECTIVE DATE: MAY 28, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **JEREMY S. NEO** hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   JEREMY S. NEO - Panel

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang

   which is entitled:

"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges".

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages connection with or arising out of the performance or use of any portion of the confidential

information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
JEREMY S. NEO	**MS. ROSEL O. ONESA, MIT**
Expert	CSPC Representative

Non-Disclosure Agreement Form

EFFECTIVE DATE: April 7, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and SHIELA DONA S. MANLAPAZ hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   SHIELA DONA S. MANLAPAZ, MIT - Adviser

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang
   which is entitled:

"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
SHIELA DONA S. MANLAPAZ, MIT	MS. ROSEL O. ONESA, MIT
Expert	CSPC Representative
DATE: May 3,2025	DATE: May 3,2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: April 10, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and MARIVIC L. PAMIZARES hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   MARIVIC L. RAMIZARES - Consultant

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang

   which is entitled:

"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges "

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential

information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
MARIVIC L. RAMIZARES **Expert**	**MS. ROSEL O. ONESA, MIT** **CSPC Representative**
DATE: MAY 7, 2025	DATE: MAY 7, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: April 10, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and ELLAMAE S. ALAMO hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   ELLAMAE S. ALAMO - Grammarian

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang

   which is entitled:

'Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	**CAMARINES SUR POLYTECHNIC COLLEGES**
ELLAMAE A. ALAMO **Expert**	**MS. ROSEL O. ONESA, MIT** **CSPC Representative**

APPENDIX G

JOINT AFFIDAVIT OF UNDERTAKING

Joint Affidavit of Undertaking

We, **MA. ANGELICA T. NAZARIA**, of legal age, single and a resident of **SAN PEDRO BUHI, CAMARINES SUR**, **DINCE A. DELIVA**_, of legal age, single and a resident of **PALOYON PROPER NABUA CAMARINES SUR**, **JOHN PATRICK A. ACBANG**, of legal age, single and a resident of **ARO-ALDAO NABUA CAMARINES SUR**_, after having been sworn to in accordance with law, do hereby take oath and state :

(i) That, we are officially enrolled for the thesis/capstone project on the topic titled **LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT** in the College of Computer Studies of Camarines Sur Polytechnic Colleges.

(ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of **Bachelor of Science in Information Technology** Degree are original and our own work, and is not plagiarized.

(iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.

(iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/ Institute.

(v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

15 JAN 2026

IN WITNESS WHEREOF, I have hereunto set my name this ____ day of ___________ 202__ in

Nabua Camarines Sur, Philippines.

(1) MA. ANGELICA T.NAZARIA
Affiant
221004981

(2) DINCE A. DELIVA
Affiant
C21103691

(3) JOHN PATRICK A. ACBANG
Affiant
22100 7056

SUBSCRIBED AND SWORN TO before me this 15 day of JAN 2026 at Nabua Camarines Sur, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. 479;
Page No.: 86;
Book No.: XCVII;
Series of 2026

ATTY. JOSELITO F. FIGURACION
Notary Public
Until December 31, 2025
Roll of Atty's. No. 52026
PTR No. 3761510 1/2/2025 Nabua Cam. Sur
IBP O.R. No. 540225 02/18/2025
MCLE Compliance No. VII-0013316 valid 04/14/28
San Francisco, Nabua, Camarines Sur

APPENDIX H

ROLE ACCEPTANCE FORM

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: March 11, 2025

To: **Shiela Dona S. Manlapaz, MIT**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in **Bachelor of Science Information Technology**, are currently enrolled in Capstone Project 1.

We are writing to humbly request your service and expertise to serve as our **Adviser** for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Ma. Angelica T. Nazaria

Jhon Patrick A. Acbang

Dince A. Deliva

To: **MS. ROSEL O. ONESA, MIT**
OIC Dean, CCS

This formally signifies that I **ACCEPT/ REJECT** the request to serve as **Adviser** of the team CODE CRAFTER.

As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Capstone Project 1 until Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the capstone project.

Conformed:

Shiela Dona S. Manlapaz, MIT
Name and Signature

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: March 11, 2025

To: **Ellamae S. Alamo**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in **Bachelor of Science Information Technology**, are currently enrolled in Capstone Project 1.

We are writing to humbly request your service and expertise to serve as our **Grammarian** for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Ma. Angelica T. Nazaria

John Patrick A. Acbang

Dince A. Deliva

To: **MS. ROSEL O. ONESA, MIT**
OIC Dean, CCS

This formally signifies that I **ACCEPT/ REJECT** the request to serve as **Grammarian** of the team CODE CRAFTER.

As Grammarian, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Capstone Project 1 until Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the capstone project.

Conformed:

Ellamae S. Alamo
Name and Signature

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: March 11, 2025

To: **Marivic L. Ramizares**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in **Bachelor of Science Information Technology**, are currently enrolled in Capstone Project 1.

We are writing to humbly request your service and expertise to serve as our **Consultant** for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Ma. Angelica T. Nazaria

John Patrick A. Acbang

Dince A. Deliva

To: **MS. ROSEL O. ONESA, MIT**
OIC Dean, CCS

This formally signifies that I **ACCEPT/ REJECT** the request to serve as **Consultant** of the team CODE CRAFTER.

As Consultant, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Capstone Project 1 until Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the capstone project.

Conformed:

Marivic L. Ramizares
Name and Signature

APPENDIX I

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	CODE CRAFTERS
Course Code	IT 4113
Subject adviser/ CaPSA	Freddie Prianes, MIT

Name and Signature	Project Role	Email address and mobile#(s)
MA. ANGELICA T. NAZARIA	Project Head/Database Designer/ Document Writer	manazaria@my.cspsc.edu.ph 09516170135
JOHN PATRICK A. ACBANG	Network Designer/ UI Designer	joacbang@my.cspc.edu.ph 09942314625
DINCE A. DELIVA	Software Engineer/Programmer	dideliva@my.cspc.edu.ph 09914961797

APPENDIX J

FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM

Proponents/Researchers:

1. MA. ANGELICA T. NAZARIA
2. DINCE A. DELIVA
3. JOHN PATRICK A. ACBANG

Proposed Thesis/ Capstone Project Title:

'From Shelves to Statistics: Leveraging Barcode Technology for Evidence-Based Collection Development Through Data Analytics of Book Collection"

Submitted by: **MA. ANGELICA T. NAZARIA** (Signature of Project Head over printed name) Date: MAY 7, 2025	**Noted by:** **JOCELLE B. MONREAL, DIT** (Signature of Subject adviser over printed name) MAY 0 7 2025 Date:
Recommending Approval: **SHIELA DONA S. MANLAPAZ, MIT** (Signature of Adviser over printed name) Date: MAY 7, 2025	**Approved:** **ROSEL O. ONESA, MIT** OIC DEAN, CCS MAY 0 7 2025 Date:

FINAL PROJECT TITLE FORM

Proponents/Researchers:

1. MA. ANGELICA T. NAZARIA
2. DINCE A. DELIVA
3. JOHN PATRICK A. ACBANG

Proposed Thesis/ Capstone Project Title:

"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges "

Submitted by: **MA. ANGELICA T. NAZARIA** (Signature of Project Head over printed name) Date: SEPTEMBER 9, 2025	**Noted by:** **FREDDIE PRIANES ,MIT** (Signature of Subject adviser over printed name) Date: SEPTEMBER 9, 2025
Recommending Approval: **SHIELA DONA S. MANLAPAZ, MIT** (Signature of Adviser over printed name) Date: SEPTEMBER 9, 2025	**Approved:** **ROSEL O. ONESA, MIT** OIC DEAN, CCS Date: SEPTEMBER 9, 2025

APPENDIX K

CAPSTONE HEARING FORM (TD, POD, FOD)

CAPSTONE PROJECT HEARING FORM

Date Filed: 05 / 27 / 25 [☒] Title Proposal [☐] Pre-Oral [☐] Final Oral

Date of Hearing: 05 / 28 / 25 Time: 1:00 - 3:00 Venue: MAC LAB

Department: **COLLEGE OF COMPUTER STUDIES (CCS)**

Research Title:

LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES

Proponent/s:

MA. ANGELICA T. NAZARIA **DINCE A. DELIVA**

JOHN PATRICK A. ACBANG

Recommended by:

SHIELA DONA S. MANLAPAZ, MIT
Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

JEREMY S. NEO
PANEL MEMBER 1

JOSEFINA H. LLAGAS, MIT
PANEL MEMBER 2

MARICRIS L. RAMIZARES, MIT
PANEL CHAIR

NOTED BY:

FREDDIE PRIANES, MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

CAPSTONE PROJECT HEARING FORM

Date Filed: 12/11/25 [☐] Title Proposal [☒] Pre-Oral [☐] Final Oral

Date of Hearing: 12/12/25 Time: 12:00 - 3:00 Venue: OPEN LAB

Department: **COLLEGE OF COMPUTER STUDIES (CCS)**

Research Title:

LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES

Proponent/s:

MA. ANGELICA T. NAZARIA **DINCE A. DELIVA**

JOHN PATRICK A. ACBANG

Recommended by:

SHIELA DONA S. MANLAPAZ, MIT
Capstone Project Adviser

CERTIFICATION

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JEREMY S. NEO, MIS
PANEL MEMBER 1

JOSEFINA H. LLAGAS, MIT
PANEL MEMBER 2

MARICRIS L. RAMIZARES, MIT
PANEL CHAIR

NOTED BY:

FREDDIE PRIANES ,MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

CAPSTONE PROJECT HEARING FORM

Date Filed: 12/19/25 [☐] Title Proposal [☐] Pre-Oral [☒] Final Oral

Date of Hearing: 12/20/25 Time: 3:00-4:00 Venue: MAC LAB

Department: **COLLEGE OF COMPUTER STUDIES (CCS)**

Research Title:

LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES

Proponent/s:

MA. ANGELICA T. NAZARIA **DINCE A. DELIVA**

JOHN PATRICK A. ACBANG

Recommended by:

SHIELA DONA S. MANLAPAZ, MIT
Capstone Project Adviser

CERTIFICATION

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PANEL MEMBER 1

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MARICRIS L. RAMIZARES, MIT
PANEL CHAIR

NOTED BY:

FREDDIE PRIANES ,MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

APPENDIX L

PANEL RSC (TD, POD, FOD)

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED

COLLEGE of COMPUTER STUDIES

Title : LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES

Alias : Team Crafter

Date : May 28, 2025, 1:00 PM | / | TD | | POD | | FOD

Secretary : Jedele V. Gaspi

CHAPTER	PAGE NO.	MANUSCRIPT RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN
	Front Page	• Change title to "Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges".	○ The title has been revised to "Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges."
1	2-6	**1.1 Project Context** • Place the number citation at the end of the statement. • Limited discussion in CSPC Library context. **1.2 Purpose and Description of the Project** • Revise and base it on the modules presented. • Spell out CSPC. **1.3 Objectives of the Project** • Revise Objectives. • Add the archiving of the books. • Add prescriptive analytics based on the collected data. • 3. Follow this format: 3.1 Integrational Testing 3.2 System Testing 3.3 Usability Testing 3.4 Unit Testing 3.5 Performance Testing • Add SO4: Deployment **1.4 Significance of the Study** • Include students. **1.5 Scope and Limitations of the Project**	○ Citations were placed at the end of statements and the discussion was expanded to include the CSPC Library context. ○ The purpose and description were revised based on the modules, with CSPC spelled out in full. ○ Objectives were revised to include book archiving, prescriptive analytics, the specified testing formats, and SO4: Deployment. ○ The significance of the study was updated to include students. ○ The scope and limitations were expanded to discuss all objectives, with

	8-10	• Discuss all specific objectives in your scope. • Avoid hanging of pages. **1.6 Project Dictionary** • Remove Barcode Technology, Inventory Management, and Shelves to Statistics. • Define the specific testing you will conduct.	page formatting corrected to avoid hanging text. ○ The project dictionary was revised to remove Barcode Technology, Inventory Management, and Shelves to Statistics, and now defines specific testing.
2	17-19	• Revise based on revision of scope/suggestion in data analytics.(synthesis and gap needed for revision) • Add RRL evaluation of AI.	○ The synthesis and gap were revised to align with data analytics scope, with additional evaluation of AI included in the RRL.
		• Remove Barcode Implementation Technology in Libraries	○ The section on Barcode Implementation Technology in Libraries was removed.
3	27	• Highlight frameworks, and AI to be used. **3.1 Overview of Current Technologies to be Used in the System** • Remove barcode technology, database management system. • Discuss what framework for web development framework.	○ Frameworks and AI applications were highlighted, with discussion of the chosen web development framework.
	30	• Revise Hardware Specification and list down all hardware. • Remove Visual Studio Code from the hardware list.	○ The hardware specification was revised, Visual Studio Code was removed from the list, and updated hardware details were listed.
	32-36	• Add flowchart of the process. **3.2.3 Program Specification** • Follow proper format. • Revise diagram. • Show module in tabular format.	○ A process flowchart was added ○ ACM formatting was applied ○ The diagram was revised

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COMPUTER STUDIES

			○ The diagram was revised
	38	**3.3 Testing Plan** • Fix and Include Test Cases.	○ The testing plan was fixed and test cases were included.
	All	**Notes** • Follow proper ACM format.	○ ACM formatting was applied

SYSTEM		
MODULE NO. (DFD)	**RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)**	**ACTION TAKEN**
	• Implement AI • Remove bar code • Notify via SMS • Add search history (frequently searched books) • More analysis	○ AI was implemented in the system ○ Revised and the barcode removed. ○ Added search history

NOTED BY:

MARICRIS L. RAMIZARES, MIT
PANEL, CHAIRMAN

JEREMY JIREH S. NEO
PANEL, MEMBER 1

JOSEFINA R. LLAGAS, MIT
PANEL, MEMBER 2

Title	:	**Learning Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges**
Alias	:	CODE CRAFTER
Date	:	December 16, 2025, 2P M – 3 PM TD RecheckPOD FOD
Secretary	:	JOHNMIR ISAAC

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)**	**ACTION TAKEN**
2	**17**	• **Synthesis** ○ Synthesize the RRLs by topic and cite the authors using this format Author [Year] (Example: Reyes [2024]. ○ After the synthesis, discuss now the similarities and differences of the RRLs with the study.	• the synthesis is corrected
3	**25-26**	• **Overview of Current Technologies to be Used in the System** ○ Section title should now be Overview of Current Technologies Used in the System ○ Still, no specific current technology used was discussed. What was discussed were the modules which are not considered as current technologies we have now.	• Change to into Overview of Current Technologies Used in the System • Specific technology used was added to the discussion
	40	• **Testing Plan** ○ Per specific objectives, you have 4 tastings namely Functional Testing, Usability Testing, Compatibility Testing and Performance Testing. But in Testing Plan you stated Functional Testing, Interface Testing and Compatibility Testing. I have clearly pointed out that these sections should be in sync.	• The testing plan is now align the specific objectives
	41	• **Types of Testing** ○ Still not fully in sync with the testing mentioned in SO3.	• it is now align in SO3
		• **Level of functionality table** ○ The description seems to still be disconnected. Please revise and follow a clear progression. Here below is an example: ▪ For Completely Functional, you can describe it like this: The system works correctly in all tasks, shows stable performance, and is ready for deployment. ▪ For Highly Functional: The system performs its main tasks correctly, with only minor	• The manuscript do not have a function and highly functionnal

		issues that do not affect normal use.	
		• **User Interface Performance Rating Table:** not sure if this is really needed for these were not mentioned in your SOs. • Per your Chapter 1, you are not involved in any ISO 9241-11 metrics.	• revised
		• **Level of Compatibility Table:** Please revise also the description and follow a clear progression. • No rating scale for Performance Testing and usability testing	• revised
4	**58**	• **Table 9: Identified Problems and Proposed Solutions** o Your table only contain 3 modules as solution. How did you came up with the other modules stated in your table for functional requirements?	• revised
	94	• Please include a tabular format of respondents and their count. I have said this during your initial POD. I even asked you how many staff Supply does office has. Please prepare a table for your respondents after the sampling technique. Make sure to include 5 IT experts	• Added a tabular format for respondent and their counts
		• **Use Case Diagram of UtiliCSPC System** o Revise the diagram. All modules/cases should be inside the rectangle.	• Use CASE Diagram has been revised
	71-73	• Figure 7 is a Context Diagram not a DFD. • **Figure 8. Data Flow Diagram Level 1** o This should be the Level 0 DFD. But please redo for what you presented is not a DFD diagram. Check this tutorial which shows the context diagram and how to create a DFD Level 1: https://www.youtube.com/watch?v=bL00UAW_ygE. Please make sure that ALL functionalities and processes is reflected in your DFD Level 0.	• Context diagram and DFD are revised
3	**38**	• All Flowchart is incomplete, unclear end and no Input from user. Some symbols were not properly used. Please use this as reference: https://www.lucidchart.com/pages/what-is-a-flowchart-tutorial	• Flowchart is revised
	90-97	• **ALL Testing Discussion:** o Not following the PIIILS discussion format. It doesn't mean that you have cited an RRL, it's already PIIILS. PIIILS means Principal Findings, Interpretation of the findings, Interpretation in context of literature, Implication, Limitation and Summary.	• Revised following the PIIILS discussion

		• **User Interface Testing** ○ Again, I don't know if this is really a part of your study for this is not mentioned in your SO3. This was already mentioned during your initial POD. Also, no proof of the actual result of testing using the tools.	• We do not have a user interface testing
		• **Compatibility Testing Results** ○ Please include the actual result of the Device Matrix Testing and Cross Browser Tools. I need to know how did you come up with such numerical result.	• We don't have comptibilty testing
	90	• No result for Usability Testing and Performance testing which are stated in your SO3. • No discussion of the deployment in CSPC.	• The result of Usablity and Performance testing has been added
5	104-107	• **Findings** ○ Must have a section intro ○ Findings must answer your EVERY Specific Objective including its sub items. • **Conclusion** ○ This should contain your conclusion for every given finding. This explains the conclusion. • **Recommendations** ○ Add a section intro	• The section intro was added
		•	

SOFTWARE PROGRAM		
MODULE NO. (DFD)	**RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)**	**ACTION TAKEN**
	• Creating staff member, utilize the CSPC email and sen an invite link to the account • Use convenient name for the system • Include dashboard • Use proper terms for labels • Fix dropdown for colleges • Full name of programs • Recommendation of books while typing • Notification for upcoming outdated books • Remove redundant buttons • Scanning ID in borrowing books • Change visualizations for AI analytics • Have at least 1 year of data • Inquire about all data needed or what they want to display and also possible important data to display and for reportorial requirements	• Creating staff and utilizing the CSPC email and sending invite link is removed from the system in reason that it was no longer needed in the system • The system's name now is CSPC LIA • Dashboard implemented • Fixed the terms for labels • Full name of programs are now fixed • Notification already implemented • Redundant buttons was removed • Scanning ID in borrowing books is removed • The visualization for AI analytics is implemented • Had 1 year of data • Implented

	• Charts can be downloaded as image • Narrative style of report in AI • Remove unnecessary modules	• Downloadable charts report is now implemented • Narratve style of reports is implemented • The unnecessary modules are removedd

NOTED BY:

MARICRIS L. RAMIZARES, MIT
PANEL, CHAIRMAN

JEREMY S. NEO
PANEL, MEMBER 1

JOSEFINA H. LLAGAS
PANEL, MEMBER 2

Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified
COLLEGE of COMPUTER STUDIES

Title : **Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges**

Alias : CODE CRAFTER

Date : DECEMBER 20, 2025, 3:00 PM – 4:00 PM ☐ TD ☐ POD ☒ FOD

Secretary : JOHNMIR ISAAC

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)**	**ACTION TAKEN**
ALL		• FIX PAGINATION • Use proper indention • Fix hanging pages • Don't insert page number on the first part of every chapter	• Page number is fixed • Hanging pages has been fixed
TITLE	i	• USE PROPER FORMAT AND FONT & SIZE FOR TITLE	• The proper font and size of tile is fixed
PANEL OF EXAMINERS	ii	• Fix date	• The date is fixed
ACKNOWLEDGEMENTS	iii	• Use proper format in name suffix	• Name suffix is edited and used a proper format
ABSTRACT	v	• Include the overall numerical results	• Numerical results is included
TOC		• fix	• Table of content is fixed
2	12	• 2.1 sub topics • Missing number citations	• 2.1 sub topics is revised • Added missing number citation
	17	Synthesis • 1 topics, 1paragraph, then discuss the similarities and differences by topic	• The synthesis is revised
3	29	Functional requirements • Use single spacing	• Fixed spacing
	31	Non-functional • Fix table format	• Fixed the table format
5	100	Findings • Align with SO • Include the numerical result of testing	• Aligned with SO • Numerical result is included
APPENDICES		• All forms must be signed	
		•	

SOFTWARE PROGRAM		
MODULE NO. (DFD)	**RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)**	**ACTION TAKEN**
	• Remove first dashboard, just retain the analytics dashboard • Notifications should be for all modules • Legends should be of the same with charts on same card • Hide details that are not usually needed, but it can be	• The first dashboard has been removed • Notification for all modules has been added • Legends is revised • Revised • Monthly circulation report is revised

	viewed by having dropdown • Fix monthly circulation report	

NOTED BY:

MARICIRIS L. RAMIZARES, MIT
PANEL, CHAIRMAN

JEREMY S. NEO
PANEL, MEMBER 1

JOSEFINA H. LLAGAS
PANEL, MEMBER 2

APPENDIX M

CONSULTATION LOGS FORM

CONSULTATION LOGS FORM

Capstone Project Title:	"Enhancing Library Management: A Data Analytics Approach to Book Used and Borrowed Tracking Using Barcode Systems at CSPC"
Proponents:	Ma. Angelica T Nazaria John Patrick A. Acbang Dince A. Deliva
Alias:	CODE CRAFTER

Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chap 2	5/19/25						
	Remarks: RRL on Data Analytics - 2 big data & 1 library						
Deadline:							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	"Enhancing Library Management: A Data Analytics Approach to Book Used and Borrowed Tracking Using Barcode Systems at CSPC"						
Proponents:	Ma. Angelica T Nazaria John Patrick A. Acbang Dince A. Deliva						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chap 1	4/14/25						[signature]
	Remarks: Update Obj. Update Dictionary						
Deadline:							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	"Enhancing Library Management: A Data Analytics Approach to Book Used and Borrowed Tracking Using Barcode Systems at CSPC"						
Proponents:	Ma. Angelica T Nazaria John Patrick A. Acbang Dince A. Deliva						
Alias:	CODE CRAFTER						
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PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chap II							SHIELA DONA S. MANLAPAZ, MIT
	Remarks: *Fix Formating * Add more topics on project context and arrange the topic from broad to specifics * Revise the dictionary * Apply this improve title From Shelves to Statistics: Leveraging Barcode Technology for Evidence-Based Collection Development Through Data Analytics of Book Circulation						
Deadline:							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	"Enhancing Library Management: A Data Analytics Approach to Book Used and Borrowed Tracking Using Barcode Systems at CSPC"						
Proponents:	Ma. Angelica T Nazaria John Patrick A. Acbang Dince A. Deliva						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
							SHIELA DONA S. [illegible]
	Remarks: - Give letter to CSPC library asking access to inventory & borrowing books - Revise Objective - align Scope to the objectives						
Deadline:							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	**"From Shelves to Statistics: Leveraging Barcode Technology for Evidence-Based Collection Development Through Data Analytics of Book Collection"**
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG
Alias:	CODE CRAFTER

Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chap II							[signature]

Remarks:
PAGE 1-4 not ALM
Remove content not appropriate for RRL
Use all RRL either in GAP or synthesis
Revise Synthesis & Gap

Deadline:

8/3/25
submit chapter 1 V2
Revise chap II ASAP
[signature]

Remarks:

Deadline:

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	**"From Shelves to Statistics: Leveraging Barcode Technology for Evidence-Based Collection Development Through Data Analytics of Book Collection"**						
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Augst	3/9/25						SHELA DONA B. MANJARES
	Remarks: ·Improve Purpose & Description ·Add data analytics as objectives ·Revise black box criteria						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	**" Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"**
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG
Alias:	CODE CRAFTER
Total # of Modules:	as approved by the CAPSA /TSA:

PROTOTYPE

Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
						Gendaye

Remarks:

Apply Real time
Activate API subscription
- Under testing double check each Functional & Non-functional
- Align the question base on required question.
- properly cite sources of question or apply cronbach alpha
- Flowchart?
- System Architecture

Deadline:

Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature

Remarks:

Deadline:

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	**" Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"**						
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
	Remarks: Good to Go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	**" Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"**						
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
	01/12/2025						[signature]
	Remarks: * Good to go. Recommended for bluebook						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	**" Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"**						
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
	Remarks: Recommended for Blue book.						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	**" Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"**						
Proponents:	MA. ANGELICA T. NAZARIA DINCE A. DELIVA JOHN PATRICK A. ACBANG						
Alias:	CODE CRAFTER						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
	Remarks: Recommended for Blackbook.						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

APPENDIX N

SECRETARY'S CERTIFICATE

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the capstone project entitled **"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges "** prepared and submitted by Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang and that the same have not been amended, modified or obliterated.

Signed:

MR. JOHNMIR R. ISAAC
Secretary

Date signed: January 4, 2024

APPENDIX O

GRAMMARIAN'S CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the capstone project entitled **"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges"** as against the set of structural rules that govern research writing in accord with the composition of sentences, phrases, and words in the English language.

Signed:

MS. ELLAMAE S. ALAMO
Grammarian

Date signed: 01/17/24

APPENDIX P

CERTIFICATE OF TRANSFER

This is to certify that the project **"Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges "** developed by by Ma. Angelica T. Nazaria, Dince A. Deliva, John Patrick A. Acbang students of College of Computer Studies (CCS) of Camarines Sur Polytechnic Colleges (CSPC), as their undergraduate capstone project, has been officially transferred to the CCS Dean's Office, Camarines Polytechnic Colleges, for further research.

This certification is issued this 30th day of January 2026 of the CCS Dean's Office for whatever legitimate it may serve.

ROSEL O. ONESA, MIT
OIC Dean, CCS
Date 01/30/26

APPENDIX Q

CERTIFICATE OF UTILIZATION

ISO 9001:2015

CAMARINES SUR POLYTECHNIC COLLEGES
Nabua, Camarines Sur

CERTIFICATE

OF UTILIZATION

This is to certify that the capstone project entitled

Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges

an Intelligent Library System developed by the researchers as part of their undergraduate capstone project, has been officially deployed, adopted, and utilized by the Camarines Sur Polytechnic Colleges (CSPC) Library.

The system has been in active use starting December 18, 2025 to present, supporting library operations through its Notification Module, Analytics Module, and Reports Module. These modules enable data-driven monitoring of library usage, automated notifications, and evidence-based report generation to support informed collection development and library decision-making.

This certification attests that the system is currently being used by the CSPC Library staff and administrators for academic and operational purposes.

Issued this 18th day of December, 2025, at
Camarines Sur Polytechnic Colleges, Nabua, Camarines Sur,
for whatever legal, academic, and institutional purpose it may serve.

Certified by:

YVES ARISTEO A. FEBRES,RL, MLIS
Unit Head, Learning and Development, CL III

JEROME L. MAMANSAG
Administrative Assistant I
CSPC Learning Resources and Development

APPENDIX R

ACM FORMAT

Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic Colleges

Intelligent Library Management

A Data Analytics–Driven Library Management System

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Traditional library management in academic institutions often relies on manual or semi-automated processes, leading to inefficiencies in circulation monitoring, reporting, and decision-making. This study presents the design and deployment of an intelligent library system for Camarines Sur Polytechnic Colleges that integrates data analytics to support evidence-based operations. The system implements real-time data processing, prescriptive analytics, automated notifications, and reporting to improve collection monitoring, overdue management, and administrative reporting. It was developed using a modular architecture composed of Analytics, Reports, and Notification modules. Data analytics techniques were applied to analyze borrowing patterns and generate actionable insights for collection development, while artificial intelligence services via Hugging Face supported prescriptive recommendations. System evaluation employed black box testing, including unit, integration, usability, and performance testing. Results show that the system satisfied functional and usability requirements, demonstrating high responsiveness, reliability, and user satisfaction. The findings confirm that data-driven library systems enhance efficiency and

reporting accuracy.

CCS CONCEPTS • Library Information System • Data Analytics • Web-based application systems

Additional Keywords and Phrases: Intelligent Library Management System, Data Analytics, Evidence-Based Collection Development, Prescriptive Analytics, Automated Notifications, Reporting Module, Digital Transformation, Academic Libraries, Black Box Testing

ACM Reference Format:
Ma.Angelica T. Nazaria, John Ptrick A. Acbang, Dince A. Deliva, 2025.Leveraging Data Analytics for Intelligent Library Management of Camarines Sur Polytechnic

1 INTRODUCTION

Academic libraries play a crucial role in supporting education, research, and lifelong learning by providing structured access to knowledge resources. However, many academic institutions continue to rely on traditional or semi-automated library management practices that limit timely data analysis and informed decision-making. The rapid growth of information and user demand has increased the need for intelligent systems capable of processing circulation data and generating actionable insights [1][2]. Libraries that lack real-time analytics often experience inefficiencies in book tracking, overdue monitoring, and collection development [3]. Reading culture and continuous learning further emphasize the importance of responsive and data-driven library services [4]. Although integrated library systems are widely adopted, their analytical capabilities are frequently underutilized, resulting in manual report generation and delayed insights [5]. Effective use of circulation and usage data is essential for optimizing acquisitions, improving resource allocation, and enhancing service delivery [6].

2 BACKGROUND

The continuous advancement of information technology has increased the demand for efficient and reliable management systems in academic libraries to support service quality, accountability, and informed decision-making [1][2]. Core library operations such as circulation monitoring, collection evaluation, and report generation remain essential; however, many academic libraries still rely on manual or partially automated systems, resulting in data inaccuracies and delayed reporting [3]. Traditional library management practices depend heavily on manual tracking of borrowing transactions and static reports, limiting timely analysis of usage patterns and collection performance. Studies indicate that data-driven and automated library systems enhance operational accuracy and support evidence-based collection development through analytics and centralized reporting mechanisms [6][7]. In institutions utilizing integrated library systems, analytical features are often underutilized, leading to inefficient decision support and reduced administrative responsiveness [5][8]. This study implements an Intelligent Library Management System for Camarines Sur Polytechnic Colleges to address these operational and analytical limitations.

3 METHODOLOGY

3.1 Software Development Methodology

The researchers employed an Agile development approach utilizing the Scrum framework to enhance collaboration and efficiency throughout the software development process. The methodology followed an iterative structure by dividing the project into manageable sprint cycles, with each sprint focused on specific objectives such as module development, system integration, and testing. Continuous feedback from librarians, staff, and end users was incorporated to refine system functionalities and ensure alignment with operational requirements.

3.1.1 Research Design

The study was developed using the Agile Scrum methodology through iterative sprint cycles focused on module development, integration, and testing. Continuous feedback from librarians, staff, and users was incorporated to ensure the system aligned with operational requirements.

3.1.2 Research Method

The researchers utilized quantitative approach was employed using a questionnaire administered to users and librarians. The research instruments, data collection procedures, sampling method, participants and respondents, and the statistical techniques used for data analysis were also described.

3.1.3 Instrument

The researchers utilized an interview guide and a structured questionnaire as the primary instruments for data collection. These tools were administered to students and library staff to obtain relevant information necessary for system development.

3.1.3 Sampling Technique

The researchers utilize a purposive sampling technique to select participants directly involved in the operation and use of the library system. This approach ensured the inclusion of individuals with relevant knowledge and experience, including librarians, staff, and students. The method was considered appropriate as it enabled the collection of data from respondents capable of providing informed insights into the system's functionality, usability, and overall performance.

3.1.4 Respondents

The study involved a total of 599 respondents drawn from different stakeholder groups, These included 389 students (64.94%), representing the largest portion of the sample, followed by **190** employees (31.72%). In addition, 12 staff members (2.00%) and 6 librarians (1.00%) participated in the study, providing institutional and operational perspectives. The respondents also included **2** IT experts (0.33%), who contributed technical insights relevant to the system's development and evaluation.

3.1.5 Statistical Analysis Tool

To examine the collected data, the researchers utilized various statistical methods to systematically organize, summarize, and interpret the results. These methods allowed for the assessment of the system's performance, the analysis of user responses, and the validation of the reliability and significance of the findings. The main statistical tools applied in the study were the Arithmetic Mean and Slovin's Formula.

4 RESULTS AND DISCUSSION

4.1 Planning

The planning phase is the first stage of the SCRUM software development process, where the researchers defined the project objectives, requirements, and development strategies. During this phase, the researchers conducted a Sprint Planning meeting to establish project goals, assign roles such as the Scrum Master, and organize the development team and stakeholders. System requirements were gathered through discussions between the researchers and the client to ensure that both functional and non-functional requirements were addressed. The researchers also identified the required hardware and software resources and selected appropriate development approaches to support efficient system development.

4.2 Testing

The Intelligent Library System was subjected to extensive black-box testing to assess its usability, unit, integration and performance across all system modules.

4.3 Unit Testing

Unit testing results indicate that all integrated analytic modules of the system successfully passed their respective test cases.

Table 1: Unit Testing Result

Integrated Modules	Result	Interpretation
Prescriptive Analytics	18%	Passed
Overdue Risk	13%	Passed
Book Reports	72%	Passed
Overall	100%	Passed

Table 2 shows that all integrated analytical modules of the system successfully passed their respective test cases. The Prescriptive Analytics module achieved an 18% coverage, the Overdue Risk module recorded 13%, and the Book Reports module attained the highest coverage at 72%. Despite the variation in test coverage percentages, all modules met their expected outcomes and obtained a "Passed" result, demonstrating correct execution of their defined unit-level functions.

4.4 Integration Testing

Integration testing results shows that all integrated modules of the Intelligent Library System successfully passed the defined integration test cases.

Table 2: Integration Testing Result

Integrated Modules	Result	Interpretation
Authentication Module	1.00	Passed
Analytics Module	1.00	Passed
Overdue Risk	1.00	Passed
SMS Module	1.00	Passed

Overall	1.00	Passed

Table 2 show that all integrated modules of the Intelligent Library System successfully passed the defined integration test cases. The Authentication Module, Analytics Module, Overdue Risk Module, and SMS Module each obtained a result rating of 1.00, indicating a Passed status for all modules. These results confirm that the system components functioned correctly and consistently when integrated, demonstrating seamless interaction and reliable data flow within the overall system environment.

4.5 Usability Testing Result

The usability evaluation indicates that the system meets its overall usability objectives, demonstrating that users were able to navigate the interface efficiently, understand system functions with minimal effort, and complete required tasks effectively and satisfactorily.

Table 3: Usability Testing Results

Metric	Result	Interpretation
User Satisfaction	4.0	Agree
User Interface Design	3.0	Neutral
Accessibility	3.76	Agree
System Responsiveness	4.1	Agree
Error and Feedback Display	3.48	Neutral
Overall	3.67	Agree

Table 3 shows the results indicate an overall mean score of 3.67 (Agree), reflecting generally positive user perceptions of the system. Strong performance was noted in System Responsiveness (4.1), User Satisfaction (4.0), and Accessibility (3.76), showing that the system is efficient, easy to use, and suitable for regular library operations. However, User Interface Design (3.0) and Error and Feedback Display (3.48) received neutral ratings, suggesting the need for improvements in visual consistency and clearer system feedback to enhance usability, especially for first-time and non-technical users.

5 CONCLUSION AND RECCOMDATIONS

This study successfully developed and implemented an Intelligent Library Management System for the CSPC Library to address inefficiencies in manual cataloging, monitoring, and reporting processes. The system automated core library operations and integrated data analytics to deliver real-time, accurate insights that support evidence-based collection development and resource management. Comprehensive black-box testing showed that the system 100% met its functional and performance requirements, demonstrating high functionality, usability, reliability, and acceptable performance across all modules. The system's deployment improved operational efficiency, reduced librarian workload, and supported sustainable library management. To sustain and enhance these benefits, it is recommended that the CSPC Library conduct periodic workflow assessments, integrate advanced modules such as predictive demand forecasting and acquisition recommendations, perform regular regression testing and performance monitoring, and ensure continuous system maintenance, security updates, and scalability planning.

ACKNOWLEDGEMENTS

The researchers extend their heartfelt gratitude to all individuals who contributed to the successful completion of this project. Sincere appreciation is given to Ms. Rosel O. Onesa, MIT, OIC Dean of the College of Computer Studies, for her unwavering support; to Ms. Shiela Dona S. Manlapaz, MIT, the project adviser, for her constant guidance, valuable insights, and encouragement throughout the development of the Intelligent Library Management System; and to Ms. Marivic L. Ramizares for her expertise and constructive recommendations as both consultant and panelist. The researchers also thank Ms. Ellamae S. Allamo for her patience and dedication in refining the manuscript, the project panelists Mr. Jeremy Jireh S. Neo, MIS, and Ms. Josefina H. Llagas for their meaningful feedback, as well as classmates and friends for their continued motivation and support. Deep appreciation is extended to their families for their unconditional love and sacrifices, to Mr. Jerome L. Mamansag for his assistance and understanding, and to Mr. Yves Aristeo A. Febres, RL, MLIS, for his willingness to provide guidance beyond his responsibilities. Above all, the researchers give thanks to God for the strength, wisdom, and perseverance that made this achievement possible.

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APPENDIX S

CERTIFICATE OF PLAGIARISM CHECK

REPUBLIC OF THE PHILIPPINES
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CERTIFICATION

Date of Release:	January 26, 2026
Submission ID:	23367:126900327
Output Title:	LEVERAGING DATA ANALYTICS FOR INTELLIGENT LIBRARY MANAGEMENT OF CAMARINES SUR POLYTECHNIC COLLEGES
Author(s):	John Patrick A. Acbang Dince A. Deliva Ma Angelica Nazaria
Program:	Bachelor of Science in Information Technology

Authenticity Report:

Similarity Index Report: **7%**

Internet Sources: *1%*

Publications: *0%*

Student Papers: *7%*

- *Interpretation:* The similarity index report means that **7%** of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: ***0%***

AI-generated only: *0%*

AI-generated text that was AI-paraphrased: *0%*

- *Interpretation:* The AI generated report means that **0%** of the output indicates a *likely* AI-generated text and *likely* AI-paraphrased text, which is ***within the acceptable AI usage*** in the field of Engineering.

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